Verified constructions, a classification programme, and conditional reductions for Gallai's path decomposition conjecture

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Abstract

Gallai conjectured that the edges of every connected graph on n vertices can be decomposed into at most $\lceil n/2 \rceil$ paths. We present a self-contained study of the conjecture organised around three kinds of results. First, exact constructions: we prove that the path number of $K_{4,n}$ equals $(n+1)/2$ for every odd $n \geq 3$, develop a structure theory and a ring-closure theorem that reduce the remaining even columns of the odd-n_1 case of the Constantinou–Ellinas problem to finitely many base instances (closing the columns $n_2 = 6$, 8 and 10 except five named bases), and prove a chain theorem for odd clique chains, giving $p(G) \leq \lfloor n/2 \rfloor$ for $b \geq 2$ blocks and $p(G) \leq \lfloor n/2 \rfloor - 1$ for $b \geq 4$, via a new decomposition lemma for complete graphs of odd order minus an arbitrarily prescribed Hamilton path. Second, a classification programme for the sharp placement problem on complete graphs K_{2k+1}: we prove the prime-family theorem that for $n = p + 1$ with $p \geq 7$ a prime congruent to 3 mod 4, every non-infinity placement regime is dead, together with corrected Fourier identities for the infinity branch, and we register the evidence structure of the exception-set candidate, whose uniqueness clause is already in tension at $k = 21$ with the propagation theorem (a withdrawn quantitative registration is corrected to the paper-level bound $|\mathcal{E}_{21}| \geq 1$). Third, conditional reductions and formalisation: an endpoint-budget framework whose budget conjecture K implies Gallai's conjecture, a conditional chain registering Gallai against an explicit eight-item premise ledger, per-domain theorem-form closure of the aligned valuation criterion for $N \in \{6, 8, 10, 12\}$ with the general-N existence half open, Lean formalisations of $p(K_{4,n}) = (n+1)/2$ (odd n), $p(K_{2k+1}) = k + 1$ and a criterion-level bound (carried under an explicit hypothesis), and a machine-account layer for a six-family covering identity, all quoted with explicit one-way (superset) reading conventions. Every claim is stated at its verified strength; open problems and residual gaps are collected at the end.

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*Contact: 672178818@qq.com. Parts of the mathematical and bibliographic work, including automated literature cross-checking and machine verification of the computational accounts, were carried out with computer assistance; the division of labour is summarised in Section 4.

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1 Introduction, notation, and the shape of the results

Let G be a finite connected simple graph on n vertices. A *path decomposition* of G is a partition of $E(G)$ into paths, and $p(G)$ denotes the minimum number of paths in such a decomposition. In his 1966 Tihany lecture, published as [15], Gallai conjectured that $p(G) \le \lceil n/2 \rceil$ for every connected G; the conjecture was communicated by Erdős and is tracked as problem #583 in [1]. It remains open, with $\lfloor 2n/3 \rfloor$ the best general upper bound, due to Dean and Kouider [11].

The case analysis of the last six decades has been organized, to a remarkable degree, by the structure of the subgraph $\mathcal{E}(G)$ induced by the even-degree vertices, together with bounds on the degrees of G itself. Lovász [16] showed that $E(G)$ decomposes into at most $\lfloor n/2 \rfloor$ paths and cycles, and settled the case $|\mathcal{E}(G)| \le 1$; Pyber [20] proved that $\mathcal{E}(G)$ a forest implies $p(G) \le \lfloor n/2 \rfloor$, with the general bound $n/2 + O(n^{3/4})$; Favaron and Kouider [14] settled Eulerian graphs of maximum degree 4; and Fan [13] proved the conjecture when every block of $\mathcal{E}(G)$ is triangle-free with maximum degree at most 3. Two strengthenings followed: Botler and Sambinelli [4] proved it for graphs whose $\mathcal{E}(G)$ is a subgraph of a graph in their class $\mathcal{G}$ (each block of $\mathcal{E}(G)$ of maximum degree at most 3, at most one triangular block per component; their main theorem is stated with Gallai's ceiling $\lceil n/2 \rceil$ rather than the floor of Theorem 1.6 below), and Fan, Hou and Zhou [12] proved it when $\mathcal{E}(G)$ contains no pair of triangles and every block of $\mathcal{E}(G)$ has maximum degree at most 3. A parallel track bounds the degrees of G itself: $\Delta \le 5$ [3] and treewidth at most 3 [5]. The class-based programme then accelerated: Zhang, Liu and Hong [21] proved the conjecture, with the sharper $\lfloor n/2 \rfloor$ bound, for all 3-degenerate graphs except K_3 and K_5^-; Lu and Yi [17] did the same for treewidth at most 4 with exceptions K_3, K_5^-, K_5; and Chen and Wu [6] settled it for block graphs. On the algorithmic side, exact path partitions on subcubic graphs are computable in polynomial time [19]. Two further general results bound the problem from above: for triangle-free graphs, Chu, Fan and Zhou [8] proved the global bound $p(G) \le 3n/5$; for the *odd semi-cliques* (Definition 1.7), the question raised by Bonamy and Perrett [3] whether $p(G) \le \lfloor n/2 \rfloor$ holds for every connected graph that is not an odd semi-clique has its first substantial progress in Chu, Fan and Zhou [9]: if the E-subgraph ($\mathcal{E}(G)$) is isomorphic to a complete graph K_m with $m \le 15$, then $p(G) \le \lfloor n/2 \rfloor + 1$. Finally, minimal path decompositions of complete bipartite graphs were studied by Constantinou and Ellinas [10], whose algorithms cover three of four parameter cases; we return to the open case below.

1.1 Blocks versus global maximum degree

The Fan-line conditions quoted above are *block-level*: they constrain the blocks of $\mathcal{E}(G)$ one by one, not $\mathcal{E}(G)$ as a whole. The distinction is not pedantic; the two readings are not equivalent, in either direction. In $K_{1,5}$ with one pendant leaf attached at each of its six vertices, the even-degree

subgraph is $\mathcal{E}(G) \cong K_{1,5}$, which has a vertex of E-degree 5, while every block of $\mathcal{E}(G)$ is a single edge (maximum degree 1): a large *global* degree can coexist with a Fan-line hypothesis satisfied block by block. Conversely, a cut vertex of $\mathcal{E}(G)$ may belong to arbitrarily many blocks each of maximum degree at most 3, so the absence of high E-degree vertices is stronger than the block condition. Accordingly, the correct reading of the twenty-year-old Fan-line hypothesis is: *every block of* $\mathcal{E}(G)$ *is triangle-free and has maximum degree at most* 3 [13], read through Theorem 2(c) of [4], with the forest case of Theorem 2(b) of [4] going back to Pyber [20]. The theorems of Section 3.1 are new precisely at the block level: they admit a *block* of $\mathcal{E}(G)$ of maximum degree at least 4.

1.2 Contributions

The paper has four mathematical parts (Parts I–III being constructive and Part IV conditional and machine-backed) and a verification apparatus. We list the contributions, with pointers to the sections that carry them.

(i) **The exact path number of $K_{4,n}$ for odd n, and a decision procedure for the even columns of Case 4 of [10]** (Section 2.1). Constantinou and Ellinas [10] partition the parameter pairs of K_{n_1,n_2} into four cases and give minimal-decomposition algorithms for three of them, leaving open Case 4 (odd n_1, $1 \le n_2 \le n_1 - 1$) with an explicit request for its "exact characteristics". We close the $n_2 = 4$ column completely:

$$p(K_{4,n}) = (n+1)/2 \qquad \text{for every odd } n \ge 3,$$

improving the best published bound for the family (the published bound $k{+}2$, where $n = 2k{+}1$, from [20, 13, 17]) by one. The value for odd $n \ge 3$ is formalized in Lean as a single theorem with axiom profile `[propext, Classical.choice, Quot.sound]` and no `sorryAx`; numeric corollaries at $n = 5, 13, 19$ follow from it without independent gaps. For the remaining even columns we prove a general counting cap, a structure theorem for optimal decompositions across the whole even range, and a *ring-closure theorem* (one explicit base decomposition of $K_{2k_0+1,2h}$ into $k_0 + 1$ paths extends, by an explicit construction, to every $k \ge k_0 + h + 1$); the column $n_2 = 6$ closes completely, the columns $n_2 = 8, 10$ close except for five named base instances, and the conjecture that $p(K_{n_1,n_2}) = \lceil n_1/2 \rceil$ throughout the even range (Conjecture 2.9 in the body) becomes a column-by-column decision procedure with $h + 1$ base instances per column.

(ii) **Odd clique chains: refinements beyond block graphs, with the domain fixed** (Section 2.4). Chen and Wu [6] proved Gallai's conjecture for all block graphs. For the subfamily of *odd clique chains* — block graphs whose block tree is a path and whose blocks are $B_i \cong K_{2j_i+1}$ with *all* j_i *odd*, i.e. blocks of order $\equiv 3 \pmod 4$, K_3 allowed — we prove

$$p(G) \le \lfloor n/2 \rfloor \quad (b \ge 2), \qquad p(G) \le \lfloor n/2 \rfloor - 1 \quad (b \ge 4),$$

where b is the number of blocks and $n = 2\sum_i j_i + 1$, along with the exact value $p(G) = \lceil n/2 \rceil$ for a single block. Two qualifications fix the domain precisely. First, *every* vertex of such a chain has even degree: a non-cut vertex of a block K_{2j_i+1} has degree $2j_i$, and a cut vertex shared by consecutive blocks has degree $2j_i + 2j_{i+1}$; hence $\mathcal{E}(G) = G$. Second, the complementary case — chains containing blocks of order 1 (mod 4) (even j_i, e.g. K_5) — is *open*: no general construction is known, and in-chain existence is machine-verified only for the block values K_5 and K_9. For $b = 3$ the -1 refinement is likewise open and reduces to a distinctness question on the middle block. The local engine is: *for every odd* j *and every Hamilton path* P *of* K_{2j+1},

the graph $K_{2j+1} - P$ decomposes into j paths; the base zigzag objects are classical [18, 2], the statement for an arbitrarily prescribed Hamilton path appears to be new.

(iii) **Fan-type theorems admitting a block of $\mathcal{E}(G)$ of maximum degree ≥ 4** (Section 3.1). The Fan line [13, 4, 12] has kept its requirement at the block level for twenty years: every block of $\mathcal{E}(G)$ has maximum degree at most 3. We prove class theorems that admit a block of $\mathcal{E}(G)$ of maximum degree at least 4: Theorems 3.1 and 3.2 (if $\mathcal{E}(G)$ is triangle-free, high E-degrees are compensated by an odd-degree condition, and an explicit alignment condition holds at one high vertex, then $p(G) \leq \lfloor n/2 \rfloor$; and the same bound with no alignment hypothesis whenever distinct high vertices have no common EV-neighbour), and a two-high-vertex theorem with a corner-case reduction. An honesty note travels with these results: our witness family $W(m)$ (m odd, ≥ 5) has $\mathcal{E}(W(m)) \cong K_{2,m}$, a single block of maximum degree m, and has treewidth 5, degeneracy 5 and maximum degree at least $m+1$; it therefore lies *outside* the degeneracy and treewidth classes of [21, 17] and outside every published class we are aware of, so our theorems and the class-based line [21, 17, 5, 3, 6] are incomparable rather than competitive. The general statement is posed as Conjecture 3.7.

(iv) **A conditional route: Gallai modulo an explicit premise ledger** (Part IV, Sections 4–C). We develop an endpoint-budget framework: the per-vertex *endpoint budget* $\beta(G, x)$ (Definition 1.17) and the *budget conjecture* K that $\beta(G, x) \leq \lceil n/2 \rceil$ always. On the proved side we establish an absorption-cost viewpoint (Remark 1.16) unifying three obstruction phenomena from the literature; that K implies Gallai's conjecture (Theorem 1.18, proved here), so K is a genuine strengthening; and a conditional chain (Theorem 1.19, stated here and carried at pointer level in the ledger subsection) assembling Gallai $\Leftarrow K \Leftarrow$ degree-two layer of $K \Leftarrow$ SC-PH $\Leftarrow$ a placement conjecture $\Leftarrow$ (E)+(R), with the bridge from the valuation line (the criterion of Section 3.2.7) carried as a parallel, redundant-for-implication route. The chapter's central deliverable is an *explicit hypothesis ledger*: an eight-item premise set, restated descriptively in Section 1.8 below. The premise set mixes three kinds of conjuncts: external steps flagged as unverified against the journal record (the all-odd support of the Lovász decomposition theorem as consumed by our proof of Theorem 1.18), open conjectures internal to the paper (BP-odd, the odd-order density conjecture, SC-PH), and paper-open items with machine backing. No premise is claimed to be proved.

(v) **Verification methodology** (Sections 4 and the formalization section, Appendices). All exact values and all decompositions cited in this paper were produced and re-checked by an independent machine pipeline (a bitmask dynamic program plus a decomposition verifier asserting that the parts are simple paths and exactly partition the edge set). The large-scale machine accounts of Part IV are quoted at their measured strength, under two standing conventions. First, the *one-way superset convention*: every machine check of the covering identity is performed in a reading that omits one definitional branch (the `v7` branch, whose machine face is frozen because the legacy candidate-generation source files are lost), giving a conservative superset $\mathrm{fam}_{\mathrm{noV7}} \subseteq \mathrm{Im} \cap L$; zero counterexamples on the superset face strictly imply zero counterexamples on the true face, and not conversely (the residual `v7`-only mass is nonzero and precisely measured on the tested domains). Second, the *accounting layers are kept separate*: the nine-domain key tables (comprising 2,412,436 keys and 98,602,396 key-weighted positions over $N \in \{6, \ldots, 14\}$) are counted key-weighted, each key once, while the per-instance accounts at $N = 8$ and $N = 9$ are instance-weighted; the two accountings are never mixed, and the $N = 14$ domain is an affine-normalized domain whose equivalence to exhaustive enumeration is unproven. Lean certificates carry their trust boundaries verbatim: theorems proved

from first principles are reported with their axiom profiles, and finite certificates checked by `native_decide` carry the kernel-reflection axiom `Lean.ofReduceBool` explicitly.

1.3 Claim-strength declaration

Six no-claim constraints bind the whole account, including this introduction; we state them verbatim. We do *not* claim:

(1) closure at $L \geq 13$ (machine evidence at larger N is extrapolation evidence, not proof; only candidate-level status is claimed there);

(2) general-N paper closure of the covering identity (the general-N existence half stays open);

(3) that the four-piece (E)+(R) landing is complete;

(4) paper closure of the general-L uniform-construction account (the capacity-starved branch of the channel-or-menu dichotomy programme);

(5) full closure of that dichotomy programme;

(6) an unconditional theorem-form statement of the criterion of Conjecture 3.26. The Lean theorem `thmA2663_upgraded` is a *criterion-level bound*: under the explicit hypotheses that the target side has the prescribed size and that val ≤ 1, it proves $g_{13} \leq d_2 + 1$ with axiom profile `[propext, Quot.sound]` and no `sorryAx`; the hypothesis val ≤ 1 itself is *not* proved by the theorem, and a per-instance criterion verified over finitely many N is not a feasibility-existence proof for general N.

1.4 Notation and preliminaries

All graphs are finite, simple and undirected. A *path* has at least one edge and no repeated vertex; a *path decomposition* of G is a set D of paths whose edge sets partition $E(G)$; $p(G)$ is the minimum size of a path decomposition. For a decomposition D and a vertex x we write $D(x)$ for the number of paths of D having x as an endpoint. A *block* of G is a maximal 2-connected subgraph or a bridge (together with its endpoints); every two blocks share at most one vertex, a *cut vertex*. G is *k-degenerate* if every subgraph has a vertex of degree at most k. The *even-degree subgraph* $\mathcal{E}(G)$ is the subgraph induced by the vertices of even degree; for $v \in V(G)$ we write $d_{\mathrm{E}}(v)$ for the degree of v inside $\mathcal{E}(G)$, the *E-degree* of v; a vertex with $d_{\mathrm{E}}(v) \geq 4$ is called *high*. Note that $d_{\mathrm{E}}(v) \geq 4$ implies in particular that v has even degree in G.

We collect the three folklore lower bounds used throughout; proofs are included for completeness.

Lemma 1.1 (odd-degree lower bound; endpoint parity)**.** *Let D be a path decomposition of G. Then:*

(i) $2|D| \geq \mathrm{odd}(G)$*, where* $\mathrm{odd}(G)$ *is the number of odd-degree vertices; in particular* $p(G) \geq \mathrm{odd}(G)/2$*;*

(ii) every odd-degree vertex is an endpoint of at least one path of D;

(iii) for every vertex v, $D(v) \equiv \deg_G(v) \pmod 2$*.*

Proof. For a path P and a vertex v, the degree of v in P is 0, 1 or 2, and it is 1 exactly when v is an endpoint of P. Summing over D, $\deg_G(v) = \sum_{P \in D} \deg_P(v) \equiv D(v) \pmod 2$, which gives (iii); (ii) is (iii) for odd $\deg_G(v)$; and (i) follows by charging each odd-degree vertex to one path ending at it, each path being charged at most twice. □

Lemma 1.2 (edge-count lower bound)**.** *If G is connected with $n \geq 2$ vertices and m edges, then $p(G) \geq \lceil m/(n-1) \rceil$.*

Proof. Each path has at most $n-1$ edges, so k paths cover at most $k(n-1)$ edges. □

Lemma 1.3 (disconnected sum)**.** *If G has components $G_1, \ldots, G_c$ on $n_1, \ldots, n_c$ vertices, then $\sum_i \lfloor n_i/2 \rfloor \leq \lfloor n/2 \rfloor$.*

Proof. Immediate from $\sum_i \lfloor n_i/2 \rfloor \leq \lfloor \sum_i n_i/2 \rfloor$ for integers. □

The three standard tools from the literature, stated in the form used later:

Theorem 1.4 (Lovász [16])**.** *Every n-vertex graph (not necessarily connected) has a decomposition of its edge set into at most $\lfloor n/2 \rfloor$ paths and cycles.*

Theorem 1.5 (Pyber [20])**.** *If $\mathcal{E}(G)$ is a forest, then $p(G) \leq \lfloor n/2 \rfloor$.*

Theorem 1.6 (Fan [13])**.** *If every block of $\mathcal{E}(G)$ is triangle-free with maximum degree at most 3, then $p(G) \leq \lfloor n/2 \rfloor$.*

Definition 1.7 (odd semi-clique; [3])**.** An *odd semi-clique* is a graph obtained from the complete graph on $2k+1$ vertices by deleting at most $k-1$ edges.

Observation 1.8 (density obstruction)**.** An odd semi-clique on $n = 2k+1$ vertices has $m \geq \binom{2k+1}{2} - (k-1) = 2k^2 + 1$ edges, so by Lemma 1.2 $p \geq \lceil (2k^2+1)/2k \rceil = k+1 = \lceil n/2 \rceil$; in particular no odd semi-clique admits a $\lfloor n/2 \rfloor$-path decomposition.

Finally we record the classical Walecki partition of the complete graph of odd order, in the form used in Section 2.4; the base objects are classical [18, 2]. Fix $j \geq 1$ and let $V = \{\infty\} \cup \mathbb{Z}_{2j}$. For $k \in \mathbb{Z}_{2j}$ let C_k be the cycle

$$C_k \;=\; \infty,\, k,\, k+1,\, k-1,\, k+2,\, k-2,\, \ldots,\, k+(j-1),\, k-(j-1),\, k+j,\, \infty$$

(indices mod $2j$); equivalently the finite vertices of C_k are $p_0 = k$ and $p_s = k + (-1)^{s+1} \lceil s/2 \rceil$ for $s = 1, \ldots, 2j-1$.

Lemma 1.9 (Walecki partition; classical, cf. [18, 2])**.** *For $k = 0, 1, \ldots, j-1$ the cycles C_k are edge-disjoint and $\bigsqcup_{k=0}^{j-1} E(C_k) = E(K_{2j+1})$.*

Proof. Each C_k is a cycle: the finite vertices $k, k \pm 1, \ldots, k \pm (j-1), k+j$ are $2j$ distinct elements covering $\mathbb{Z}_{2j}$. We show every edge lies in exactly one C_k, k taken mod j. Write $\delta(x,y)$ for the circular distance of $x, y \in \mathbb{Z}_{2j}$, i.e. the minimum of $\pm(x-y)$ mod $2j$ in $\{0, \ldots, j\}$.

Reading off consecutive differences in the sequence p_s (as above, $p_{s+1} - p_s = \pm(s+1)$), one checks that the finite edges of C_k are exactly the pairs $\{k+t,\, k-t\}$ for $t \in \mathbb{Z}_{2j} \setminus \{0, j\}$ and the pairs $\{k+t,\, k-t+1\}$ for $t \in \mathbb{Z}_{2j}$, together with the two ∞-edges $\{\infty, k\}$, $\{\infty, k+j\}$. (In each of the two forms an edge is produced twice, once from each endpoint, so it is counted once; $t = 0$ and $t = j$ would give loops in the first form and are excluded.) Hence C_k has, among finite edges, two edges of each circular distance $1, \ldots, j-1$ and one edge of distance j.

Now fix an edge $e = \{x, y\}$ of K_{2j+1} and let $d = \delta(x,y)$.

- If $y = x + j$ (the diameter case, $d = j$): for j odd, e has the form $\{k+t, k-t+1\}$, which forces $2k \equiv 2x + d - 1 \pmod{2j}$, a unique solution mod j (the right side is even); for j even, e has the form $\{k+t, k-t\}$ with $2t \equiv d = j$, forcing $2k \equiv 2x + j \pmod{2j}$, again a unique solution mod j. So e lies in exactly one C_k.

- If $d < j$ is even: $e = \{x, x+d\} = \{k+t, k-t\}$ forces $2k \equiv 2x + d \pmod{2j}$, unique mod j.
- If $d < j$ is odd: $e = \{x, x+d\} = \{k+t, k-t+1\}$ forces $2k \equiv 2x + d - 1 \pmod{2j}$, unique mod j.
- If $e = \{\infty, x\}$: $e \in C_k$ iff $x \in \{k, k+j\}$, i.e. $k \equiv x$ or $x + j \pmod{2j}$, a unique class mod j.

In each case e lies in exactly one of $C_0, \ldots, C_{j-1}$, and each C_k has $2j+1$ edges, so the $j(2j+1) = \binom{2j+1}{2}$ edges are partitioned. □

1.5 Notation for the machine chapters

The machine chapters (Section 4 and the formalization material of Section C) work with finite combinatorial instances; we fix the notation once here, transcribed so that the chapter statements are readable without further apparatus.

Definition 1.10 (instances)**.** An *instance* is $I = (N, P, T, d)$: $\mathbb{Z}_N := \mathbb{Z}/N\mathbb{Z}$; $P \subseteq \mathbb{Z}_N$ with $|P| = p$; $T = (t_1, t_2, t_3)$, three pairwise distinct elements of $\mathbb{Z}_N$; $d = (d_1, d_2, d_3)$ with $d_i \geq 1$; and the capacity surplus $\beta := p - d_1 - d_2 - d_3 \geq 1$ (capacity admission: $d_1 \leq p - [t_1 \in P]$). The shifts are $\delta_{12} := t_1 - t_2$, $\delta_{13} := t_1 - t_3$, $\delta_{23} := t_2 - t_3 \pmod{N}$. The instance is *aligned* (*so1*) when, with $g := \gcd(\delta_{12}, N)$ and $L := \operatorname{ord}(\delta_{12})$, one has $\delta_{13} \equiv k\,\delta_{12} \pmod{N}$ for the canonical $k \in [2, L)$; equivalently, t_1, t_2, t_3 lie on one coset ("line") of $\langle \delta_{12} \rangle$.

Definition 1.11 (legal pairs and the value functional)**.** For $C_1 \subseteq P \setminus \{t_1\}$ put $B_{12} := C_1 - \delta_{12}$ and $Q(C_1) := P \setminus (C_1 \cup (B_{12} \cap P) \cup \{t_2\})$. A *legal pair* is (C_1, C_2) with $C_1 \subseteq P \setminus \{t_1\}$, $|C_1| = d_1$, $C_2 \subseteq Q(C_1)$, $|C_2| = d_2$. With $F_{13} := (C_1 - \delta_{13}) \cap P$ and $F_{23} := (C_2 - \delta_{23}) \cap P$, the *value* of a legal pair is the three-term count

$$\operatorname{val}(C_1, C_2) := \big|F_{13} \setminus (C_1 \cup C_2)\big| + \big|F_{23} \setminus (C_1 \cup C_2 \cup F_{13})\big| + \big|\{t_3\} \cap P \setminus (C_1 \cup C_2)\big|.$$

The *min-val* of the instance is the minimum of val over all legal pairs.

The *so1 (aligned) family* is the family of aligned instances; it is the family on which the paper layer of the conditional chain is formulated, and the machine accounts of Section 4 sweep exactly the statement "every so1 instance with $\beta \geq 2$ has min val ≤ 1" on the explicitly tested domains (the sweep domain $\beta \geq 1$ contains the $\beta \geq 2$ slice). The candidate form of that statement is registered as Conjecture 3.26 in Section 3.2.7; its general-N existence half is Conjecture 3.27.

1.6 Two reinsertion lemmas from the literature

The Fan-type arguments of Section 3.1 and the budget framework of Section 1.8 consume two deletion–reinsertion lemmas, which we state in the form used throughout; both follow the restatements in [4] (Lemmas 6 and 7 there), and we record our provenance boundaries with them.

Lemma 1.12 (addible edge; [13],[4])**.** *Let $uv \in E(G)$ and let D' be a path decomposition of $G - uv$. If $D'(v) > |\{w \in N_{G-uv}(u) : D'(w) = 0\}|$, then G has a path decomposition D with $|D| = |D'|$, $D(u) = D'(u) + 1$, $D(v) = D'(v) - 1$, and $D(w) = D'(w)$ for all other w. One says that uv is addible towards u.*

Remark 1.13 (transcription, downgrade and machine evidence). The proof of Lemma 1.12 in [13] could not be consulted; the statement follows the restatement in [4] (Lemma 6) and has not been independently checked against the original. A strengthening of the lemma that additionally requires

the underlying re-arrangement to preserve the endpoint profile of D' pointwise is false in general (a five-vertex counterexample is machine-certified); the lemma as stated imposes no such requirement and has no known counterexample, and the Lean formalization used in Section 4 carries the four-clause form obtained by deleting the final pointwise clause $D(w) = D'(w)$ for the remaining vertices — no application in this paper consumes that clause. The machine evidence is direction-sensitive and has been direction-corrected: the bidirectional rescan is complete at $n \leq 6$, all 1,701,141,150 ordered instances at $n = 6$ (the fixed-orientation half exactly reproducing the historical census of 850,570,575), with zero violations (Section 4.1).

Lemma 1.14 (star absorption; [13],[4])**.** *Let $B = \{uw_1, \ldots, uw_h\}$ be a set of h edges incident to u, let D' be a path decomposition of $G - B$ in which every neighbour of u in G has $D' \geq 1$, and let uw be a specified edge of B. Then there is a set $A \subseteq B$ with $uw \in A$ and $|A| \geq \lceil h/2 \rceil$ such that the edges of A can be added back one by one, each by Lemma 1.12 towards u, preserving $|D|$ and increasing $D(u)$ by one at each step.*

Remark 1.15. The hypothesis of Lemma 1.14 is read over the neighbours of u *in* G, so that it includes the star vertices w_i themselves, which are not neighbours of u in $G - B$; this stronger reading is the one verified in every application below. As with Lemma 1.12, the proof in [13] (Lemma 3.6 there) could not be consulted, and the transcription from the restatement in [4] (Lemma 7) has not been independently checked against the original.

1.7 The absorption-cost accounting and the budget conjecture

Several proofs in the literature — Lovász' deletion–reinsertion, Fan's star absorption, and the analysis of minimal counterexamples — fail at the same type of point: one needs a decomposition of a reduced graph in which *specified vertices hold sufficiently many path endpoints*, and the count of available endpoints is not controlled by the usual hypotheses. The following elementary accounting fact, quoted from the project's verification work, quantifies the obstruction class once and is used only as a viewpoint in this paper.

Remark 1.16 (absorption cost). Let G be a graph, D a path decomposition of G, v a vertex not incident with any edge of G (v may be a new vertex), and $N = \{w_1, \ldots, w_d\} \subseteq V(G)$ a set of d distinct vertices; let $G^+ = G + \{vw : w \in N\}$. Build the bipartite graph $B(D)$ with left side the d edges vw of the star and right side the paths of D, where vw is adjacent to P iff P has w (or v) as an endpoint; let μ^* be a maximum matching of $B(D)$. Within the operation class — (i) attach a star edge vw to the w-end of a path of D ending at w; (ii) pair two star edges vw, vw' into the single path w–v–w'; (iii) let a star edge be a single-edge path — the minimum number of new paths required to extend D to a decomposition of G^+ is exactly $\lceil (d - \mu^*)/2 \rceil$, and the resulting $D(v)$ can be made at least μ^*. The equality holds within the operation class and is *not* a lower bound for arbitrary decompositions of G^+. Three obstruction phenomena in the literature are instances of "the cost exceeds the budget": the Lovász-style vertex reinsertion gap (the unproved strengthening of Theorem 1.4 to path-only decompositions with endpoint control), the parity wall for even-sized stars in the Fan engine (a star of even size leaves its centre even, so no endpoint at the centre is forced), and the simplicity wall of Lemma 1.12 (a host endpoint may exist but be unusable because the path through it already contains the reinsertion target).

1.8 The shape of the conditional route

The conditional chapters measure, at each vertex, the one quantity that the classical deletion–reinsertion arguments leave uncontrolled: the number of path endpoints a decomposition can be made to hold at a specified vertex.

Definition 1.17 (endpoint budget)**.** For a connected graph G and an even-degree non-isolated vertex x of G, the *endpoint budget* $\beta(G, x)$ is the minimum cardinality of a path decomposition D of G in which x is an endpoint of at least two paths, i.e. $\beta(G, x) := \min\{|D| : D \text{ a path decomposition of } G,\ D(x) \geq 2\}$. The *budget conjecture* K is the statement that $\beta(G, x) \leq \lceil n/2 \rceil$ for every connected G on n vertices and every even-degree non-isolated vertex x.

Theorem 1.18. *Conjecture K implies Gallai's conjecture: every connected n-vertex graph G has $p(G) \leq \lceil n/2 \rceil$.*

Proof. If G has an even-degree non-isolated vertex x, then K provides a decomposition with at most $\lceil n/2 \rceil$ paths and we are done. Otherwise G is all-odd, so n is even, and Theorem 1.4 gives a decomposition into p paths and c cycles with $p + c \leq n/2$. In a mixed path-cycle decomposition the parity identity of Lemma 1.1(iii) still holds (cycles contribute even degree at each vertex), so every vertex is an endpoint of at least one path and $2p \geq n$; hence $p \geq n/2$, so $p = n/2$, $c = 0$, and the decomposition is pure, giving $p(G) \leq n/2 = \lceil n/2 \rceil$. □

Thus K is a strengthening of Gallai's conjecture, not a consequence of it. The remaining links of the conditional chain — from the degree-two layer of K through the odd semi-clique pair-hosting conjecture to the placement problems — are assembled descriptively below and carried at ledger level in Section C.2; here we restate, descriptively, the eight-item premise set behind the conditional-chain claim of Section 1.2; each link's verification level, premises and residual gaps are registered at pointer level in the limitations subsection.

Theorem 1.19 (conditional chain; descriptive registration)**.** *There is a chain of implications, each available at paper level or registered at pointer level with its verification status recorded, leading from the conjunction of the eight-item premise set listed in this subsection (with the valuation criterion of Conjecture 3.26 entering, if at all, only through the bridge premises as a parallel, redundant-for-implication route) to the conclusion $p(G) \leq \lceil n/2 \rceil$ for every connected G on n vertices. The theorem is the existence of the conditional route, not the truth of any premise: every premise is an open conjecture, an unverified external step, or a paper-open item with machine backing, none is claimed proved, and no unconditional progress on Gallai's conjecture is claimed.*

Each item below is an *open* conjecture, an *unverified external step*, or a *paper-open item with machine backing*; none is claimed proved.

- *(external, flagged unverified against the journal record)* the all-odd branch of the Lovász decomposition theorem (Theorem 1.4) as consumed by the proof that the budget conjecture implies Gallai's conjecture: if G is all-odd, the theorem gives p paths and c cycles with $p + c \leq n/2$, and the parity identity forces $p = n/2$, $c = 0$.
- *(external-origin, open)* BP-odd: every connected graph H on an odd number $2k+1$ of vertices with $|E(H)| \leq 2k^2$ satisfies $p(H) \leq k$ — the odd-order, tightness-special case of Question 1.1 of [3].
- *(internal, open)* the odd-order density conjecture at $k \geq 4$: every tight connected graph on $2k+1$ vertices ($p = \lceil (2k+1)/2 \rceil$) is an odd semi-clique; proved at $k \leq 3$ by exhaustive census, machine-certified at $k = 4$ and supported by directed search at $k = 5$, open at $k \geq 6$.
- SC-PH: every odd semi-clique on $2k + 1$ vertices hosts any two specified non-isolated vertices on distinct paths of a $(k + 1)$-path decomposition.

- *(open)* the surviving layers of the budget conjecture K: the degree-two cut-vertex branch (computational level, awaiting theorization) and the degree-≥ 4 general-layer residual (a recorded dead zone of 42 instances, discharged at witness level only). **This premise is open as stated**: the recorded instances have witness certificates, but no general paper proof exists, and the premise must be quoted as an open hypothesis wherever the chain is used.
- *(paper-open, machine-backed)* general solvability of the placement problem on the (E)-side, cyclic domain included: zero counterexamples at $k \leq 7$ over all 188 types (and machine accounts beyond), paper proof open.
- *(paper-open)* the (R)-side on the $3f \geq k$ domain: computational at $k \leq 7$ (zero counterexamples), paper proof open.
- *(bridge, open)* the valuation-line bridge conditions: three registered open conjectures of the difference-board analysis, quoted here descriptively and carried at pointer level in the ledger.

We record explicitly what the ledger is and is not. The chain above is *not* assembled into a single master theorem of the form "premises $\wedge$ criterion $\Rightarrow$ Gallai" in this paper: the bridge premises (the last three items) are registered at pointer level only, their full definitions residing in the project archive, so the paper certifies their provenance rather than their re-adjudicability. What the ledger delivers is checkability: any future resolution of any single premise propagates along the described chain through paper-level implications whose verification status is recorded. No premise is claimed to be within reach of the machine layer except as recorded, and the conditional route must not be cited as progress towards Gallai's conjecture beyond its registered content.

1.9 Reproducibility

The machine pipeline behind every computational claim in this paper — the exact-value computations, the decomposition verifiers, the per-instance sweeps of Section 4, and the Lean development reported there and in Section C — will be released together with the paper: the verification scripts, the certificate and ledger files, the Lean sources with their axiom profiles, and the per-theorem verification status table. The release makes every machine-checked number in this paper independently re-runnable and every Lean theorem independently re-compilable.

2 Constructions: $K_{4,n}$ and the ring closure; odd clique chains

2.1 The exact path number of $K_{4,n}$ for odd n

Let $K_{4,n}$ have bipartition $A = \{a_0, a_1, a_2, a_3\}$ and $B = \{b_0, \ldots, b_{n-1}\}$. For n odd, $n = 2k + 1$, the four vertices of A have odd degree and the vertices of B have degree 4, so $\mathrm{odd}(K_{4,n}) = 4$ and Lemma 1.1 gives only $p \geq 2$: the odd case is genuinely harder than the even one, where the edge bound alone is tight. Write $n' = n + 4 = 2k + 5$ for the total order; Gallai's target is $\lceil n'/2 \rceil = k + 3$, and the best published bound is $k + 2$, since $\mathcal{E}(K_{4,n})$ is the independent set B, a forest, so Theorem 1.5 applies, and the same bound follows from [13] and from [17], as $\mathrm{tw}(K_{4,n}) = 4$. Note that [21] does not apply: $K_{4,n}$ has minimum degree 4 and is not 3-degenerate.

For context, Constantinou and Ellinas [10] partition the parameter pairs (n_1, n_2) of K_{n_1,n_2} into four cases and give minimal-decomposition algorithms for three of them; in their Case 2 (even n_1, $1 \leq n_2 \leq n_1 - 1$) the minimum is $n_1/2$. Their Case 4 (odd n_1, $1 \leq n_2 \leq n_1 - 1$) is left open, with the explicit request that its "exact characteristics need to be first understood". Our main theorem settles the $n_2 = 4$ column of Case 4; Section 2.3 settles the $n_2 = 6$ column, proves a general structure

theorem and a ring-closure theorem for all even columns, and reduces the remaining columns to finitely many base instances.

2.1.1 Lower bound

Lemma 2.1. *Every path in $K_{4,n}$ uses at most 8 edges if $n \geq 5$, at most 6 edges if $n = 3$, and at most 2 edges if $n = 1$. Consequently, for odd $n \geq 3$, $p(K_{4,n}) \geq (n+1)/2$, and $p(K_{4,1}) \geq 2$.*

Proof. Paths in a bipartite graph alternate between A and B. A path containing α vertices of A and β of B has $e = \alpha + \beta - 1$ edges with $|\alpha - \beta| \leq 1$. Since $|A| = 4$ we have $\alpha \leq 4$ and $\beta \leq \alpha + 1 \leq 5$, so $e \leq 8$ for $n \geq 5$ (where $|B| \geq 5$); for $n = 3$, $\beta \leq 3$ gives $e \leq 6$. With $m = 4n$ edges partitioned into p paths, $8p \geq 4n$ gives $p \geq n/2$, i.e. $p \geq (n+1)/2$ for odd $n \geq 5$; and $6p \geq 12$ gives $p \geq 2 = (3+1)/2$ for $n = 3$. For $n = 1$ the graph is a 4-star and every path through the centre uses at most two edges, so $p \geq \lceil 4/2 \rceil = 2$. □

2.1.2 Structure of optimal decompositions

Theorem 2.2. *Let $n = 2k + 1 \geq 3$ and let $D = \{P_1, \ldots, P_{k+1}\}$ be a path decomposition of $K_{4,n}$ with $k + 1$ paths. Then:*

(1) *every path of D contains all four vertices of A; each $a \in A$ is an endpoint of exactly one path of D and an internal vertex of exactly k paths;*

(2) *the multiset of path lengths (edge counts) is one of $[7, 7, 7, 7, 8^{k-3}]$ ($k \geq 3$), $[6, 7, 7, 8^{k-2}]$ ($k \geq 2$), or $[6, 6, 8^{k-1}]$ ($k \geq 1$; for $k = 1$ this reads $[6, 6]$);*

(3) *a 6-edge path has vertex type $4A3B$ (both endpoints in A), a 7-edge path has type $4A4B$ (one endpoint in each class), and an 8-edge path has type $4A5B$ (both endpoints in B).*

Proof. (1) Fix $a \in A$, of degree $2k+1$. Let r_a be the number of paths of D in which a is an endpoint and s_a the number in which a is internal. Counting the degree of a path by path, $r_a + 2s_a = 2k+1$, so r_a is odd, $r_a \geq 1$, and $r_a + s_a = (2k + 1 + r_a)/2 \leq k + 1$ (there are only $k + 1$ paths) forces $r_a \leq 1$. Hence $r_a = 1$ and $s_a = k$. So each $a \in A$ is an endpoint of exactly one path and internal in exactly k paths, i.e. a occurs in all $k + 1$ paths.

(2)+(3) Assume first $n \geq 5$. By Lemma 2.1 each path has $e_i \leq 8$ edges, and $\sum_i e_i = 4n = 8k+4$, while $(k + 1) \cdot 8 = 8k + 8$; hence the deficits $\delta_i = 8 - e_i$ sum to 4. A path with α vertices of A and β of B, $\alpha + \beta = e + 1$ and $|\alpha - \beta| \leq 1$, has an endpoint in A exactly when $\alpha = \beta + 1$. Enumerating the possibilities with $\alpha \leq 4$: an 8-edge path ($\delta = 0$) has type $(\alpha, \beta) = (4, 5)$ and 0 endpoints in A; a 7-edge path ($\delta = 1$) has type $(4, 4)$ and exactly 1; a 6-edge path ($\delta = 2$) has type $(4, 3)$ or $(3, 4)$, so at most 2; a 5-edge path ($\delta = 3$) has type $(3, 3)$, exactly 1; a 4-edge path ($\delta = 4$) has type $(3, 2)$ or $(2, 3)$, at most 2. (Larger deficits cannot occur since $\sum \delta_i = 4$.) Let $f(\delta)$ denote these maxima $0, 1, 2, 1, 2$. By (1) the total number of endpoints lying in A is exactly $\sum_a r_a = 4$, while it is at most $\sum_i f(\delta_i)$; over all partitions of $\sum \delta_i = 4$ the maximum of $\sum_i f(\delta_i)$ is 4, attained exactly at the profiles $(2, 2)$, $(2, 1, 1)$, $(1, 1, 1, 1)$ (all other partitions give ≤ 2). Hence equality holds throughout: the deficit multiset is one of the three listed, and each path attains its $f(\delta_i)$, which for $\delta = 2$ forces the type $(4, 3)$; this gives (2) and (3). For $n = 3$ the cap is 6: two paths with $\sum e_i = 12$ force both to be 6-edge paths of type $(4, 3)$ with $2 + 2 = 4$ endpoints in A, consistent with all three claims. □

In particular, in any minimal decomposition each $a \in A$ has its $2k+1$ incident edges partitioned into k "pairs" (the two B-edges flanking a inside a path) plus one "single" (the edge of the path on which a is an endpoint), and the four resulting pair-plus-single systems each partition B. This

is the language in which we audit the constructions below, and it answers, for the column $n_2 = 4$, the request of [10] for the exact characteristics of Case 4.

2.1.3 Upper bound: a closed-form construction for large n

Proposition 2.3. *For every $k \geq 6$, writing $n = 2k + 1$, the graph $K_{4,n}$ has a path decomposition into $k + 1$ paths.*

Proof. Split B into a *core* $S = \{s_0, \ldots, s_6\}$ of size 7 and a *rim* $R = \{r_0, \ldots, r_{m-1}\}$ of size $m = 2k-6$ (even, and $m \geq 6$); $|S| + |R| = 2k + 1 = n$. We use four *gadget paths* of 7 edges each, on S only:

$$G_1 = a_0\, s_0\, a_1\, s_1\, a_2\, s_2\, a_3\, s_3, \qquad G_2 = a_1\, s_2\, a_0\, s_3\, a_2\, s_0\, a_3\, s_1,$$
$$G_3 = a_2\, s_4\, a_1\, s_5\, a_3\, s_6\, a_0\, s_1, \qquad G_4 = a_3\, s_4\, a_0\, s_5\, a_2\, s_6\, a_1\, s_3,$$

and $k - 3$ *cyclic paths* of 8 edges each, on R only, with indices mod m:

$$Q_t \;=\; b_{r(2t)}\, a_0\, b_{r(2t+1)}\, a_1\, b_{r(2t+2)}\, a_2\, b_{r(2t+3)}\, a_3\, b_{r(2t+4)}, \qquad t = 0, \ldots, k-4.$$

The paths are legal. Each G_i is an alternating sequence of 4 distinct A-vertices and 4 distinct elements of S, hence a simple path of 7 edges. Each Q_t uses five consecutive indices $2t, \ldots, 2t + 4$ mod m; since $m = 2k - 6 \geq 6 > 5$ these are distinct, so Q_t is a simple path of 8 edges.

The gadget paths partition $A \times S$. For each $a \in A$, list the pairs of S-edges flanking a (or the single edge, when a is an endpoint) in each of the four G_i:

$$a_0 :\ \{s_0\};\ \{s_2, s_3\};\ \{s_6, s_1\};\ \{s_4, s_5\} \qquad \text{partition } S;$$
$$a_1 :\ \{s_0, s_1\};\ \{s_2\};\ \{s_4, s_5\};\ \{s_6, s_3\} \qquad \text{partition } S;$$
$$a_2 :\ \{s_1, s_2\};\ \{s_3, s_0\};\ \{s_4\};\ \{s_5, s_6\} \qquad \text{partition } S;$$
$$a_3 :\ \{s_2, s_3\};\ \{s_0, s_1\};\ \{s_5, s_6\};\ \{s_4\} \qquad \text{partition } S.$$

Each pair $\{s', s''\}$ corresponds to the two edges as', as'' used by that path at the internal position of a, and each singleton to the endpoint edge. Since the four groups partition S for every a, every edge of $A \times S$ is used by exactly one G_i.

The cyclic paths partition $A \times R$. For fixed a_c, $c \in \{0, 1, 2, 3\}$, the path Q_t uses at a_c the pair of edges $\{a_c r(2t + c),\ a_c r(2t + c + 1)\}$ (indices mod m). As t runs over $0, \ldots, k - 4$, the even residues $2t$ cover all $k - 3$ even elements of $\mathbb{Z}_m$ and the odd residues $2t + 1$ cover all $k - 3$ odd elements; shifting by c, each element of $\mathbb{Z}_m$ occurs exactly once, in exactly one pair, across the $k - 3$ paths. Hence the $m = 2(k - 3)$ edges $a_c R$ are partitioned, for each c independently.

Count. $4 + (k - 3) = k + 1$ paths, as claimed. □

2.1.4 Small cases

Proposition 2.4. *Explicit decompositions into $k+1$ paths exist for $n = 3, 5, 7, 9, 11$ ($k = 1, \ldots, 5$). The path lists below are machine-verified to partition $E(K_{4,n})$; write b_j for the j-th vertex of B.*

$= 3$; *lengths* $[6, 6]$*):* $a_0 b_0 a_1 b_1 a_2 b_2 a_3$; $\quad a_1 b_2 a_0 b_1 a_3 b_0 a_2$.

$(n = 5$; $[6, 7, 7]$*):* $a_0 b_0 a_1 b_1 a_2 b_2 a_3 b_3$; $\quad a_1 b_2 a_0 b_4 a_3 b_0 a_2 b_3$; $\quad a_2 b_4 a_1 b_3 a_0 b_1 a_3$.

$n = 7$; $[7, 7, 7, 7]$*):* $a_0 b_0 a_1 b_1 a_2 b_2 a_3 b_3$; $\quad a_1 b_2 a_0 b_3 a_2 b_0 a_3 b_1$; $\quad a_2 b_4 a_1 b_5 a_3 b_6 a_0 b_1$; $\quad a_3 b_4 a_0 b_5 a_2 b_6 a_1 b_3$.

$= 9$; $[7, 7, 7, 7, 8]$*):* $b_0 a_0 b_1 a_1 b_2 a_2 b_3 a_3 b_4$; $\quad a_0 b_3 a_1 b_0 a_2 b_1 a_3 b_2$; $\quad a_1 b_4 a_0 b_5 a_2 b_6 a_3 b_0$; $\quad a_2 b_7 a_1 b_5 a_3 b_8 a_0 b_2$; $a_3 b_7 a_0 b_6 a_1 b_8 a_2 b_4$.

1; $[7,7,7,7,8,8]$): $b_0a_0b_1a_1b_2a_2b_3a_3b_4$; $b_2a_0b_3a_1b_0a_2b_1a_3b_5$; $a_0b_4a_1b_5a_2b_6a_3b_0$; $a_1b_6a_0b_7a_2b_8a_3b_2$; $a_2b_9a_1b_7a_3b_{10}a_0b_5$; $a_3b_9a_0b_8a_1b_{10}a_2b_4$.

The reason $k = 4, 5$ are listed separately is that the rim R of Proposition 2.3 is too small there; the single 8-edge paths absorb the shortfall. All lists conform to the length profiles of Theorem 2.2.

2.1.5 The main theorem and its corollaries

Theorem 2.5. *For every odd $n \geq 3$, $p(K_{4,n}) = (n+1)/2 = \lceil n/2 \rceil$. Also $p(K_{4,1}) = 2$.*

Proof. Combine the lower bound of Lemma 2.1 with the upper bounds of Proposition 2.3 ($k \geq 6$) and Proposition 2.4 ($k \leq 5$). For $n = 1$, the lower bound of Lemma 2.1 matches the two-path decomposition $a_0b_0a_1 \mid a_2b_0a_3$. □

Corollary 2.6 (Gallai comparison)**.** *For odd $n \geq 3$, $K_{4,n}$ has $n + 4 = 2k + 5$ vertices and Gallai's conjecture demands $p \leq k + 3$; in fact $p = k + 1$, with slack 2. There is no counterexample here.*

Proof. Immediate from Theorem 2.5: the value $(n+1)/2 = k+1$ is two fewer than the $k+3$ paths the conjecture allows. □

Corollary 2.7 (comparison with published bounds)**.** *The best published bound for this family was $k + 2$: it follows from Pyber's theorem [20] (equally from [13], and from [17] since $\mathrm{tw}(K_{4,n}) = 4$). Theorem 2.5 improves it by one and shows that the $\lfloor n'/2 \rfloor$ bound of the treewidth-4 theorem [17] is not tight on this family.*

Proof. Immediate: Theorem 1.5 (equally [13, 17]) gives $k + 2$ for the family, while Theorem 2.5 attains $k + 1$; the $\lfloor n'/2 \rfloor = k + 3$ bound of [17] exceeds the exact value by 2. □

Remark 2.8 (unified formula for the whole family). Combining Theorem 2.5 with the known even-case values (Case 2 of [10] for even $n \geq 6$, and the small values $p(K_{4,2}) = 2$, $p(K_{4,4}) = 3$, which we also re-verified by machine), the path number of $K_{4,n}$ is

$$p(K_{4,n}) = \lceil n/2 \rceil + [n \leq 4] \quad \text{for all } n \geq 1,$$

i.e. $\lceil n/2 \rceil$ except for the three small values $p(K_{4,1}) = p(K_{4,2}) = 2$ and $p(K_{4,4}) = 3$.

2.2 The remaining columns of Case 4

For odd n_1, Case 4 of [10] is the range $1 \leq n_2 \leq n_1 - 1$. The state of affairs by column:

n_2	$p(K_{n_1,n_2})$, n_1 odd	status
1	$\lceil n_1/2 \rceil$	star decomposition, classical
2	$(n_1 + 1)/2$	counting cap (as in Lemma 2.1) + [20]
3	$(n_1 + 3)/2$	Lemma 1.1 + [20]
4	$(n_1 + 1)/2 = \lceil n_1/2 \rceil$	Theorem 2.5 (this paper)
6	$(n_1 + 1)/2$ for $n_1 \geq 7$	Corollary 2.13 (Sec. 2.3)
even ≥ 8	$\lceil n_1/2 \rceil \leq p \leq \lfloor (n_1 + n_2)/2 \rfloor$	bases settled for $n_2 = 8, 10$; rest open (Sec. 2.3)

Conjecture 2.9. *For all odd n_1 and even n_2 with $2 \leq n_2 < n_1$: $p(K_{n_1,n_2}) = \lceil n_1/2 \rceil$.*

The evidence: the universal counting cap gives $p \geq \lceil n_1/2 \rceil$ in the whole even range (Theorem 2.10 below; each path uses at most $2n_2$ edges, and $m = n_1n_2$); the even analogue (Case 2 of [10]) has exactly the value $n_1/2 = \lceil n_1/2 \rceil$; and $p(K_{5,2}) = p(K_{5,4}) = 3 = \lceil 5/2 \rceil$ by exact computation. The structural tools developed in the next subsection reduce the conjecture to finitely many instances per column, of which all but five named ones are now settled.

2.3 Beyond the fourth column: ring closure and the state of Conjecture 2.9

In this subsection $n_1 = 2k+1$ is odd and $n_2 = 2h$ is even with $k \geq h \geq 1$ — exactly the range of Conjecture 2.9. We write $A = \{a_0, \ldots, a_{2h-1}\}$ for the side of size n_2 (each of its vertices has degree $2k+1$, odd) and $B = \{b_0, \ldots, b_{2k}\}$ for the side of size n_1 (each of degree $2h$, even); note that this swaps the roles of A and B relative to the earlier parts of this section, where the 4-vertex side was called A. Conjecture 2.9 predicts $p(K_{2k+1,2h}) = k+1$.

Theorem 2.10 (general counting cap)**.** *For all $k \geq h \geq 1$, $p(K_{2k+1,2h}) \geq k+1$.*

Proof. Paths in a bipartite graph alternate between the two sides. A path containing α vertices of A and β vertices of B has $\alpha+\beta-1$ edges with $|\alpha-\beta| \leq 1$; since $\alpha \leq 2h$ we get $\beta \leq \alpha+1 \leq 2h+1$, so every path has at most $2h+(2h+1)-1 = 4h$ edges. If p paths partition the $m = (2k+1)\cdot 2h$ edges, then $4hp \geq 2h(2k+1)$, i.e. $p \geq \lceil (2k+1)/2 \rceil = k+1$. □

Theorem 2.11 (structure of optimal decompositions, general h)**.** *Let $k \geq h \geq 1$ and let D be a path decomposition of $K_{2k+1,2h}$ with exactly $k+1$ paths. Then:*

(1) *every path of D contains all $2h$ vertices of A; each $a \in A$ is an endpoint of exactly one path of D and an internal vertex of exactly k paths;*

(2) *every path has $e \in \{4h-2, 4h-1, 4h\}$ edges, and writing $\delta = 4h-e \in \{2,1,0\}$, δ equals the number of endpoints of the path lying in A;*

(3) *with x, y, z the numbers of paths of deficit $\delta = 2, 1, 0$, respectively, we have $x+y+z = k+1$ and $2x+y = 2h$; the total number of path-ends lying in B equals $2z+y = 2(k-h+1)$;*

(4) *every $b \in B$ is an endpoint of an even number of paths of D.*

Proof. (1) Fix $a \in A$, of degree $2k+1$; let r_a be the number of paths of D ending at a and s_a the number having a as an internal vertex. Counting the degree of a path by path gives $r_a + 2s_a = 2k+1$, so r_a is odd and $r_a \geq 1$; since a occurs in $r_a + s_a$ of the $k+1$ paths, $r_a + s_a \leq k+1$, i.e. $(2k+1+r_a)/2 \leq k+1$, forcing $r_a = 1$ and $s_a = k$. So a occurs in all $k+1$ paths, and $\sum_{a\in A} r_a = 2h$.

(2)+(3) Every path contains all $2h$ vertices of A and alternates, so if it contains β vertices of B then $|\beta - 2h| \leq 1$; accordingly its two endpoints lie both in A (when $\beta = 2h-1$), one in each class ($\beta = 2h$), or both in B ($\beta = 2h+1$), and $e = 2h+\beta-1 \in \{4h-2, 4h-1, 4h\}$ with $\delta = 4h-e$ equal to the number of A-endpoints. Summing over D, $\sum e = |E| = 2h(2k+1)$, so $\sum \delta = 4h(k+1) - 2h(2k+1) = 2h$. Counting A-endpoints two ways gives $2x+y = \sum_{a\in A} r_a = 2h$; combined with $x+y+z = k+1$ this yields $z = k+1+x-2h$ and $2z+y = 2(k+1)-2h = 2(k-h+1)$.

(4) For $b \in B$, of degree $2h$, the count of (1) reads $r_b + 2s_b = 2h$, so r_b is even. □

Theorem 2.12 (ring closure)**.** *Let $h \geq 1$ and suppose that for some $k_0 \geq h$ the graph $K_{2k_0+1,2h}$ has a path decomposition D_0 into k_0+1 paths. Then for every $k \geq k_0+h+1$ the graph $K_{2k+1,2h}$ has a path decomposition into $k+1$ paths; consequently $p(K_{2k+1,2h}) = k+1$ for all $k \geq k_0+h+1$, by Theorem 2.10.*

Proof. Write $m = 2(k-k_0)$, an even number satisfying $m \geq 2h+2$, in particular $2h+1 < m$. Split the B-side of $K_{2k+1,2h}$ into $B_0 = \{b_0, \ldots, b_{2k_0}\}$, of size $2k_0+1$, which carries D_0, and m *fresh* ring vertices $b_{2k_0+1}, \ldots, b_{2k}$; write $r(i) = 2k_0 + 1 + (i \bmod m)$, so that the ring vertices are $\{b_{r(i)} : i \in \mathbb{Z}_m\}$. The new decomposition consists of the paths of D_0 unchanged (they use only edges into B_0) and $k-k_0 = m/2$ *ring paths*, one for each $t \in \{0, \ldots, m/2-1\}$:

$$Q_t \;=\; b_{r(2t)}\, a_0\, b_{r(2t+1)}\, a_1\, b_{r(2t+2)}\, a_2 \cdots a_{2h-1}\, b_{r(2t+2h)} \qquad \text{(all ring indices mod } m\text{)}.$$

The paths are legal. Q_t uses the $2h+1$ consecutive ring indices $2t, \dots, 2t+2h$; since $2h+1 < m$ these are distinct, so Q_t is a simple path with $4h$ edges, all of them edges between A and the ring.

The ring paths partition $A \times R$, where R is the ring. Fix a_c, $c \in \{0, \dots, 2h-1\}$. The path Q_t uses at a_c the pair of edges $\{a_c b_{r(2t+c)}, a_c b_{r(2t+c+1)}\}$. As t runs over $0, \dots, m/2-1$, the index $2t+c$ runs over all elements of $\mathbb{Z}_m$ congruent to c modulo 2, and $2t+c+1$ over the complementary class (the two parity classes are well defined because m is even); hence the $m/2$ pairs are pairwise disjoint with union $\mathbb{Z}_m$, i.e. each of the m edges from a_c to the ring is used exactly once, and this for each c independently.

Ring degrees. A ring vertex $b_{r(j)}$ occurs at a_c exactly when $j \in \{2t+c, 2t+c+1\}$ for some t, and for each fixed c there is exactly one such t; so each ring vertex occurs once at each a_c, i.e. exactly $2h$ times in all — its degree — each time through a different edge.

Count. The edges used by D_0 lie in $A \times B_0$ and those of the ring paths in $A \times R$, disjoint sets; D_0 partitions $A \times B_0$, of size $2h(2k_0+1)$, by hypothesis, and the ring paths partition $A \times R$, of size $2hm$, by the paragraph above. The total number of paths is $(k_0+1)+(k-k_0) = k+1$, so this is a decomposition of $K_{2k+1,2h}$ into $k+1$ paths, and Theorem 2.10 makes the value exact. □

Corollary 2.13 (the $n_2 = 6$ column)**.** *For every odd $n_1 \geq 7$, $p(K_{n_1,6}) = (n_1+1)/2$; equivalently, $p(K_{2k+1,6}) = k+1$ for all $k \geq 3$.*

Proof. By Theorem 2.10 it suffices to exhibit $(k+1)$-path decompositions for the four bases $k = 3, 4, 5, 6$. For $k = 3$ ($K_{7,6}$, with $A = \{a_0, \dots, a_5\}$, $B = \{b_0, \dots, b_6\}$) the following four paths, of deficits $[2, 2, 1, 1]$ in the sense of Theorem 2.11(3), are machine-verified to partition the 42 edges (Appendix A):

$$a_0b_0a_1b_1a_2b_2a_3b_3a_4b_4a_5; \quad a_1b_2a_0b_3a_2b_0a_3b_5a_5b_6a_4;$$
$$a_2b_4a_0b_5a_1b_6a_3b_1a_4b_0a_5b_2; \quad a_3b_4a_1b_3a_5b_1a_0b_6a_2b_5a_4b_2.$$

For $k = 4, 5, 6$ ($K_{9,6}$, $K_{11,6}$, $K_{13,6}$) machine-verified $(k+1)$-path witnesses are archived, each with the length profile forced by the ledger of Theorem 2.11(3) ($\sum \delta = 6$), and each checked by two independent implementations. The base $k_0 = 3$, fed into Theorem 2.12, covers every $k \geq 7$; the pipeline additionally re-derived the ring construction explicitly for $k = 7, \dots, 31$ and obtained overlapping closure chains from the bases $k_0 = 4, 5$ (Appendix A). □

The state of Conjecture 2.9. Theorem 2.12 upgrades Conjecture 2.9 from a statement about infinitely many graphs to a *column-by-column decision procedure*: in the column $n_2 = 2h$, Theorem 2.10 shows that it suffices to exhibit, for each of the $h+1$ *base* values $k \in [h, 2h]$, one explicit $(k+1)$-path decomposition of $K_{2k+1,2h}$; the base $k_0 = h$ then closes the whole tail $k \geq 2h+1$ for free, and every further base lowers the closure line. Applying the procedure as far as the current machine pipeline reaches gives:

n_2	status of $p(K_{n_1,n_2})$, n_1 odd	remaining bases
2	$(n_1+1)/2$; proved (counting cap + [20]; the pipeline also holds a constructive closure chain from the base $k_0 = 1$)	none
4	$(n_1+1)/2$; Theorem 2.5	none
6	$(n_1+1)/2$ for $n_1 \geq 7$; Corollary 2.13	none
8	bases $k = 4,5,6$ exact: $p(K_{9,8}) = 5$, $p(K_{11,8}) = 6$, $p(K_{13,8}) = 7$; tail $k \geq 9$ closed by ring closure	$k = 7,8$
10	bases $k = 5,6,7$ exact: $p(K_{11,10}) = 6$, $p(K_{13,10}) = 7$, $p(K_{15,10}) = 8$; tail $k \geq 11$ closed	$k = 8,9,10$
even ≥ 12	open; reduced to $h+1$ bases per column	$k \in [h, 2h]$

In total this adds ten exact values beyond Theorem 2.5 (four in the column $n_2 = 6$, three in the column $n_2 = 8$, three in the column $n_2 = 10$), every one attained by an explicitly archived and independently verified witness, and two constructively closed tails ($k \geq 9$ for $n_2 = 8$ from the base $K_{9,8}$, $k \geq 11$ for $n_2 = 10$ from the base $K_{11,10}$). No instance in the range is known to violate Conjecture 2.9.

What remains open are exactly the five middle-window bases $K_{15,8}$, $K_{17,8}$, $K_{17,10}$, $K_{19,10}$, $K_{21,10}$. On these the deterministic search pipeline has burnt more than 14 million search nodes (for the $n_2 = 8$ pair) and about 0.9 million nodes (for the $n_2 = 10$ triple), over every ledger-feasible deficit multiset of Theorem 2.11(3), without a hit and without exhausting a single schedule; these negatives are truncations, not exclusions. The following observation records the data-level regularities; it is a *computational observation*, not a theorem.

Observation 2.14 (middle-window cost wall; computational)**.** The following regularities of the deterministic, fixed-seed search pipeline are data, not proved statements. (1) *Hit-cost trajectory.* Within a column, the first hit at level k is cheap for $k \leq 2h-3$ — $5, 7, 43$ nodes at $k = 4,5,6$ in the column $n_2 = 8$, and $6, 13, 182$ nodes at $k = 5,6,7$ in the column $n_2 = 10$ — and then jumps off a cliff inside the *middle window* $k \in [2h-2, 2h]$: at $k = 2h-1$ in the column $n_2 = 8$ (i.e. $K_{15,8}$) the strongest schedule burnt more than 4.8×10^6 nodes without a hit, and at $k = 2h-2$ in the column $n_2 = 10$ (i.e. $K_{17,10}$) more than 1.19×10^5 nodes; no schedule was run to exhaustion. The two columns share the same cost anatomy — cheap head bases, a free tail by Theorem 2.12, a hard middle window — with the cliff at $k = 2h-1$ for $n_2 = 8$ and, on current evidence one notch earlier, at $k = 2h-2$ for $n_2 = 10$. (2) *Deficit-multiset signal.* Every hit at $k \leq 2h-3$ lies in a $z = 0$ multiset of Theorem 2.11(3), at the first schedule tried; the unique $z = 0$ multisets of the window — $(x, y, z) = (0, 8, 0)$ at $K_{15,8}$ and $(1, 8, 0)$ at $K_{17,10}$ — are precisely the types that resisted the largest search effort, which makes them the strongest current candidates for non-realizable ledgers in the finitary question "which ledger-feasible multisets (x, y, z) occur in some $(k+1)$-path decomposition?". The only proved non-realizable multiset so far is $(x, y, z) = (3, 0, 1)$, i.e. deficit list $[2, 2, 2, 0]$, at $K_{7,6}$: a single BB-type path has two distinct B-endpoints $u \neq v$, each forced to be an endpoint of an even, hence at least 2, number of paths (Theorem 2.11(4)), contradicting $2z + y = 2$.

2.4 Odd clique chains: refinements beyond block graphs

A *block graph* is a graph whose blocks are complete graphs. Chen and Wu [6] proved Gallai's conjecture for all block graphs. Here we study *odd clique chains*: block graphs whose blocks are $B_i \cong K_{2j_i+1}$ with j_i odd (equivalently, blocks of order 3 mod 4), and whose block tree is a path

(every block has at most two cut vertices). We write b for the number of blocks and $n = 2\sum_i j_i + 1$ for the order; note n is odd and $\lfloor n/2 \rfloor = \sum_i j_i$. Blocks that are K_3 ($j_i = 1$) are allowed.

The engine is the following statement about complete graphs of odd order minus a Hamilton path.

Theorem 2.15. *Let $j \geq 1$ be odd and let P_0 be any Hamilton path of K_{2j+1}. Then $K_{2j+1} - P_0$ has a path decomposition into j paths, each with exactly $2j-1$ edges. Consequently $p(K_{2j+1} - P_0) = j$, the value of Lemma 1.2.*

Proof. For $j = 1$ the graph $K_3 - P_0$ is a single edge, a decomposition into $j = 1$ paths, and the value j matches Lemma 1.2; so assume $j \geq 3$ odd. The automorphism group of K_{2j+1} is transitive on Hamilton paths, so it suffices to exhibit one Hamilton path P^* such that $K_{2j+1} - P^*$ decomposes into j paths; a general P_0 is then the image of P^* under a permutation of vertices, which transports the decomposition.

Write $V = \{\infty\} \cup \mathbb{Z}_{2j}$, $\beta = (j+1)/2$. Let $C_0, \ldots, C_{j-1}$ be the Walecki cycles of Lemma 1.9. For a vertex $v = k + t$ of $\mathbb{Z}_{2j}$, its two neighbours inside C_k are read off from the sequence: they are $\{\infty, k+1\}$ if $t = 0$; $\{k+j+1, \infty\}$ if $t = j$; and $\{k-t+1,\ k-t\}$ otherwise. The two *corners* at a designated *centre* c of C_k are the two edges of C_k incident to c.

We designate one centre in each cycle and one arm-pair $\{y_i, y_i + 1\}$ per corner, as follows (all arithmetic mod $2j$):

1. in C_{j-1}: centre $c_1 = 2j - 1$ (a vertex of type $t = j$ there), arm-pair $\{0, \infty\}$ via the reading above: the neighbours of $2j-1$ in C_{j-1} are $\{2j, \infty\} = \{0, \infty\}$;
2. in $C_{\beta-1}$: centre $c_2 = \beta - 1$ (type $t = 0$), arm $\{\infty, \beta\}$;
3. for $i = 3, \ldots, j$: in C_{k_i}, where $k_i = i - 3$ if $3 \leq i \leq \beta + 1$ and $k_i = i - 2$ if $\beta + 2 \leq i \leq j$, take centre $c_i = 2k_i - y_i$ with arm $\{y_i, y_i + 1\}$, $y_i = \beta + i - 3$.

Corner legality. Cases 1 and 2 are immediate from the neighbour reading. For case 3, write $c_i = k_i + t_i$ with $t_i = k_i - y_i \equiv -\beta \equiv (3j-1)/2$ for $i \leq \beta + 1$ and $t_i \equiv 1 - \beta \equiv (3j+1)/2$ for $i \geq \beta + 2$. Both values lie in $\mathbb{Z}_{2j} \setminus \{0, j\}$ (they equal 0 never, and equal j only in the void case $j = 1$, $i \geq \beta + 2 = 3$), so the neighbour reading gives $N_{C_{k_i}}(c_i) = \{k_i - t_i + 1,\ k_i - t_i\} = \{y_i + 1,\ y_i\}$, as required.

The prescribed Hamilton path. Let

$$P^* \;=\; 0,\ 2j-1,\ \infty,\ \beta-1,\ \beta,\ c_3,\ \beta+1,\ c_4,\ \beta+2,\ \ldots,\ c_i,\ \beta+i-2,\ c_{i+1},\ \ldots,\ c_j,\ \beta+j-2.$$

The edge set of P^* is the union of the j corner arm-pairs (adjacent corners share the arm vertex $a_i = \beta + i - 2$), i.e. $2j$ edges. Its vertex set is $\{0, \infty\} \cup \{\beta, \ldots, \beta + j - 2\} \cup \{c_1, \ldots, c_j\}$: the arm interval $[\beta, \beta + j - 2] = [(j+1)/2, (3j-3)/2]$ is disjoint from the centre set $\{1, \ldots, (j-3)/2\} \sqcup \{(j-1)/2\} \sqcup \{(3j-1)/2, \ldots, 2j-2\} \sqcup \{2j-1\}$, every centre interval below $(j+1)/2$ resp. above $(3j-3)/2$ being separated from the arm interval (for $j = 3$ the first centre interval is empty), and all $2j+1$ listed values are distinct and cover V. Hence P^* is a Hamilton path with $2j$ edges.

The partition. The designated centre of C_{j-1} is c_1, of $C_{\beta-1}$ is c_2, and of C_{k_i} is c_i for $i \geq 3$; the index map $i \mapsto k_i$ satisfies $\{j-1, \beta-1\} \cup \{0, \ldots, \beta-2\} \cup \{\beta, \ldots, j-2\} = \{0, \ldots, j-1\}$ bijectively, so each cycle loses exactly one corner and the corners are edge-disjoint. By Lemma 1.9,

$$E(K_{2j+1}) \;=\; E(P^*) \;\sqcup \bigsqcup_{k=0}^{j-1} \big(E(C_k) \setminus \{\text{the two corner edges at } c_k\}\big),$$

and each C_k minus the two edges at its centre is a path on the remaining $2j$ vertices with $2j-1$ edges. This gives j paths decomposing $K_{2j+1}-P^*$. Finally $p(K_{2j+1}-P_0)\geq\lceil j(2j-1)/2j\rceil=j$ by Lemma 1.2, so the value is exactly j. □

Remark 2.16. The construction is a Walecki-type zigzag, and the base decomposition objects are classical [18, 2]. The statement for an *arbitrarily prescribed* Hamilton path, as needed to chain blocks of order 3 mod 4 (the 1 mod 4 case is the even-j residual, left open below and in Section C), appears to be new: the Oberwolfach problem and its variants concern 2-regular factors and do not cover path factors, and we could not find this statement in the Hamilton decomposition literature. The even-j residual is recorded as an open problem in Section C. The formula was verified by an independent computer check for all odd $j\leq 121$; this anchor is a certificate, not part of the proof.

The chaining argument needs the following accounting of the decomposition produced by Theorem 2.15.

Lemma 2.17 (slot accounting)**.** *Let $D=\{Q_1,\dots,Q_j\}$ be a decomposition of $K_{2j+1}-P_0$ into j paths of $2j-1$ edges each (as in Theorem 2.15), where the ends of P_0 are w,w'. Then:*

(i) *each Q_i misses exactly one vertex, and the j missed vertices are distinct;*

(ii) *the missed vertices all lie among the internal vertices of P_0 (the even-degree vertices of $K_{2j+1}-P_0$), and w,w' are never missed;*

(iii) *each of w,w' is an endpoint of exactly one path of D; every internal vertex v of P_0 is either missed (endpoint of no path) or an endpoint of exactly two paths; and the total number of path-ends is $2j$.*

Proof. Each Q_i has $2j-1$ edges, hence spans $2j$ of the $2j+1$ vertices and misses exactly one. For $v\in V$, let k_v be the number of paths missing v and s_v its total endpoint count; let a_v be the number of paths containing v internally, so that the paths containing v are exactly a_v+s_v and $j=k_v+a_v+s_v$. Lemma 1.1(iii) gives $s_v\equiv\deg(v)\pmod 2$, where $\deg_{K_{2j+1}-P_0}(v)=2j-1$ for $v\in\{w,w'\}$ and $2j-2$ otherwise; and $2\,a_v+s_v=\deg(v)$. If v is internal to P_0 and $k_v=0$: $2a_v+s_v=2j-2$ and $a_v+s_v=j$ give $s_v=2$; if $k_v=1$: $a_v+s_v=j-1$ gives $s_v=0$. If $v=w$ and $k_v=1$: $2(j-1-s_v)+s_v=2j-1$ is absurd; so $k_v=0$, $a_v+s_v=j$, $2a_v+s_v=2j-1$ give $s_v=1$. This proves (ii) and (iii). For (i): each Q_i misses exactly one vertex, so it suffices that no vertex is missed twice. But $a_v+s_v=j-k_v$ and $2a_v+s_v=\deg(v)$ give $s_v=2(j-k_v)-\deg(v)$; for $k_v\geq 2$ this is $s_v\leq 2(j-2)-\deg(v)\leq 2j-4-(2j-2)=-2<0$ (and $=-3$ at w,w'), impossible. Hence $k_v\leq 1$ for every v, and the j missed vertices are distinct. The total $2j$ of path-ends is automatic. □

We also need the elementary splicing operation. If P and P' are paths with $V(P)\cap V(P')=\{w\}$ and w is an endpoint of both, then $P\cdot P'$ (the concatenation) is a path, and replacing P,P' by $P\cdot P'$ reduces the size of a decomposition by one, with all other path-end counts unchanged except that w loses two endpoints. We call this a *splice at w*.

Theorem 2.18. *Let G be a connected graph whose blocks are $B_1,\dots,B_b$ with $B_i\cong K_{2j_i+1}$, all j_i odd, and whose block tree is a path. Let $n=2\sum_i j_i+1$. Then:*

(i) *if $b=1$, $p(G)=\lceil n/2\rceil=\sum_i j_i+1$;*

(ii) *if $b\geq 2$, $p(G)\leq\lfloor n/2\rfloor=\sum_i j_i$;*

(iii) *if $b\geq 4$, $p(G)\leq\lfloor n/2\rfloor-1=\sum_i j_i-1$.*

Proof. (1) For $b = 1$, $G = K_{2j+1}$: Lemma 1.2 gives $p \geq \lceil j(2j+1)/2j \rceil = j+1$; conversely a Hamilton path P together with the j-path decomposition of $K_{2j+1} - P$ from Theorem 2.15 gives $p \leq j+1$.

Assume $b \geq 2$ and order the blocks along the block tree as $B_1, \ldots, B_b$, with $B_i \cap B_{i+1} = \{w_i\}$; the w_i are distinct cut vertices, B_1, B_b are end blocks with one cut each, and each middle block B_i $(2 \leq i \leq b-1)$ has cut set $W_i = \{w_{i-1}, w_i\}$.

Per-block decompositions. In each block B_i, choose a Hamilton path P_i whose endpoint set is W_i (for an end block, one endpoint is its cut vertex and the other is any non-cut vertex). By Theorem 2.15, relabelled, $B_i - P_i$ decomposes into j_i paths $Q_{i,1}, \ldots, Q_{i,j_i}$. By Lemma 2.17 applied to (B_i, P_i): each cut vertex of B_i is an endpoint of P_i and of *exactly one* $Q_{i,\cdot}$, and no other path ends at a cut vertex. Thus each block offers exactly two path-ends at each of its cut vertices, one from P_i and one from a single $Q_{i,\cdot}$; call the latter the *Q-end of B_i at w*.

Splicing. At a cut vertex w_i joining B_i and B_{i+1}, any path of B_i ending at w_i and any path of B_{i+1} ending at w_i meet in exactly $\{w_i\}$ (distinct blocks share only their cut vertex), so they can be spliced. Each splice reduces the path count by one. Starting from the disjoint union of the per-block decompositions we have $\sum_i (j_i + 1) = \sum_i j_i + b$ paths.

(2) Perform the following b splices:

- at w_1: splice P_1@w_1 with P_2@w_1, and splice the Q-end of B_1 at w_1 with the Q-end of B_2 at w_1 (two splices; all four paths exist and are distinct within their blocks, the two of B_1 being P_1 and its unique cut-Q, the two of B_2 being P_2 and its unique Q-end at w_1);
- at each w_i, $2 \leq i \leq b-1$: splice the (currently available) w_i-end of the P-chain accumulated so far with P_{i+1}@w_i (one splice each).

The P-chain after the splices at $w_1, \ldots, w_i$ is a single path contained in $B_1 \cup \cdots \cup B_{i+1}$ and ending at w_{i+1}: indeed each splice consumes one endpoint of the chain (the one at the cut just crossed) and one endpoint of P_{i+1}, retaining P_{i+1}'s other endpoint, which is w_{i+1}. All splices are between paths intersecting only at their common cut vertex, hence legal. Total splices: $2 + (b-2) = b$, so

$$p(G) \;\leq\; \sum_i (j_i + 1) - b \;=\; \sum_i j_i \;=\; \lfloor n/2 \rfloor .$$

(3) For $b \geq 4$, add one further splice at the far end w_{b-1}: splice the Q-end of B_{b-1} at w_{b-1} with the Q-end of B_b at w_{b-1}. The Q-end of B_{b-1} at w_{b-1} is still available: in step (2) the block B_{b-1} only lost the endpoint P_{b-1}@w_{b-2} (consumed by the P-chain splice at w_{b-2}), not any Q-end at w_{b-1}; and B_b's two w_{b-1}-paths (P_b and its cut-Q) had only P_b@w_{b-1} consumed. The two spliced paths lie in different blocks and meet only at w_{b-1}. Total splices: $b+1$, giving

$$p(G) \;\leq\; \sum_i (j_i + 1) - (b+1) \;=\; \sum_i j_i - 1 \;=\; \lfloor n/2 \rfloor - 1. \qquad \square$$

Remark 2.19 (comparison with [6] and the computational record). (1) Theorem 2.18(ii)–(iii) are strict refinements of [6] on this subfamily: block graphs in general are only granted $\lceil n/2 \rceil$ there, while odd clique chains admit $\lfloor n/2 \rfloor$, and $\lfloor n/2 \rfloor - 1$ from four blocks on. The odd clique chains are far from arbitrary block graphs: *every* vertex has even degree (a non-cut vertex of a block K_{2j_i+1} has degree $2j_i$, and a cut vertex has degree $2j_i + 2j_{i+1}$; hence $\mathcal{E}(G) = G$), which is what the extra splicing exploits.

(2) For $b = 3$ the extra splice of (iii) needs the Q-ends of the middle block at its two cuts to be *distinct* paths (for $b \geq 4$ this is not required, by the proof above). We record that the

chain of three K_5's — here $j = 2$, so this example lies outside the domain of Theorem 2.18 and illustrates only the distinctness question above — admits a 5-path decomposition, i.e. $p \le \sum j_i - 1 = 5$, exhibited by:

$$a\text{–}b\text{–}c\text{–}d\text{–}w_1\text{–}e\text{–}f\text{–}g\text{–}w_2\text{–}h\text{–}i\text{–}j\text{–}k; \quad d\text{–}a\text{–}c\text{–}w_1\text{–}f\text{–}w_2\text{–}e; \quad e\text{–}g\text{–}w_1\text{–}w_2\text{–}k\text{–}h\text{–}j; \quad a\text{–}w_1\text{–}b\text{–}d; \quad k\text{–}i\text{–}w_2\text{–}j,$$

with blocks $B_1 = \{w_1, a, b, c, d\}$, $B_2 = \{w_1, w_2, e, f, g\}$, $B_3 = \{w_2, h, i, j, k\}$; each line is a simple path and the five paths partition the 30 edges. We note that this contradicts the value $p = 6$ for this graph recorded in the computational archive of the underlying project; the record is refuted by the printed decomposition (Appendix A). The general $b = 3$ case of the -1 refinement is open and reduces to the distinctness question above.

(3) The even-j residual of Theorem 2.15 (chains containing blocks of order $4t+1$) is open; existence is verified by computer for blocks K_5 and K_9 within chains, but no general construction is known (Section C).

2.5 Formalization and independent-recheck status of this section

Two statements of this section are fully formalized in Lean, with no admitted axioms beyond the standard three: for every odd $n \ge 3$, $p(K_{4,n}) = (n+1)/2$ (declaration `pNumK4n_odd_all`; hypothesis $3 \le n$ with n odd; an axiom audit returns exactly `[propext, Classical.choice, Quot.sound]`), and for every $k \ge 1$, $p(K_{2k+1}) = k+1$ (declaration `pCompleteOdd`, via the edge-count lower bound and the explicit Walecki-type construction). The odd clique chain theorem (Theorem 2.18) and its engine (Theorem 2.15) have no Lean formalization and rest on paper proofs; the machine anchors cited above cover the underlying decompositions, not the chaining argument.

An independent recheck script, written from scratch and run for this paper, verified: the ring-closure construction of Theorem 2.12 over all 280 parameter pairs ($h = 1, \dots, 14$ and, per h, the 20 even ring sizes m from $2h+2$ to $2h+41$; simplicity, edge-exactness and the $4h$-edge length profile at each pair), and the $K_{4,n}$ construction of Proposition 2.3 over all 95 parameter values $k = 6, \dots, 100$ (asserting $k+1$ paths, simplicity, and the exact $A \times B$ edge partition with multiplicity 1). These are construction-level checks: they cover the printed constructions and the small-case path lists of Proposition 2.4 and Corollary 2.13, and do not re-verify the search-based negatives of Observation 2.14 or the splicing argument of Theorem 2.18.

3 Fan-type theorems, the placeability classification programme, and criterion-level results

3.1 Fan-type theorems beyond the block-level degree bound

Throughout this subsection G is a connected graph, $n = |V(G)|$, and $d_{\mathrm{E}}(v)$ denotes the E-degree of v. Recall that a vertex with $d_{\mathrm{E}} \ge 4$ has even degree in G; call such a vertex *high*, and call a vertex with $1 \le d_{\mathrm{E}} \le 3$ *low*. All published results in the Fan line [13, 4, 12] require the blocks of $\mathcal{E}(G)$ to have maximum degree at most 3 (the triangle conditions of the three differ: [13] asks every block of $\mathcal{E}(G)$ to be triangle-free, [12] that $\mathcal{E}(G)$ contain no pair of triangles, and the class $\mathcal{G}$ of [4] allows one triangular block per component). This block condition is implied by, but does not imply, the absence of high vertices: a cut vertex of $\mathcal{E}(G)$ may belong to arbitrarily many blocks of maximum degree at most 3 (in $K_{1,5}$ with one pendant leaf attached at each of its six vertices, $\mathcal{E}(G) = K_{1,5}$ has a vertex of E-degree 5 while every block is a single edge); what is new below is therefore a

block of $\mathcal{E}(G)$ of maximum degree at least 4, not a high vertex as such. Theorem 3.1 allows high vertices, under an odd-degree compensation condition (A2) together with an alignment condition (A4) at one high vertex; Theorem 3.2 removes the alignment condition when distinct high vertices have no common EV-neighbour, and Remark 3.3 weakens separation to a parity condition on the profiles of common EV-neighbours; Theorem 3.4 treats the case of exactly two high vertices that share EV-neighbours, and Corollary 3.5 collects the at-most-two-high layer, leaving only a corner open there, subject to five simultaneous constraints at every common EV-neighbour of the two high vertices; the band-free tightness theorem (Remark 3.8) empties this corner on the constructed family proper whenever the counting gap is at most two.

Theorem 3.1 (alignment at a high vertex)**.** *Let G be a connected graph such that*

(A1) $\mathcal{E}(G)$ is triangle-free;

(A2) every high vertex v has odd $d_{\mathrm{E}}(v)$, and every EV-neighbour w of v satisfies $d_{\mathrm{E}}(w) \le 3$;

(A4) there is a high vertex u such that every even vertex $x \notin \{u\} \cup N_{\mathcal{E}}(u)$ satisfies $d_{\mathrm{E}}(x) - |N_{\mathcal{E}}(x) \cap N_{\mathcal{E}}(u)| \le 3$, where $N_{\mathcal{E}}(x)$ denotes the set of EV-neighbours of x.

Then $p(G) \le \lfloor n/2 \rfloor$.

Here $N_{\mathcal{E}}(x)$ denotes the set of EV-neighbours of x, i.e. its neighbours inside $\mathcal{E}(G)$.[1]

Proof. Let $F = \{uv : v \in N_{\mathcal{E}}(u)\}$ be the E-star at u, $|F| = k$, and let $\tilde{G} = G - F$. We first analyse the parity and structure of $\tilde{G}$.

Step 1 (structure of $\tilde{G}$). Every edge of F joins two even-degree vertices of G, so odd-degree vertices of G lose no edges and stay odd. The vertex u loses k edges, an odd number, so u is odd in $\tilde{G}$. Each $v \in N_{\mathcal{E}}(u)$ loses exactly one edge, so v is odd in $\tilde{G}$. Every other vertex keeps its degree. Hence $V(\mathcal{E}(\tilde{G})) = \{x \notin \{u\} \cup N_{\mathcal{E}}(u) : \deg_G(x) \text{ even}\}$, and since $\mathcal{E}(\tilde{G})$ is an induced subgraph of $\mathcal{E}(G)$, it is triangle-free by (A1).

Moreover every $x \in V(\mathcal{E}(\tilde{G}))$ has d_{E} in $\tilde{G}$ equal to $d_{\mathrm{E}}(x) - |N_{\mathcal{E}}(x) \cap N_{\mathcal{E}}(u)| \le 3$ by (A4). Thus *every* vertex of $\mathcal{E}(\tilde{G})$ has E-degree at most 3, so every block of $\mathcal{E}(\tilde{G})$ is triangle-free of maximum degree at most 3. Applying Theorem 1.6 to each component of $\tilde{G}$ (the hypotheses are component-wise) and summing with Lemma 1.3,

$$p(\tilde{G}) \;\le\; \textstyle\sum_c \lfloor n_c/2 \rfloor \;\le\; \lfloor n/2 \rfloor\,, \tag{1}$$

where n_c are the component orders. Fix a decomposition $\tilde{D}$ of $\tilde{G}$ with $|\tilde{D}| \le \lfloor n/2 \rfloor$.

Step 2 (hosts). The neighbours of u in $\tilde{G}$ are precisely its odd-degree neighbours in G; they are untouched and odd in $\tilde{G}$, so by Lemma 1.1(ii) each has $\tilde{D} \ge 1$.

Step 3 (reinsertion of the addible part). Apply Lemma 1.14 with $B = F$, $h = k$, and specified edge uv_1 (any fixed element of F). The hypothesis of Lemma 1.14 is satisfied: every neighbour of u has $\tilde{D} \ge 1$ by Step 2 (this holds whether the hypothesis is read over the neighbours of u in $\tilde{G}$ or in G, since the star vertices $v \in N_{\mathcal{E}}(u)$ are odd in $\tilde{G}$, each having lost a single edge, and have $\tilde{D} \ge 1$ by Lemma 1.1(ii)). We obtain $A \subseteq F$ with $|A| \ge \lceil k/2 \rceil$ which can be added back towards u, producing a decomposition D_1 of $\tilde{G} \cup A$ with $|D_1| = |\tilde{D}|$ and $D_1(u) \ge \tilde{D}(u) + \lceil k/2 \rceil \ge 1 + \lceil k/2 \rceil$, the bound $\tilde{D}(u) \ge 1$ coming from Lemma 1.1(ii) applied to u, which is odd in $\tilde{G}$.

[1] The numbering skips (A3): a condition so numbered in an earlier draft of this material was absorbed into the separation hypothesis of Theorem 3.2, and the remaining labels were kept unchanged.

Step 4 (reinsertion of the remainder). Let $L = F \setminus A$; $|L| = \lfloor k/2 \rfloor = (k-1)/2$. Process the edges of L one at a time. Suppose $j-1$ edges of L have been reinserted, giving a decomposition D_j of $\tilde{G} \cup A \cup (\text{first } j-1 \text{ edges of } L)$ with

$$D_j(u) \;\geq\; 1 + \lceil k/2 \rceil - (j-1).$$

Let $uv \in L$ be the next edge. We apply Lemma 1.12 *towards* v, i.e. with the roles of u and v in the statement exchanged: the condition required is

$$D_j(u) \;>\; \big|\{w \in N_{G_j}(v) : D_j(w) = 0\}\big|,$$

where G_j is the current graph, and the operation then yields a decomposition D_{j+1} of $G_j + uv$ with $D_{j+1}(v) = D_j(v) + 1$, $D_{j+1}(u) = D_j(u) - 1$, and $|D_{j+1}| = |D_j|$.

We bound the right-hand side. If $D_j(w) = 0$ then w has even degree in G_j by Lemma 1.1(iii). We classify the neighbours of v and their endpoint counts. Odd-degree vertices of G are never incident with a star edge, keep their degree, and remain odd throughout; they always have $D \geq 1$. A star vertex $v' \in N_{\mathcal{E}}(u)$, $v' \neq v$, is *not a neighbour of v at all*: the vertices v and v' are both even-degree vertices of G, so the edge vv' would close the triangle u, v, v' in $\mathcal{E}(G)$, contradicting (A1). (This is just as well: a star vertex whose edge has been reabsorbed towards u by Lemma 1.14 may well have $D = 0$, each absorption step lowering $D(v')$ by one.) The vertex u *is* a neighbour of v and has regained at least $\lceil k/2 \rceil \geq 3$ edges, so $D_j(u) \geq 3 > 0$. The only remaining neighbours of v that were even in G and lie outside $\{u\} \cup N_{\mathcal{E}}(u)$ are the members of $N_{\mathcal{E}}(v) \setminus \{u\}$, of which there are $d_{\mathrm{E}}(v) - 1 \leq 2$. Hence

$$\big|\{w \in N_{G_j}(v) : D_j(w) = 0\}\big| \;\leq\; \big|\{w \in N_{\mathcal{E}}(v) : w \neq u\}\big| \;=\; d_{\mathrm{E}}(v) - 1 \;\leq\; 2,$$

using that v is low, i.e. $d_{\mathrm{E}}(v) \leq 3$ by (A2). At the last step $j = (k-1)/2$ we have $D_j(u) \geq 1 + \lceil k/2 \rceil - (j-1) = 1 + (k+1)/2 - (k-3)/2 = 3 > 2$, and for earlier steps the budget is only larger. So each edge of L is addible towards its star vertex v, free of charge.

After all k edges are reinserted we obtain a decomposition of G with $|D| = |\tilde{D}| \leq \lfloor n/2 \rfloor$, completing the proof. □

The alignment condition is not the only route through a high vertex; the following theorem needs no alignment hypothesis at all, at the price of a separation condition between high vertices.

Theorem 3.2 (separated high vertices)**.** *Assume* (A1), (A2)*, and moreover that any two distinct high vertices of G have no common EV-neighbour. Then $p(G) \leq \lfloor n/2 \rfloor$.*

Proof. We prove the stronger statement, for all graphs in the class connected or not, by strong induction on $|E(G)|$. If there is no high vertex, every E-degree is at most 3 and Theorem 1.6 applies component-wise, so Lemma 1.3 gives the bound. Otherwise pick a high vertex u and let $k = d_{\mathrm{E}}(u) \geq 4$, which is odd by (A2); let F be the E-star at u and $\tilde{G} = G - F$. Every star vertex has odd degree in $\tilde{G}$ (it loses the single edge uv), and u has odd degree in $\tilde{G}$; the even-degree vertices of $\tilde{G}$ are precisely the even-degree vertices of G outside $\{u\} \cup N_{\mathcal{E}}(u)$, and for such a vertex x its E-degree in $\tilde{G}$ equals $d_{\mathrm{E}}(x) - |N_{\mathcal{E}}(x) \cap N_{\mathcal{E}}(u)|$; when x is high the subtraction term is 0, because distinct high vertices have no common EV-neighbour. Hence $\tilde{G}$ again satisfies (A1), (A2) and the separation condition, so the induction hypothesis gives a decomposition $\tilde{D}$ of $\tilde{G}$ with $|\tilde{D}| \leq \lfloor n/2 \rfloor$. The neighbours of u in $\tilde{G}$ are its odd-degree neighbours in G, so each has $\tilde{D} \geq 1$ (Lemma 1.1(ii)); the star vertices are odd in $\tilde{G}$, each having lost a single edge, so they too have $\tilde{D} \geq 1$ by Lemma 1.1(ii), and the hypothesis of Lemma 1.14 holds read over G; Lemma 1.14 returns an addible set $A \subseteq F$

with $|A| \geq \lceil k/2 \rceil$ towards u, giving $D_1(u) \geq 1 + \lceil k/2 \rceil$. Each edge of $L = F \setminus A$, $|L| = \lfloor k/2 \rfloor$, is then reinserted towards its star vertex v via Lemma 1.12: the vertices w with $D_j(w) = 0$ are even in G_j, and as in the proof of Theorem 3.1 they are confined to $N_{\mathcal{E}}(v) \setminus \{u\}$, of size at most $d_{\mathrm{E}}(v) - 1 \leq 2$; at the last step $D_j(u) \geq 3 > 2$. This rebuilds a decomposition of G with $\lfloor n/2 \rfloor$ paths. □

Remark 3.3. The proof of Theorem 3.2 goes through, verbatim up to one parity check, with separation replaced by the following weaker condition: for any two distinct high vertices x, z and any set S of high vertices disjoint from $\{x, z\}$, the number of common EV-neighbours w of x and z with $N_{\mathcal{E}}(w) \cap H \cap S = \emptyset$ is even (equivalently: for each pair x, z, the common EV-neighbours are split by their "profile" $\pi(w) = N_{\mathcal{E}}(w) \cap H$ into classes of even size). Indeed this condition is again inherited after a star deletion, because a vertex of $\mathcal{E}(\tilde{G})$ is precisely an even vertex seeing no member of $\{u\} \cup N_{\mathcal{E}}(u)$, so the counts for $\tilde{G}$ with S' are the original counts for $S = \{u\} \cup S'$.

Separation and its profile relaxation still leave open the case of high vertices that do share EV-neighbours. The next theorem settles the case of exactly two high vertices except for a corner recorded afterwards.

Theorem 3.4 (two high vertices)**.** *Assume* (A1) *and* (A2)*, let u, x be the only high vertices of G, and suppose $t_{ux} := |N_{\mathcal{E}}(u) \cap N_{\mathcal{E}}(x)| \geq 1$. If either*

(i) *some common EV-neighbour v_1 of u and x has $d_{\mathrm{E}}(v_1) = 2$, or*

(ii) *x has a private EV-neighbour y^*, i.e. $y^* \in N_{\mathcal{E}}(x) \setminus N_{\mathcal{E}}(u)$, with $d_{\mathrm{E}}(y^*) = 1$,*

then $p(G) \leq \lfloor n/2 \rfloor$.

Proof. Let F be the E-star at u, $k = d_{\mathrm{E}}(u)$, and delete one further edge e^* at x: $e^* = xv_1$ in (i), $e^* = xy^*$ in (ii), giving $\tilde{G} = G - F - e^*$. In $\tilde{G}$ the vertices u and x are odd, and so is every former star vertex (each lost the single edge uv); in (ii) the far endpoint y^* of e^* is odd as well (it lost xy^*). The remaining even vertices — in (i) the vertex v_1 (which lost both uv_1 and xv_1 and stays even) together with the far even vertices and the private EV-neighbours of x; in (ii) the far even vertices other than y^* and the private EV-neighbours of x other than y^* — all have E-degree at most 3 in $\tilde{G}$: the private ones by (A2) at x, the far ones because u and x are the only high vertices, and v_1 because $d_{\mathrm{E}}(v_1) = 2$. Hence $\mathcal{E}(\tilde{G})$ is triangle-free, being an induced subgraph of $\mathcal{E}(G)$, of maximum E-degree at most 3, so $\tilde{G}$ lies in the class of Theorem 1.6 and has, component-wise with Lemma 1.3, a decomposition $\tilde{D}$ with $|\tilde{D}| \leq \lfloor n/2 \rfloor$. In (i) reinsert e^* towards v_1 (Lemma 1.12: $\tilde{D}(x) \geq 1$ since x is odd, while every neighbour of v_1 in $\tilde{G}$ is odd except x — the even-degree neighbours of v_1 in G are exactly u and x — so the passing count is 0); this activates v_1. In (ii) no activation is needed: every star vertex has lost exactly one edge and is odd. Then Lemma 1.14 (hypothesis read over G, as in the proof of Theorem 3.1) returns $A \subseteq F$ with $|A| \geq \lceil k/2 \rceil$ addible towards u, and each edge of $L = F \setminus A$ is reinserted towards its star vertex as in the proof of Theorem 3.1 (passing count $\leq d_{\mathrm{E}}(v) - 1 \leq 2$, budget ≥ 3 at the last step). In (ii) the edge xy^* is reinserted towards y^*, whose neighbours except x are all odd, so this step is legal at any position of the sequence. All $k + 1$ deleted edges are restored with $|D| = |\tilde{D}| \leq \lfloor n/2 \rfloor$. □

Corollary 3.5. *If G satisfies* (A1)–(A2) *and has at most two high vertices, then $p(G) \leq \lfloor n/2 \rfloor$, unless G has exactly two high vertices u, x with $t_{ux} \geq 1$, every common EV-neighbour of u and x of E-degree* 3*, no private EV-neighbour of E-degree* 1 *on either side, and such that, for every common EV-neighbour v of u and x, writing z for the third EV-neighbour of v, F for the E-star at u and $\tilde{G} = G - F - \{xv\}$, all of the following hold: $p(\tilde{G}) = \lfloor n/2 \rfloor$; $\beta(\tilde{G}, x) = \lfloor n/2 \rfloor + 1$ (cf. Section 1.7); z is buried (i.e. $D(z) = 0$) in every optimal decomposition of $\tilde{G}$; in every optimal decomposition $\tilde{D}$*

of $\tilde{G}$, either $\tilde{D}(v) = 0$, or $\tilde{D}(v) \geq 2$ and at least $\tilde{D}(v)$ private EV-neighbours of x have $\tilde{D} = 0$; and $\tilde{G}$ has at most one component of odd order.

Proof. $t_{ux} = 0$ is Theorem 3.2 (with only two high vertices the separation condition reduces to $t_{ux} = 0$). The two alternatives are Theorem 3.4. For the remaining graphs fix a common EV-neighbour v, write z for its third EV-neighbour and $\tilde{G} = G - F - \{xv\}$; then $\mathcal{E}(\tilde{G})$ again has maximum E-degree ≤ 3, and each of the following five routes closes the case for this v, the star at u being absorbed afterwards exactly as in the proof of Theorem 3.4:

(a) $p(\tilde{G}) \leq \lfloor n/2 \rfloor - 1$; take an optimal decomposition $\tilde{D}$. If $\tilde{D}(x) \geq 3$ we are on route (b). Otherwise $\tilde{D}(x) = 1$ (it is odd by Lemma 1.1(iii), x being odd in $\tilde{G}$), and since $\deg_{\tilde{G}}(x) = \deg_G(x) - 1 \geq 5$, the unique path ending at x continues through x; cutting it at x adds one path (at most $\lfloor n/2 \rfloor$ in all) and raises the endpoint count at x to 3 — route (b);

(b) a decomposition $\tilde{D}$ with $|\tilde{D}| \leq \lfloor n/2 \rfloor$ and $\tilde{D}(x) \geq 3$: then xv is addible towards v, the passing count at v being at most 1 (z is the only neighbour of v in $\tilde{G}$ that can be dead; all other neighbours are odd);

(c) such a decomposition with $\tilde{D}(x) = 1$ and $\tilde{D}(z) \geq 1$: the passing count at v drops to 0;

(d) such a decomposition with $\tilde{D}(v) \geq 2$ larger than the number of dead private EV-neighbours of x: reinsert xv towards x (the passing count at x sees only private EV-neighbours of x, the other neighbours of x being active), so that v remains active and the full star, including uv, is absorbable;

(e) $\tilde{G}$ has at least two components of odd order: the component-wise Theorem 1.6, summed with Lemma 1.3, gives route (a).

The budget arithmetic of the reinsertion skeleton behind routes (a)–(d), and the formal statement that the exception of this corollary is exactly the simultaneous failure of all five routes at *every* common EV-neighbour, are given in Appendix B. The stated exception is exactly the case in which, for every common EV-neighbour v, none of the five routes applies.

Band closure of the corner. One further narrowing is available in the constructed family of Remark 3.8. Route (a) can fail there only when $\tilde{G}$ carries a *deadlock band* (an even vertex of $\tilde{G}$ with more corridor neighbours than terminal units; see Remark 3.8 for the definition), because a band-free $\tilde{G}$ in the family has $p(\tilde{G}) = \mathrm{odd}(\tilde{G})/2$; and the counting theorem forces $\tilde{G}$ to be band-free whenever the counting gap $\lfloor n/2 \rfloor - \mathrm{odd}(\tilde{G})/2$ — which is at least 1 in this situation — is at most 2, on the family proper (the constructed family without cross-private pairs). Hence for every family-proper instance with counting gap at most 2, route (a) applies unconditionally: the corner is empty at theorem level, not merely computationally; the cross-private instances are backed at probe level by the machine certificates of Appendix A. For larger gaps the corner is blocked only by the excess bound recorded in Remark 3.8. □

Corollary 3.6. *If G satisfies (A1) and (A2) and has exactly one high vertex, then $p(G) \leq \lfloor n/2 \rfloor$.*

Proof. Let u be the unique high vertex. Every vertex other than u is low, so every even vertex $x \notin \{u\} \cup N_{\mathcal{E}}(u)$ satisfies $d_{\mathrm{E}}(x) - |N_{\mathcal{E}}(x) \cap N_{\mathcal{E}}(u)| \leq d_{\mathrm{E}}(x) \leq 3$; this is condition (A4) of Theorem 3.1. (Alternatively, with only one high vertex the separation hypothesis of Theorem 3.2 holds vacuously.) □

We now formulate the statement with neither the alignment nor the separation condition, which is the form investigated in the computational study behind this paper and which all our computations support.

Conjecture 3.7. *Let G be a connected graph such that $\mathcal{E}(G)$ is triangle-free, and every vertex v with $d_{\mathrm{E}}(v) \geq 4$ has odd $d_{\mathrm{E}}(v)$ and only EV-neighbours w with $d_{\mathrm{E}}(w) \leq 3$. Then $p(G) \leq \lfloor n/2 \rfloor$.*

Remark 3.8 (where the general argument stops). The class (A1)–(A2) is not preserved by the only deletion the engine supports. Deleting the E-star at a high vertex u turns every star vertex odd; an even vertex x outside the star keeps its degree but loses the E-degrees contributed by star vertices, and its E-degree in $\tilde{G} = G - F$ equals $d_{\mathrm{E}}(x) - t_x$ with $t_x = |N_{\mathcal{E}}(x) \cap N_{\mathcal{E}}(u)|$. If x is a high vertex with t_x odd and $d_{\mathrm{E}}(x) - t_x \geq 4$, then x sits in $\mathcal{E}(\tilde{G})$ at an even E-degree, the class is lost, and neither Theorem 1.6 nor the induction applies to $\tilde{G}$. Concretely, in the double star on $\{u, x, v_1, w_1, \ldots, w_4, y_1, \ldots, y_4\}$ with edges $uv_1, uw_1, \ldots, uw_4, xv_1, xy_1, \ldots, xy_4$ and pendant leaves attaching so that the eleven listed vertices have even degree, both u and x are high with $d_{\mathrm{E}} = 5$, $t_{ux} = 1$ is odd, and after deleting the star at u the vertex x has E-degree 4, even. This graph (and its even-order one-leaf extension, on which our exact programs give $p = 5 \leq 11$) satisfies (A1)–(A2) but none of (A4), the separation condition, or the profile condition of Remark 3.3; a minimal counterexample to Conjecture 3.7, if it exists, must have a high vertex with such a bad partner at every high vertex (Proposition 3.9). No counterexample is known: exact dynamic programs verify $p \leq \lfloor n/2 \rfloor$ on all tested instances satisfying (A1)–(A2) (166 instances in the computational study behind this paper, a further 413 in Appendix A, zero violations), and the family above sits far from the bound ($p = \mathrm{odd}(G)/2$).

The "one endpoint unit" obstruction recorded in an earlier draft of this material was an artefact: the reinsertion bookkeeping it refers to restores $k + 2$ edges after deleting $k + 1$, i.e. counts the patched edge xv_1 twice (an activation step towards v_1 and a final step towards x). Once x's endpoint loss is recognised as cost-free — $|D|$ is the only invariant, and the argument of Theorem 3.4 involves no induction — the obstruction disappears for all two-high graphs except the corner recorded in Corollary 3.5.

The exceptional graphs of Corollary 3.5 are subject to five simultaneous constraints, of which only the first two were recorded when the corner was first isolated: the burial of the third EV-neighbour z and of v (or of enough private neighbours of x) in *every* optimal decomposition of $\tilde{G}$ (with $\tilde{G}$ as in Corollary 3.5), and the parity of the components of $\tilde{G}$, are each individually sufficient to reopen the reinsertion argument (the edge xv can then be added towards v at zero passing cost, or a second deletion $G - \tilde{G}$ becomes unnecessary). No instance satisfying even the original two constraints was found: across 207 constructed instances of the dirty class—including the entire range where the counting gap $\lfloor n/2 \rfloor - \mathrm{odd}(\tilde{G})/2$ attains its minimum 1, which the degree capacity of the two far vertices forces to a single parameter-free shape—a decomposition of $\tilde{G}$ with at most $\lfloor n/2 \rfloor - 1$ paths was always produced, so the corner never materializes in computation.

The band-free tightness statement is now a theorem for the *constructed family* (the two-high graphs satisfying the invariants of our construction pipeline: no $\mathcal{E}$-edge joins two far even vertices of $\tilde{G}$; the odd-neighbour blocks of u and of x are of leaf type (an odd number of pendant leaves) or gadget type, possibly mixed; every far vertex of $\tilde{G}$ has degree at most 4; the 207 instances above and the 125 decision-level instances of Appendix A all lie in it). Writing, for an even vertex f of $\tilde{G}$, $Y(f)$ for its *corridor* neighbours — the degree-two even vertices adjacent to both x and f — and $N(f)$ for its neighbour set in $\tilde{G}$, call f a *deadlock band* if it has more corridor neighbours than terminal units, where the terminal units of f are its remaining incident edges (in symbols, $|Y(f)| > |N(f)| - |Y(f)|$). If no even vertex is a deadlock band, then $\tilde{G}$ admits a *paired* decomposition: exactly $\mathrm{odd}(\tilde{G})/2$ paths with all and only the odd vertices of $\tilde{G}$ as endpoints.

The proof is a five-step pairing scheme—self-contained odd-neighbour blocks, forced fusions of the corridors and of the unit at the common neighbour, an injective pairing of corridor ends with distinct terminal units at every far vertex (which is precisely the negation of the band condition), and a sorted offset matching at x that never joins two corridors of the same far vertex, possible because a far vertex of the family has degree at most 4 and hence at most two corridors. Every resulting trail visits x once and leaves each far vertex for good, so no trail closes. A single band, on the other hand, costs exactly one path: spending the free edge of x on a surplus corridor makes that corridor a double endpoint and lets its partner ride through x to a terminal, matching the exact value $p(\tilde{G}) = \mathrm{odd}(\tilde{G})/2 + 1$ computed by dynamic program. Consequently the corner of Corollary 3.5 is empty whenever the counting gap is at most 2, unconditionally on the constructed family proper, with the cross-private extension covered at probe level by the machine certificates of Appendix A; for larger gaps it remains blocked only by the excess bound excess $\leq \lfloor y/2 \rfloor$, where excess $:= p(\tilde{G}) - \mathrm{odd}(\tilde{G})/2$ and y is the number of corridors — verified on all 235 computed instances of the family, the provable bound being $p(\tilde{G}) \leq \mathrm{odd}(\tilde{G})/2 + 2B - 1$ for B bands, with a single band forcing exactly $+1$. The restriction to the constructed family is now known to be sharper than necessary in two directions and irreducible in a third. Far–far edges of the even subgraph and far vertices carrying several parity leaves, both excluded by the construction, can be admitted: the corridor ends at a far vertex f admit an injective pairing with terminal units and far–far ends whenever $|Y(f)| \leq t(f) + d(f)$ (where $t(f)$ is the number of terminal units of f and $d(f)$ its far–far degree), and the matching at x survives arbitrary corridor multiplicities because a far vertex of the class has at most three even neighbours, so a rearrangement argument replaces the sorted offset. What cannot be admitted are arbitrary odd-neighbour blocks: a block whose component has exactly two odd vertices admits a paired decomposition if and only if it admits a Hamilton path between them, and the cycle-with-one-pendant and double-triangle blocks have none, so $\tilde{G}$ can be untight while carrying no band at all—the band-free tightness statement is a phenomenon of the constructed family, not of the class, and these cycle blocks are the first known counterexamples to the general band-free statement. (Same-side pairs of private neighbours, a third candidate obstruction, are excluded by the triangle-free condition (A1) already.) The counting gap of the dirty class also assumes that every even vertex lies in the skeleton; blocks with even interior vertices shift $\lfloor n/2 \rfloor$ and must be recomputed directly (see Appendix A for the machine certificates).

Proposition 3.9. *Let G be a minimal counterexample to Conjecture 3.7. Then every high vertex u of G admits a far high vertex x with $|N_{\mathcal{E}}(x) \cap N_{\mathcal{E}}(u)|$ odd and $d_{\mathrm{E}}(x) - |N_{\mathcal{E}}(x) \cap N_{\mathcal{E}}(u)| \geq 4$. In particular, by Theorem 3.4 a minimal counterexample has at least three high vertices, unless it is one of the two-high exceptional graphs of Corollary 3.5 (empty on the constructed family proper whenever the counting gap is at most two, by the band-free tightness material of Remark 3.8; no instance was found in our computations, in or beyond the family).*

Proof. Read "minimal" as minimal with respect to the number of edges, and let u be a high vertex of G. We show that if u has no bad partner, then $p(G) \leq \lfloor n/2 \rfloor$, contradicting the choice of G; hence u has a bad partner. Suppose, then, that no far even vertex x satisfies t_x odd and $d_{\mathrm{E}}(x) - t_x \geq 4$, where $t_x = |N_{\mathcal{E}}(x) \cap N_{\mathcal{E}}(u)|$. Let F be the E-star at u and $\tilde{G} = G - F$. As in Step 1 of the proof of Theorem 3.1, the even-degree vertices of $\tilde{G}$ are precisely the far even vertices of G, $\mathcal{E}(\tilde{G})$ is an induced subgraph of $\mathcal{E}(G)$ and hence triangle-free, and a far vertex x has E-degree $d_{\mathrm{E}}(x) - t_x$ in $\tilde{G}$. Thus every high vertex of $\tilde{G}$ is a far even vertex x of G with $d_{\mathrm{E}}(x) - t_x \geq 4$; by our assumption t_x is even for such an x, and since $d_{\mathrm{E}}(x) \geq 4$ makes x high in G, (A2) gives $d_{\mathrm{E}}(x)$ odd, so the E-degree $d_{\mathrm{E}}(x) - t_x$ of x in $\tilde{G}$ is odd. Moreover the EV-neighbours w in $\tilde{G}$ of such an x satisfy $d_{\mathrm{E}}(w) - t_w \leq d_{\mathrm{E}}(w) \leq 3$, the last inequality by (A2) applied to the high vertex x in G. Hence $\tilde{G}$ satisfies (A1) and (A2). Each component of $\tilde{G}$ is a connected graph satisfying (A1)–(A2) with fewer

edges than G (as u is high, $F \neq \emptyset$), so by minimality each admits a decomposition with at most $\lfloor n_c/2 \rfloor$ paths, and Lemma 1.3 gives a decomposition $\tilde{D}$ of $\tilde{G}$ with $|\tilde{D}| \leq \lfloor n/2 \rfloor$. Now the reinsertion argument of Steps 3 and 4 of the proof of Theorem 3.1 uses (A4) only through the existence of such a $\tilde{D}$: the star vertices are odd in $\tilde{G}$, so the hypothesis of Lemma 1.14 holds read over G exactly as in Step 3, and each edge of $F \setminus A$ is addible towards its star vertex against a passing count of at most $d_{\mathrm{E}}(v) - 1 \leq 2$ (by (A2)) and a budget of at least 3. The full star is restored with $|D| = |\tilde{D}| \leq \lfloor n/2 \rfloor$ paths — a contradiction. □

We also record a structural consequence of (A1) on odd orders, used below to place the sharpness family of Proposition 3.11 relative to the class.

Proposition 3.10. *Let G be connected of odd order n with $\mathcal{E}(G)$ triangle-free. Then $m \leq (n-1)^2/2$; equivalently, no odd semi-clique has triangle-free $\mathcal{E}(G)$.*

Proof. Suppose $c = \binom{n}{2} - m \leq (n-3)/2$. Let S be the set of even-degree vertices, $s = |S|$; since n is odd, a vertex has odd degree iff its complement-degree is odd, so each vertex outside S has positive complement-degree. Let t be the number of complement edges inside S. Since $G[S] = \mathcal{E}(G)$ is triangle-free, Mantel's theorem gives $\binom{s}{2} - t \leq \lfloor s^2/4 \rfloor$, i.e. $t \geq \binom{s}{2} - \lfloor s^2/4 \rfloor$. Counting complement degrees over S and over its complement separately, $2c \geq 2t + (n-s)$, so $2t \leq s - 3$. If s is even this reads $s(s-2)/2 \leq s-3$, i.e. $s^2 - 4s + 6 \leq 0$; if s is odd, $(s-1)^2/2 \leq s-3$, i.e. $s^2 - 4s + 7 \leq 0$; both are impossible. (Note $s \geq 1$: the number of odd-degree vertices is even while n is odd.) □

We record the sharpness of the triangle-free requirement. By Proposition 3.10 this failure is a density phenomenon: on odd orders, high density and (A1) are mutually exclusive, so the following family is forced.

Proposition 3.11. *For every $k \geq 2$, the graph $G_k = K_{2k+1} - (k-1)K_2$ (delete a matching of size $k-1$) satisfies $\mathcal{E}(G_k) \cong K_3$, hence violates (A1), and $p(G_k) \geq k+1 = \lceil n/2 \rceil > \lfloor n/2 \rfloor$. Moreover $p(G_k) = k+1$ for the tested values $k \in \{2, 3, 4, 5, 6, 8, 10\}$, with explicit decompositions machine-verified.*

Proof. The $k-1$ matched vertices have degree $2k-1$ and the three unmatched vertices have degree $2k$; the latter are pairwise adjacent, so $\mathcal{E}(G_k)$ is a triangle on them. The edge count is $m = \binom{2k+1}{2} - (k-1) = 2k^2 + 1$, so Lemma 1.2 gives $p \geq \lceil (2k^2+1)/2k \rceil = k+1$; equivalently G_k is an odd semi-clique and Observation 1.8 applies. This proves the necessity claim for all k. For the tested values, explicit decompositions into $k+1$ paths (obtained from the Walecki cycle decomposition of K_{2k+1}, spreading the matching edges over distinct cycles, cutting each affected cycle at its matching edge and splitting the untouched cycle into a path plus a single edge) were verified by machine to partition $E(G_k)$ exactly; for $k = 2$ the value $p(K_5 - M_1) = 3$ is also certified by exact dynamic programming. □

Remark 3.12. The exactness claim in Proposition 3.11 is stated only for tested k: the existence, for all k, of a Walecki-type cycle decomposition of K_{2k+1} in which a given $(k-1)$-matching is spread over distinct cycles is plausible but we do not prove it.

Next, the witness family $W(m)$, which motivates Theorem 3.1: it lies in no published class we are aware of, yet satisfies all hypotheses.

Lemma 3.13. *For every odd $m \geq 5$, the graph $W(m)$ of Figure 1 satisfies $\mathcal{E}(W(m)) \cong K_{2,m}$ exactly, hence satisfies (A1) and (A2) with $d_{\mathrm{E}}(a_1) = d_{\mathrm{E}}(a_2) = m$ and $d_{\mathrm{E}}(b_j) = 2$ for all j.*

For odd $m \geq 5$, let $W(m)$ have vertex set $\{a_1, a_2\} \cup \{b_1, \ldots, b_m\} \cup \{\ell_1, \ell_2\} \cup \{h, h'\} \cup \{x_1, \ldots, x_6\}$ and edge set

$$E(K_{2,m}) \cup \{a_1\ell_1,\, a_2\ell_2\} \cup \{b_1h,\, b_1h'\} \cup \{hx_1,\, hx_2,\, h'x_1,\, h'x_2\} \cup E(K_6),$$

where K_6 has vertex set $\{x_1, \ldots, x_6\}$. Precisely: h and h' each have neighbourhood $\{b_1, x_1, x_2\}$.

Figure 1: The witness family $W(m)$.

Proof. The degrees are: $\deg(a_i) = m + 1$, even; $\deg(b_j) = 2$ for $j \geq 2$ and $\deg(b_1) = 4$, all even; $\deg(\ell_i) = 1$, $\deg(h) = \deg(h') = 3$, and $\deg(x_i) = 5 + 2 = 7$ for $i \in \{1, 2\}$, $\deg(x_i) = 5$ for $i \in \{3, \ldots, 6\}$, all odd. Hence the even-degree vertex set is exactly $\{a_1, a_2\} \cup \{b_1, \ldots, b_m\}$, and the induced subgraph is $K_{2,m}$: triangle-free, with $d_{\mathrm{E}}(a_i) = m$ odd and all EV-neighbours of a_i of E-degree $2 \leq 3$. Note that the two degree-3 vertices h, h' attach to b_1 *in pairs* (both h and h'), keeping b_1 even; attaching a single such vertex would flip b_1 to odd degree and destroy the structure. □

Proposition 3.14. *For every odd $m \geq 5$, $p(W(m)) \leq \lfloor n/2 \rfloor$ with $n = m + 12$.*

Proof. By Lemma 3.13 the graph satisfies (A1), (A2). We verify (A4) at $u = a_1$: the only even-degree vertex outside $\{a_1\} \cup \{b_1, \ldots, b_m\}$ is a_2, and, since $N_{\mathcal{E}}(a_1) = N_{\mathcal{E}}(a_2) = \{b_1, \ldots, b_m\}$, $d_{\mathrm{E}}(a_2) - |N_{\mathcal{E}}(a_2) \cap N_{\mathcal{E}}(a_1)| = m - m = 0 \leq 3$; every vertex outside this set other than a_2 has odd degree, hence is not in $\mathcal{E}(\tilde{G})$ after the deletion and is irrelevant (its E-degree in $\tilde{G}$ is 0). Thus (A4) holds, and Theorem 3.1 gives $p(W(m)) \leq \lfloor (m + 12)/2 \rfloor$. For $m \in \{5, 7, 9\}$ explicit decompositions into exactly $\lfloor n/2 \rfloor = 8, 9, 10$ paths were in addition machine-verified. □

We now compare Theorem 3.1 with the published record. The following two statements make the incomparability precise; both witnesses have exactly computed path numbers.

Proposition 3.15 (escape)**.** *Let $m \geq 5$ be odd and let $W(m)$ be as in Figure 1. Then $\mathcal{E}(W(m)) \cong K_{2,m}$, whence Theorem 3.1 gives $p(W(m)) \leq \lfloor n/2 \rfloor$ with $n = m + 12$. No published theorem known to us implies this bound:*

- *every member of the Fan line [13, 4, 12] requires every block of $\mathcal{E}(G)$ to have maximum degree at most 3; the graph $\mathcal{E}(W(m)) \cong K_{2,m}$ is 2-connected, hence a single block of maximum degree $m \geq 5 > 3$, so this requirement fails (the block level, not the global value of $\Delta(\mathcal{E})$, is the discriminating parameter — compare $K_{1,5}$ with one pendant leaf at each of its six vertices, where $\Delta(\mathcal{E}) = 5$ while every block of $\mathcal{E}$ is a single edge);*
- *$W(m)$ contains a K_6, so $\mathrm{tw}(W(m)) \geq 5$ and its degeneracy is at least 5, which excludes [17] (treewidth ≤ 4), [5] (treewidth ≤ 3) and [21] (3-degenerate);*
- *$\Delta(W(m)) = \max(m+1, 7) \geq 7$, which excludes [3] ($\Delta \leq 5$) and [7] ($\Delta \leq 6$ with an independence condition);*
- *$\Delta(W(m)) \geq 7$, so $W(m)$ is not subcubic and the polynomial-time algorithm of [19] for subcubic graphs does not apply;*
- *$\mathcal{E}(W(m)) \cong K_{2,m}$ contains a 4-cycle, so it is not a forest and [20] does not apply;*
- *$W(m)$ is not Eulerian, so [14] does not apply; it is not a block graph, so [6] does not apply.*

For $m \in \{5, 7, 9\}$ every membership and non-membership claim above, together with explicit decompositions into $\lfloor n/2 \rfloor$ paths, is machine-certified.

Proposition 3.16 (proper inclusion in both directions)**.** *The class of Theorem 3.1 neither contains nor is contained in the Botler–Sambinelli class $\mathcal{G}$ [4] or the Fan–Hou–Zhou class [12].*

(i) *Let G be $K_{2,5}$ with one pendant leaf attached to each of the two degree-5 vertices. Then $\mathcal{E}(G) \cong K_{2,5}$ is triangle-free, the two degree-5 vertices have odd E-degree 5 with all E-neighbours of E-degree 2, so G satisfies (A1)–(A2); but the graph $\mathcal{E}(G) \cong K_{2,5}$ is 2-connected, hence a single block of maximum degree $5 > 3$, so G lies in none of [13, 4, 12]. Moreover $p(G) = 3 \leq \lfloor 9/2 \rfloor$ by exact computation. (For the record, G is 2-degenerate, so its bound is also implied by [21]; this witness is used only for the incomparability with the Fan line.)*

(ii) *Let G' be a triangle $\{t_1, t_2, t_3\}$ together with two vertices w_1, w_2 adjacent only to t_1 and to three pendant leaves each, so that $\mathcal{E}(G')$ consists of the triangle $\{t_1, t_2, t_3\}$ plus the two bridge edges t_1w_1, t_1w_2. Every block of $\mathcal{E}(G')$ has maximum degree at most 3 and its single component contains exactly one block with triangles, so G' lies in $\mathcal{G}$ and in the Fan–Hou–Zhou class; yet the triangle in $\mathcal{E}(G')$ violates (A1), so Theorem 3.1 is silent on G'. Note G' is not one of the degenerate exceptions of the "Gallai-or-SET" alternative of [4]. Exactly, $p(G') = 4 \leq \lfloor 11/2 \rfloor$ by exact computation, while the main theorem of [4] gives only $p(G') \leq \lceil 11/2 \rceil = 6$ on its class (the Botler–Sambinelli bound is stated with Gallai's ceiling, as noted in Section 1); the Fan–Hou–Zhou membership of G' rests on the block structure of $\mathcal{E}(G')$ alone and is unaffected.*

Remark 3.17 (formalization and provenance status of the Fan-type chain). The Lean formalization of the statements of this subsection is in progress: in the current Lean sources the declarations corresponding to Theorem 3.1, Theorem 3.2 and their supporting reinsertion lemmas (`astar`, `astar_separated`, `fan6_use`, `fan7_use`, `fan_class`) carry admitted ("`sorry`") placeholders — five in total — so none of them may be quoted as machine-verified or formally proved, and the paper cites the chain as paper proofs with formalization in progress. One research-technical lemma of the formalization archive (`fan6_nf`, a pointwise-profile strengthening of Lemma 1.12) was refuted by a machine counterexample, is sealed as a research record, and is never cited in this paper; the Lean formalization of Lemma 1.12 uses the four-clause form obtained by deleting the final pointwise clause $D(w) = D'(w)$ for the remaining vertices, and no application in this paper consumes that clause. On the provenance side: the statements of Lemma 1.12 and Lemma 1.14 follow the restatement in [4] (Lemmas 6 and 7 there); the proofs in [13] could not be consulted and the transcription has not been independently checked against the original. The machine evidence is direction-corrected: the bidirectional rescan of Lemma 1.12 is complete at $n \leq 6$, all 1,701,141,150 ordered instances at $n = 6$ (the fixed-orientation half exactly reproducing the historical census of 850,570,575), with zero violations.

3.2 The placeability classification programme on K_{2k+1}

3.2.1 The layer domain

Fix the layer domain: the carrier is K_{2k+1} on $V = \mathbb{Z}_{2k} \cup \{\infty\}$; a *type* is an isomorphism class of k-edge loopless graphs without isolated vertices; and for such a type F we say that F is *sharp at layer k* ($\mathrm{sharp}_k(F)$) if there is an injection $\phi : V(F) \to V$ whose k image-edge classes hit $\mathbb{Z}_k$ exactly, where the edge classes are given by the closed form

$$\mathrm{class}(\{u, v\}) = \lfloor (u + v)/2 \rfloor \bmod k, \qquad \mathrm{class}(\{\infty, y\}) = y \bmod k. \tag{2}$$

A placement realizing $\mathrm{sharp}_k(F)$ is a *witness*. Write

$$\mathcal{E}_k := \{F : \neg\,\mathrm{sharp}_k(F)\}$$

for the *exception set* at layer k, and $K_n@k$ for the sharpness question for K_n at layer k. The values $k = \binom{n}{2}$ are the *triangular layers*; $K_3@3$, $K_4@6$ and $K_5@10$ are sharp (three positive controls), so $n \geq 6$ is the exact lower bound for exceptions among complete graphs.

3.2.2 The first four complete-graph layers

The exact status of the first four critical layers is as follows; tiers are quoted verbatim and travel with every use of the numbers.

1. $K_6@15$ **(canonical).** K_6 is not sharp at $k = 15$, and $\mathcal{E}_{15} = \{K_6\}$ exactly — the only layer at which the uniqueness side is closed. The evidence is four-fold: three independent depth-first implementations exhaust the same 145,923-node search tree without witness and without cap, and a direct combinatorial exhaustion of all $\binom{31}{6} = 736{,}281$ six-subsets of the carrier (the image-edge set of a K_6 placement depends only on the six chosen vertices) finds zero placements with 15 distinct classes.

2. $K_7@21$ **(machine universal-closure candidate, not independently audited).** K_7 is not sharp at $k = 21$, closed on three independent lines: an engine exhaustion of 2,868,963 nodes without witness and without cap; a regime census of 3,071,958 configurations; and an infinity-census of 3,472,896 configurations (all zero hits).

3. $K_8@28$ **(machine universal-closure candidate, not independently audited).** K_8 is not sharp at $k = 28$: a nine-line census over 354,484,845 configurations with zero hits, zero truncation, and no sieve dependence (the sieve-excluded regimes were also actually scanned and are empty).

4. $K_9@36$ **(machine universal-closure candidate, not independently audited).** K_9 is not sharp at $k = 36$: a configuration-level true enumeration over 18,385,772,680 configurations (nine regime lines summing to $\binom{71}{8} = 10{,}639{,}125{,}640$ — the Vandermonde identity is machine-checked — plus the infinity line $\binom{36}{8} \cdot 2^8 = 7{,}746{,}647{,}040$, covered in four shards with an exact non-overlapping-partition assertion), every line zero hits, doubly cross-validated against the layered-arithmetic verdict below.

Remark 3.18 (a registered prediction that must not be read as a measurement). An earlier registration predicted for $K_8@28$ "surviving regimes only $\{q = 0, q = 7\}$"; the phrase is a registered prediction, not a measured residual. The actual census found zero surviving regimes on every line, and the registered phrase must not be quoted as the census outcome.

3.2.3 The characterization candidate and the propagation bound

The four layers above are exactly what the following candidate predicts. In its literal form: for every $k \geq 1$,

$$\mathcal{E}_k \;=\; \{\, K_n \;:\; n \geq 6,\ \tbinom{n}{2} = k \,\}, \tag{3}$$

equivalently the three-part decomposition: (i) $K_n@\binom{n}{2}$ is not sharp for all $n \geq 6$ (sharp at $n = 3, 4, 5$, making $n \geq 6$ the exact lower bound); (ii) $\mathcal{E}_k$ is empty off the triangular layers; (iii) on triangular layers the exception is exactly the complete graph.

Remark 3.19 (registry-layer status of (3); travels with any citation). Three registration-layer facts bind every use of (3). *First*, two competing candidate shapes are on record — the literal shape (3) and the structural shape "$\mathcal{E}_k$ = the types containing a K_v component" — and the two have never been adjudicated against each other. *Second*, the literal uniqueness side (iii) stands in tension with the propagation theorem (a super-type is sharp only if each of its components is placeable; contrapositively, an unplaceable component makes every *same-layer* type containing it unsharp): since K_7 is unsharp at layer 21, at $k = 21$ the literal shape would force $|\mathcal{E}_{21}| = 1$, while the structural shape predicts a larger set. An earlier registration claimed the stronger quantitative form $|\mathcal{E}_{21}| \geq 740{,}226$ via the map $T \mapsto K_7 \sqcup T$ on the 14-edge types; that claim was withdrawn on review — $K_7 \sqcup T$ has $21 + 14 = 35$ edges and is *not* a layer-21 type, so the construction has no objects in its claimed quantifier domain (and cross-layer inheritance of unplaceability has known counterexamples in the positive direction). The honest paper-level reading is $|\mathcal{E}_{21}| \geq 1$ (the type K_7 itself, under both shapes), and uniqueness of the minimal core survives as an open question for $k \geq 21$. *Third*, the candidate carries mixed candidate/profile load and is not independently audited; its closed faces are: side (ii) to $k \leq 16$ (including the $k = 17$ full-type sweep of 53,394,755 types, every type sharp, tier machine universal-closure candidate, not independently audited), side (i) to $n = 13$ (tower plus census at $n = 6..9$ and the layered-arithmetic deaths of K_{11}@55, K_{12}@66, K_{13}@78), and side (iii) only at $k = 15$; cumulative over $k = 2..17$, 69,698,732 types were scanned with exactly one counterexample (K_6@15). Open faces: layers $k \geq 18$ (other than 17 itself), tower layers $n \geq 14$, uniqueness of the minimal core for $k \geq 21$, and the prime-family infinity segment (Section 3.2.4).

3.2.4 The prime-family theorem at its non-infinity scope

Theorem 3.20 (prime family, non-infinity subdomain)**.** *Let $n = p+1$ with $p \geq 7$ a prime congruent to* 3 mod 4*, and let $k = \binom{n}{2}$ (which is even, divisible by p, with $p \mid n-1$). Then every non-infinity regime of K_n@k is dead on paper: the arithmetic sieve forces $q \equiv eo \equiv 0 \pmod{p}$, leaving exactly the allowed regimes $\{(n,0),(1,p),(p,1)\}$, and each of the three dies by the order-2 criterion, whose feasibility equation reduces to "n must be a perfect square" — contradicting $p \geq 7$.*

Proof. Write e and o for the numbers of even- and odd-degree placed vertices, $q = eo$ for the number of odd–odd pairs, and Σu for the sum of the half-values $u_i = (y_i \pm k)/2$-type representatives of the finite placed vertices. Two identities hold for any placement: the total class sum reads $k(k-1)/2 + k \cdot t$ with t the number of half-sums in the upper half-range, and twice the sum of the half-sums over all pairs equals $(n-1)\Sigma u - q$ (each odd–odd pair loses one in the halving). Since the class multiset must be exactly $\mathbb{Z}_k$, summing the classes gives $(n-1)\Sigma u = k(k-1) + 2kt + q$. Reducing mod a prime p with $p \mid k$ and $p \mid (n-1)$ annihilates the first two terms, forcing $q \equiv 0 \pmod{p}$. With $e + o = n - 1 = p + 1$ we get $q = eo \equiv e(1-e) \pmod{p}$; since p is prime, $p \mid e$ or $p \mid e - 1$, and $1 \leq e \leq p$ forces $e \in \{1, p\}$ (the regimes $(1,p),(p,1)$; the case $e = p+1$ i.e. $o = 0$ is the regime $(n,0)$, where $q = 0$ trivially). These are exactly the three allowed regimes.

Now k is even, so the order-2 (second-character) criterion applies: writing $x_E = \sum(-1)^{u_i}$ over even points and $x_O = \sum(-1)^{u_j}$ over odd points, feasibility requires $x_E^2 + 2x_E x_O - x_O^2 = e - o$, equivalently $2x_E^2 - (x_E - x_O)^2 = e - o$, with $|x_E| \leq e$, $|x_O| \leq o$, $x_E \equiv e \pmod 2$, $x_O \equiv o \pmod 2$. *Regime* $(n,0)$: $o = 0$, so $x_O = 0$ and $x_E^2 = n$: n must be a perfect square. *Regime* $(1,p)$: $x_E = 1$ (from $x_E \equiv 1$, $|x_E| \leq 1$), so $(x_O - 1)^2 = p + 1 = n$: again n must be a perfect square. *Regime* $(p,1)$: $x_O = \pm 1$, so $(x_E + x_O)^2 = n$: the same conclusion. For $p \geq 7$, $n = p+1$ lies strictly between $3^2 = 9$'s neighbours $(\lceil\sqrt{8}\rceil)^2$ and the next squares: concretely $(p-1)/2 + 1$ is not an integer square root situation — $p+1$ is even and $(\frac{p}{2})^2 < p+1 < (p+2)^2$ with no square in between only for $p \in \{3\}$; since $p \geq 7$, n is not a perfect square (if $n = (m-1)(m+1)$ with $m \geq 2$, then $p = (m-1)(m+1)$

is divisible by both $m-1$ and $m+1$, so $m=2$ and $p=3$, contradiction). All three allowed regimes die. □

Remark 3.21 (scope of Theorem 3.20; binding). Three boundary sentences travel with Theorem 3.20. *First*, it covers the non-infinity regimes only: the infinity branch of the prime family is *open* — there is no paper proof that infinity-mode cannot be sharp for $n=p+1$ with $p\equiv 3 \pmod 4$, $p\geq 7$ (the machine route through the layered arithmetic is blocked at the first prime layer $d=p$ for $p\geq 19$, where the fresh-enumeration volume $\binom{3p-1}{2p-1}$ exceeds executability; $p=7$ and $p=11$ have been completed, the latter closing the infinity branch of K_{12}@66). The tower's infinity-mode deaths cover only $n=5..13$. Consequently the extrapolations registered in the source material — "$\mathcal{E}$ contains the infinite family $\{K_{p+1}\}$" and "seed propagation across $n=6..12$" — are *withdrawn*: the theorem gives the non-infinity subdomain only, is a sufficient-family statement and not a classification, and no infinite-family lower bound for $\mathcal{E}$ follows from it until the infinity branch is closed. *Second*, the full per-n verdicts require the infinity branch separately even where the non-infinity side is dead ($n=8$ is closed by census; $n=12$ by the completed layered enumeration; the remaining prime-family members are open). *Third*, the machine companion table for $n\leq 200$ lists 28 full-death points, of which 23 have $n-1$ prime (matching the theorem) and 5 are composite points $\{112, 116, 124, 160, 188\}$ — the theorem is a sufficient family, not a classification of the death set.

3.2.5 The corrected Fourier identity of the regime-general character equation

The layered arithmetic used above has a closed-form face via the characters of $\mathbb{Z}_k$. For a nontrivial character ψ of $\mathbb{Z}_k$, write, for a placement S with finite points parametrized as $y=2v$ (even) and $y=2w+1$ (odd),

$$\Psi_E=\textstyle\sum_i \psi(v_i),\quad \Psi_{E,2}=\sum_i \psi(2v_i),\quad \Psi_O=\sum_a \psi(w_a),\quad \Psi_{O,2}=\sum_a \psi(2w_a).$$

Theorem 3.22 (corrected Fourier identity)**.** *Let ψ be a nontrivial character of $\mathbb{Z}_k$.*

(i) *(non-infinity placements) If a placement is sharp, then*

$$\tfrac{1}{2}\big(\Psi_E^2-\Psi_{E,2}\big)+\psi(1)\cdot\tfrac{1}{2}\big(\Psi_O^2-\Psi_{O,2}\big)+\Psi_E\Psi_O \;=\; 0.$$

(ii) *(infinity placements) If a placement with $\infty\in S$ is sharp, then the infinity-class contribution is*

$$\rho_\psi \;=\; \sum_i \psi(y_i \bmod k) \;=\; \Psi_{E,2} \;+\; \psi(1)\cdot\Psi_{O,2}, \tag{4}$$

i.e. both sides are doubled: the odd-side contribution is the doubled half-value sum $\Psi_{O,2}=\sum_a\psi(2w_a)$, not Ψ_O. Equivalently, the infinity-mode equation takes the "$i\leq j$" form

$$\tfrac{1}{2}\big(\Psi_E^2+\Psi_{E,2}\big)+\psi(1)\cdot\tfrac{1}{2}\big(\Psi_O^2+\Psi_{O,2}\big)+\Psi_E\Psi_O \;=\; 0.$$

Proof. (i) The classes of even–even pairs are class$(\{2v_i,2v_j\})=v_i+v_j$, of odd–odd pairs class$(\{2w_a+1,2w_b+1\})=w_a+w_b+1$, and of mixed pairs class$(\{2v_i,2w_a+1\})=v_i+w_a$ (2). Summing ψ over all unordered pairs with distinct classes, using the algebraic identity $\sum_{i\neq j}\psi(x_i+x_j)=\Psi^2-\Psi_2$ folded to $i<j$ (which produces the $-\Psi_{E,2}$ resp. $-\Psi_{O,2}$ halves and the factor $\psi(1)$ on the odd side from the $+1$ shift), the sharpness hypothesis (class multiset $=\mathbb{Z}_k$ exactly) makes the total vanish.

(ii) The infinity edges $\{\infty,y\}$ have classes $y \bmod k$, so their character contribution is $\sum_i\psi(y_i \bmod k)$. For an even point $y=2v$ this is $\psi(2v)$; for an odd point $y=2w+1$ it is $\psi(2w+1)=\psi(2w)\psi(1)$. Summing gives $\rho_\psi=\Psi_{E,2}+\psi(1)\Psi_{O,2}$ (4). Substituting into (i) and merging the doubled contributions — each half-value sum now appears both through its square and through its doubled self-sum — turns the "$i<j$" form into the "$i\leq j$" form stated. □

Remark 3.23 (why the corrected form is quoted; the falsified predecessor). An earlier draft of the identity stated the infinity contribution as $\rho_\psi = \Psi_{E,2} + \psi(1)\Psi_O$, omitting the odd-side double angle. That form is false: on the K_3 configuration $\{\infty, 1, 3\}$ at $k = 3$ with $\psi(a) = \omega^a$, $\omega^3 = 1$, $\omega \neq 1$ (a sharp placement, the classes being $1, 0, 2$), the original form gives the residual $\omega + 2\omega^2 = -2 - \omega \neq 0$, while the corrected form gives 0. This was verified independently in exact $\mathbb{Z}[\omega]$ integer arithmetic, and the corrected identity (4) was verified identically on all infinity placements at $k = 3, 6, 10$ ($15 + 220 + 4{,}845$ placements; the original form fails on 8, 178 and 4,514 of them respectively). Two accompanying sentences bind every citation of the layered-arithmetic machine results: *all machine results (the tower counting rules and the censuses) are independent of the character-equation formula* — the machine enumeration counts infinity classes by their actual residues $2a$ and $2a + 1$, which is the corrected bookkeeping — and the erroneous formula, wherever it existed, appeared only in the project's internal working notes, and in no form in any version of this paper.

3.2.6 The revival-depth law and the falsification of the forall-k target

We now turn to the polar-corner family of the placeability programme. For $k \geq 5$ let $T_k = (k-1) \cdot K_2$ (the *polar-corner family*): the placement engine places the $k-1$ edges one by one, a *state* $\sigma = (U, C)$ at slot j has j components placed with occupied points U ($|U| = 2j$) and occupied classes C ($|C| = j$), free points $F = \mathbb{Z}_{2k} \setminus U$, free classes $C_f = \mathbb{Z}_k \setminus C$, and $\rho := m - j = (k-1) - j$ unplaced components. The state is *dead* if $\mathrm{Span}(F) \cap C_f = \emptyset$ (no further K_2 can be placed); a *d-revival* of a dead state is the retraction of some d placed components (a set $W \subseteq U$ of points with $|W| = 2d$ and a set $Y \subseteq C$ of classes with $|Y| = d$) after which the remaining $\rho + d$ components can all be placed legally; d^* is the minimum d admitting a revival.

Theorem 3.24 (revival-depth lower bound)**.** *Let $T_k = (k-1) \cdot K_2$ and let $\sigma = (U, C)$ be a dead state at slot j with $\rho = (k-1) - j$ unplaced components. If σ has a d-revival, then $\rho + d \leq 3d$, i.e. $\rho \leq 2d$, i.e. $d^* \geq \lceil \rho/2 \rceil$.*

Proof. Five steps, using only the placement semantics and the death identity.

(1) A d-revival retracts d placed components; since every component of T_k is a K_2, the retracted point set W has $|W| = 2d$, and since each placed component uses exactly one class, the retracted class set Y has $|Y| = d$.

(2) After the retraction, the $\rho + d$ components of the suffix must be placed on the points $F \cup W$ using pairwise-distinct classes from $C_f \cup Y$; the placements are pairwise point-disjoint with pairwise distinct classes, i.e. they form a rainbow matching of size $\rho + d$ on the board $(F \cup W,\ C_f \cup Y)$.

(3) Every edge e of this matching either lies entirely in F (an "F-internal" edge) or meets W; this dichotomy is exhaustive.

(4) An F-internal edge has its class in $\mathrm{Span}(F)$ (both endpoints in F); but the death identity gives $\mathrm{Span}(F) \cap C_f = \emptyset$, so its class lies in Y. Distinct matching edges have distinct classes, so the number of F-internal edges is at most $|Y| = d$.

(5) An edge meeting W consumes at least one point of W, and matching edges are disjoint, so their number is at most $|W| = 2d$.

Steps (3)–(5) cover all edges of the matching, so $\rho + d \leq d + 2d = 3d$, i.e. $\rho \leq 2d$. □

Remark 3.25 (the companion machine law and the paper-ization gap). The reverse inequality $d^* \leq \lceil \rho/2 \rceil$ is a *machine law, not a theorem*: it holds with zero violations over 4,273 case slots (4,252 pool dead ends plus 21 polar minima; deduplicated 4,268), but no paper proof of the upper-bound direction is known, and the paper-ization of this law is registered as an open gap. The companion deficit law $\min\text{-deficit}(d) = \max(0, \rho - 2d)$ is likewise a machine law.

Consequence (falsification): the long-standing target statements K_1 and the invariant refinement [INV-BT″-2] in $\forall k$ form are *false*. The polar-corner minima at $k = 23, \ldots, 29$ have $\rho = 5$, so by Theorem 3.24 no 2-revival exists; the machine channels agree (slot-pair channels 0 out of 136 and 0 out of 153 candidates, full (W, Y)-space sampling 0 out of 3,000), and the former 60-million-node "double OPEN" search walls at $k = 23/24$ are structural, decided false in constant time by the four-layer pruning of which Theorem 3.24 is the last layer, with $d^* = 3$ witnesses verified. The replacement targets are the stratified statements: $d^* = \lceil \rho/2 \rceil$ on the polar family (lower bound theorem, upper bound machine law).

3.2.7 The criterion-level results: stratified statements only

The remaining material of this section concerns criterion-level results: statements about necessary conditions, their completeness on tested domains, and their stratified theorem-form closures. A recurring discipline binds all of them, which we state once here: *a min-val≤ 1-type criterion verified by per-instance machine scans over finitely many N is not an existence proof for general N*, and no sentence below may be read as one.

The closed-form criterion family for the packing residual. For the $\mathbb{Z}_7$ packing problem at the top layer $\Sigma d = |S|$, define the power-sum moment conditions $P_T \pmod 7$ and their integer upgrades $\widetilde{P}_T$: there exist $C \subseteq S$ with $|C| = \Sigma d$ and pairwise disjoint key sets $K_i \subseteq \mathbb{Z}_7 \setminus \{0\}$, $|K_i| = d_i$, matching all power sums up to degree T. Necessity is immediate (a feasible packing makes the two sides the same multiset), and at the top layer with $m = \Sigma d = |S| \leq 6$, sufficiency of P_m follows from Newton's identities ($t \leq 6$ invertible in $\mathbb{F}_7$) forcing equality of characteristic polynomials, hence of multisets: *completeness at $T = |S|$* is a paper theorem with machine backing (3,872 top-layer points, zero counterexamples). On the census domain (120 orbits $\times$ 330 demand vectors = 39,600 points, exhausted), the second integer moment $\widetilde{P}_2$ and the third mod-7 moment P_3 agree with the oracle on every point — 0 misclassifications out of 39,600, zero false positives on all 6,649 feasible points — and the residual core collapses $424 \to 0$ in this domain (coverage ladder $1{,}133 \to 1{,}519 \to 1{,}557$). Out-of-domain evidence is a sample, not an enumeration, quoted only at the corrected scale: 1,200,000 draws yield 33,018 valid sample points (31,242 after deduplication; sampling support 933,450; $\approx 3.5\%$ of support). Two boundary sentences bind any citation: the $T = 3/T = 2$ closure is a domain-exhaustion statement (the census domain), not a theorem — the only paper-level completeness is at $T = |S| \leq 6$ — and the integer moments are not affinely equivariant, so cross-domain citations use the mod-7 version. A convex-unreachability companion holds in its corrected scope: if an infeasible point is a rational convex combination of feasible colour-count vectors (38/38 explicit certificates, exactly re-checked), then no criterion that is a relaxation of the natural fractional LP can exclude it — its original "any-rank Chvátal–Gomory" listing clause is refuted by rank-1 CG single cuts (9/38 of the very certificates) and is never quoted.

The stratified valuation criterion. The valuation criterion in its aligned family reads: *for every instance with the three anchor offsets on a common coset of the subgroup generated by the first anchor difference, and with capacity slack $\beta \geq 2$, the minimum of the valuation* $\mathrm{val}(C_1, C_2)$ *over admissible pairs is* ≤ 1; the machine scan domain is $\beta \geq 1$. We register the criterion and its existence half as formal objects:

Conjecture 3.26 (valuation criterion, candidate form)**.** *Every so1 instance with $\beta \geq 2$ has* $\min \mathrm{val} \leq 1$.

The statement is a *candidate*: no paper proof exists, its falsification clause is an explicit so1 instance with $\beta \geq 2$ and $\min \mathrm{val} > 1$, and every machine account of this line is a directed sweep of exactly that clause, with zero hits on the tested domains. The general-N existence half is carried separately:

Conjecture 3.27 (the residual existence gap)**.** *For every N and every so1 instance with $\beta \geq 2$,* $\min \mathrm{val} \leq 1$.

The status of Conjecture 3.26 is stratified, and only this stratified form may be cited:

- *Per-N closure in theorem form* for $N \in \{6, 8, 10, 12\}$: a three-tier finite search, proved complete and terminating on paper (a witness-bound theorem gives the per-instance bound $B(I) = d_2 + 1$ and a uniform $B(N) = N - 2$; a completeness theorem shows the three searched families cover every witness), with zero residue on 588/30,760/634,960/7,798,903 instances respectively.
- *The general-N existence half of Conjecture 3.27 is open*: whether for all N and all instances with $\beta \geq 2$ an admissible pair with $\mathrm{val} \leq 1$ exists. The C_1 side is closed for all N by a constructive one-leak theorem — for every instance and every admissible d_1 there is C_1 with at most one leak, by an explicit three-case construction over the cycle decomposition (full-cycles, partial cycles, and the cycle containing the first anchor) — so the obstruction is entirely on the C_2 side (reachability of a near-closed board under the capacity identity). The phrase "theorem-level closure of the criterion" is an explicit non-claim, including at larger N.

The covering identity as a candidate. The strongest structural statement of the valuation line is the following six-family identity, quoted strictly at candidate tier.

Proposition 3.28 (covering identity; paper candidate assembled and independently re-audited; not mechanized)**.** *For every instance $I = (N, P, T, d)$ with $P \subseteq \mathbb{Z}_N$, $|P| \geq 4$, $T = (t_1, t_2, t_3)$ pairwise distinct, $d \in \mathbb{Z}^3_{\geq 0}$, $\Sigma d_i = |P| - 1$, $d_i \leq |P| - [t_i \in P]$: let $W = P \setminus \{t_1\}$, $k = d_2 + d_3 + [t_1 \notin P]$ (so $|W| - k = d_1$), $\delta_{12} = (t_1 - t_2) \bmod N$, $\delta_{13} = (t_1 - t_3) \bmod N$, and $L = \{R \subseteq W : |R| = k\}$. Let* Im *be the set of position-sets produced by the eight-branch family generator, and let $C_{12}, C_{13}, \mathrm{RX}, O3_{12}, O3_{13}, V_7$ be the six covering families defined via the per-orbit run statistic rc_δ (full containment counts 1), cross-reachability, full-orbit removal, and the image of the legacy candidates routine. Then*

$$\mathrm{Im} \cap L \;=\; (V_7 \cap L) \cup (C_{12} \cap L) \cup (C_{13} \cap L) \cup (\mathrm{RX} \cap L) \cup (O3_{12} \cap L) \cup (O3_{13} \cap L).$$

Corollary (boundary). $R \in L \setminus \mathrm{Im}$ implies $\mathrm{rc}_{12}(R) \geq 3$, $\mathrm{rc}_{13}(R) \geq 3$, not cross-reachable, not $O3_{12}$, not $O3_{13}$, and $R \notin V_7$.

Remark 3.29 (tier and riders of Proposition 3.28). The identity is a paper candidate (both directions have paper proofs and the package passed an independent from-scratch re-derivation audit); it is *not* a mechanized or formalized theorem, and *no disjointness of the six families is claimed.* The machine-checkable face is the five-family identity without the V_7 term: all machine checks are run in the *no-V_7 reading* (the V_7 branch omitted, giving a conservative superset $\mathrm{fam}_{\mathrm{no}V_7} \subseteq \mathrm{Im} \cap L$, a one-way implication toward the true face), and the V_7 branch's machine face is frozen — the legacy candidate-generation source files were lost with no recoverable copy — while the V_7 term itself is definitional. The residual V_7-only mass is nonzero and precisely measured: 340 positions at $N = 8$ (1,266 superset-uncovered minus 926 true-face uncovered, both sides measured) and 2,267,655 instance-weighted at $N = 14$ (52,572,256 minus 50,304,601). The supporting lemma

list ($L1$–$L9$) is paper-proved; $L1/L9$ are definitional (V_7 outputs), and the count-difference identity inside $L8$ counts whole *orbits*, not points (a point-count reading mismatches in 136/936 enumerated configurations). One reverse clause of the run-statistic calculus must be stated one-way: "rc = 2 implies a union of two disjoint cyclic pieces; the converse only gives rc $\leq$ 2". The three micro-lemmas (single interval $\Rightarrow$ rc = 1; two disjoint intervals $\Rightarrow$ rc $\leq$ 2; subadditivity) carry paper proofs plus two rounds of randomized machine checks with zero violations (12,558/2,500/12,000 first pass; 4,000/1,109/4,000 audit pass).

The falsification-scale facts of the exchange ledger. The single-swap ledger statement — that an uncovered position admits an in-family companion at swap distance 2 — does *not* hold universally. On the fixed random pool (900 instances, 300 each at $N \in \{10, 12, 14\}$; 1,501 uncovered-witness positions in the no-V_7 superset reading): a distance-2 companion in the family image exists for 1,469/1,501 (97.87%) of witnesses, and within the five-family domain for 1,447/1,501 (96.40%). Failing witnesses: 54/1,501 (3.60%), concentrated in 12/900 instances (1.33%); all 12 lie in $N \in \{12, 14\}$ with $|P|$-spectrum $\{5, 6, 9, 9, 10, 11, 11, 12, 12, 12, 12, 14\}$ — 10 of 12 biased to high $|P|$, 2 outside the dense band. The cause split of the 54 is 32 structural-side (no in-family companion exists) against 22 ledger-side (companions exist but none satisfies the ledger constraints). The distance-$\leq$ 4 binary form also fails on the same pool: 26/54 have a distance-4 five-family companion, 28/54 do not. Two accounting disciplines bind every quotation: the instance-level (12/900) and witness-level (54/1,501) figures are different measurement layers and are never interchanged, and the 96.40% figure is a no-V_7-superset-reading statistic not directly comparable to true-reading protocols.

Remark 3.30 (the bridge guards). The bridge between the μ-sufficiency question (whether satisfaction of all μ-inequalities implies feasibility on the difference board) and the valuation criterion carries three explicit guards, all of which must accompany any quotation. *(a)* The equivalence "abstract theorem $\iff$ μ-sufficiency on the difference board" holds only under the $(H2)/n \leq p - 1$ guard; the unguarded form is falsified by an explicit $\mathbb{Z}_{14}$ counterexample ($P = \{0, 1, 2\}$, $T = (0, 1)$, $d = (1, 2)$: all μ-inequalities hold, the board is infeasible). *(b)* The identification of the abstract existence gap with the general-N existence half above is valid only as a $k = 3$-layer monotone-reduction pointer; as a literal identity it fails on four axes (the β-band, the so-slice, the k-domain, and the closure grade). *(c)* The $k \geq 4$ layer is not touched by the criterion at all — no criterion exists in the current language for $k \geq 4$, a language-level gap rather than an unexecuted proof. Consequently: a min-val$\leq$ 1-type criterion verified by per-instance machine scans over finitely many N is not an existence proof for general N, and μ-inequality checks over finitely many N do not constitute an existence theorem. Necessity of the μ-inequalities is strictly proved (feasible placements satisfy all of them), the sufficiency direction is false on general bipartite boards (an explicit 2×2 counterexample whose colouring violates the difference rule), and sufficiency on the difference board is open (zero violations in $N = 6..14$ machine accounts; no general-N paper proof). Finally, the μ $k = 2$ stratified existence question — the main open wall of this line — is open in its paper face (machine face closed on seven domains for the $k = 2$ layer; recorded obstruction: no base-to-step mechanism), and it has *no logical dependency* on the family-image characterization: characterizing the family image is not the realize-layer existence.

Remark 3.31 (what the criterion layer does and does not give). The criterion-level results of this subsection share one logical shape: each is a necessary condition, or a family of necessary conditions, whose completeness is established on explicitly enumerated domains, or whose per-N verification is theorem-form on the listed N. None of them yields, unconditionally and for general parameters, the existence of the object the criterion is meant to witness: the valuation criterion does not give feasible

pairs for general N (the existence half is open), the moment criteria do not give packings outside the census domain (and the top-layer completeness is bounded at $|S| \leq 6$), and the covering identity characterizes an image, not a realizability theorem. In particular the conclusion of the conditional chain consuming these criteria remains conditional: no unconditional existence statement, and hence no unconditional progress on the underlying decomposition conjecture, is claimed anywhere in this section.

4 Computational methodology: exhaustive accounts, audits, and trust boundaries

Every machine-backed statement in this paper is quoted at a measured strength, and this section documents the apparatus that produces the measurements. There are four layers. First, the exact-value pipeline behind the constructive parts (Section 4.1), together with a self-contained recheck script released with the paper (Section 4.2). Second, the re-verification account of the conditional chain's machine layer (Section 4.3). Third, the Lean development: what is formalized, and where its trust boundaries lie (Section 4.4). Fourth, the key-table ledger behind the covering identity (Proposition 3.28) and the independent audit it passed (Section 4.5). Formalization status is itemized declaration by declaration in Section C.

4.1 The exhaustive construction pipeline

All exact values and all decompositions cited in the constructive parts were produced and re-checked by one pipeline: a bitmask exact dynamic program over edge subsets computes exact values of $p(G)$ (and of the endpoint budget $\beta(G, x)$ of Definition 1.17, with an endpoint constraint); a backtracking exact-cover solver with longest-path-first enumeration handles larger instances; and an independent `verify_decomposition` routine asserts that a witness consists of simple paths exactly partitioning the edge set. Randomized steps use fixed seeds, and every number is reproducible from the scripts and certificate files released with the paper. In the per-theorem verification table (Appendix A), "exact" means computed by an exact algorithm (dynamic program or exhaustive cover), "witness" means a machine-verified decomposition, and "census" means that all graphs of the stated class were processed.

Three rows of that table indicate its granularity.

- The odd-column value $p(K_{4,n}) = (n+1)/2$ (Proposition 2.3) was re-derived independently for every $n = 3, \ldots, 31$ by a structured exact search over $K_{4,n}$, recovering $(n+1)/2$ each time; the closed-form construction was verified for $k = 6, \ldots, 301$ (i.e. $n = 13, \ldots, 603$), with witness lists machine-verified. The printed small-case witnesses of Proposition 2.4 are machine-verified by the same pipeline (edge-exact partition, simplicity).
- The witnesses behind Corollary 2.13 of the $n_2 = 6$ column ($K_{7,6}$, $K_{9,6}$, $K_{11,6}$, $K_{13,6}$) were each verified by two independent implementations (the pipeline verifier and an adjacency-matrix checker); the ring construction was re-derived for $k = 7..31$ from the base $k_0 = 3$, with overlapping chains from $k_0 = 4, 5$.
- The negatives on the five open bases (Observation 2.14) are *truncations, not exclusions*: more than 14 million search nodes (column $n_2 = 8$) and about 0.9 million nodes (column $n_2 = 10$) were spent over every ledger-feasible deficit multiset, without a hit and without exhausting a single schedule. The conjectural status of the five bases is therefore unchanged by these searches, in either direction.

4.2 A self-contained independent recheck script

The repository releases a recheck script, `checks.py`, written from scratch during the independent review of the paper: it imports none of the project's research code, consists of ten check functions built from assertions only, writes its results to `check_results.json`, and exits nonzero if any assertion fails. Running `python3 checks.py` from the repository root executes all ten checks in about three seconds (wall clock on a laptop-class machine; measured 3.1 s for the release version). The coverage is:

- *Ring closure* (Theorem 2.12): all 280 parameter pairs ($h = 1, \ldots, 14$; per h the 20 even ring sizes m from $2h+2$ to $2h+41$), checking path simplicity and that the paths partition the edge set exactly once.
- *The $K_{4,n}$ construction* (Proposition 2.3): all 95 parameter values $k = 6, \ldots, 100$, asserting $k+1$ paths, simplicity, and the exact $A \times B$ edge partition with multiplicity 1.
- *The odd-clique local engine* (Theorem 2.15): 61 odd values $j = 1, 3, \ldots, 121$, checking that the removed arcs form a Hamilton path, that the remaining j paths have the prescribed lengths, and that the partition is exact. Scope: odd j only; even j is not verified by the script (it is open; Remark 2.19).
- *The non-infinity prime-family arithmetic* (Theorem 3.20): 86 primes $p \equiv 3 \pmod 4$ up to 991, checking the sieve's allowed regimes and the perfect-square obstruction. This is a finite recheck of the elementary steps; it does not substitute for the paper proof and covers the non-infinity regimes only.
- *The K_9@36 census ledger*: the six recorded regime lines' configuration counts against their closed combinatorial forms, the histogram sums, the zero recorded hits, and the total-domain arithmetic $18{,}385{,}772{,}680 = \binom{71}{8} + \binom{36}{8} \cdot 2^8$. This is ledger and artifact arithmetic; the census itself is not rerun.
- *The non-infinity death table*: 28 full-death points for $n \le 200$, of which 23 have $n-1$ prime (matching Theorem 3.20) and the composite points are $\{112, 116, 124, 160, 188\}$. Again artifact accounting, not a rerun of the table's feasibility searches.
- *The corrected Fourier identity* (Theorem 3.22): the K_3 configuration $\{\infty, 1, 3\}$ at $k = 3$ with $\psi(a) = \omega^a$, reproducing both the failure of the superseded form (residual $-2 - \omega$) and the vanishing of the corrected form (4), in exact $\mathbb{Z}[\omega]$ integer arithmetic.
- *Block versus global degree* (Section 1.1): the $K_{1,5}$-with-pendant-leaves witness, confirming that $\mathcal{E}(G) \cong K_{1,5}$ has a vertex of E-degree 5 while every block of $\mathcal{E}(G)$ is a single edge (maximum degree 1).
- *The Lean placeholder scan*: a comment-stripped scan of the Lean sources for `sorry`, `admit` and `axiom` declarations, finding exactly five occurrences, listed in Section 4.4.
- *Snapshot integrity*: SHA-256 verification of the archived source snapshot (20 files plus the pinned artifacts). In the review environment the same check additionally reported the six post-review-edit live files (the expected repair edits to the working notes and four section files, not frozen deliverables); that comparison required the live working tree, which is not part of the released repository, so the released script reports the field as empty.

One reading rule binds the released result file: a PASS records that the intended recheck succeeded, and for two of the ten checks success means *reproducing a recorded error or distinction* (the falsified Fourier form; the block-versus-global-degree witness). A PASS is an execution record, not a certificate of global correctness.

Census methodology note (K_9@36). The largest configuration-level account behind the classification programme is conducted line by line: for each regime line of K_9@36, the configurations are enumerated in full and summarized by a closed-form count checked against the enumeration, with a histogram-sum completeness assertion per line. The nine regime lines sum to $\binom{71}{8} = 10{,}639{,}125{,}640$ (the Vandermonde identity is machine-checked against the per-line counts); the infinity line, $\binom{36}{8} \cdot 2^8 = 7{,}746{,}647{,}040$, is covered in four shards of 1,936,661,760 each under an exact non-overlapping-partition assertion; every line has zero hits; and the total domain is 18,385,772,680 configurations. The released checks verify the ledger arithmetic of this account (per-line closed forms, histogram sums, recorded hits, and the total); the census itself is not rerun by them. The tier of the verdict — machine universal-closure candidate, not independently audited — is the one recorded in Section 3.2.2, and the layered-arithmetic verdict of the same section provides a mechanism-level cross-check.

Complete-walk counts on the flicker line. The GEN existence programme contributes a counting account of a different kind: exact enumeration of zero-dodge complete walks on the skeleton template family, $w_1 = r - 5$, by depth-first search with node budgets and cap tracking, cross-checked across independent implementations. On the odd tiers the exact counts are 158, 1,733, 11,466, 81,660, 484,578, 3,670,846 for $r = 19, 21, 23, 25, 27, 29$ — six tiers, never presented as one closed sequence; the $r = 29$ count is exact but class-bound-conditioned and is quoted only with that caveat. One further exact single point, 23,933,468 at $r = 31$, stands apart at exhaustive grade ($j = 0$: 22,607,250; $j = 1$: 1,326,218). At $r = 33$ there is no count: the search volume is a quantified hard wall, and a segmented long run reports none. The $w_1 = r - 4$ side is zero for all $r \geq 19$ and the even-r side with $w_1 = r - 5$ is zero by the parity theorem — both theorem-level, with the four theorem-relevant cells ($r \in \{20, 22\}$, both w_1) genuinely exhausted at zero; the theorem clears the zero-dodge side only, complete walks with a positive zero-defect existing at $r = 20, 22$ in the record and remaining untouched.

4.3 The re-verification account of the conditional chain's machine layer

The conditional chain of Section 1.7 (Theorem 1.19) was subjected to an independent re-derivation review: a from-scratch re-derivation of its load-bearing implications, conducted before reading the recorded proofs and then reconciled against them, accompanied by fresh machine checks re-implemented from the definitions.

Implications. Of the 34 load-bearing implications of the chain, 31 were independently re-derived as valid. The remaining 3 are wording- and registration-level only: an overstated equivalence in the density remark of the chain (the unconditional direction is one-way), an under-registered residue inside the degree-two layer of the budget conjecture (recorded in Section C.2), and one stale supporting statistic. None reverses an implication direction, and no mathematical hole was found in any implication.

Machine checks, at the downgraded caliber. Six independent machine-check groups accompanied the review. *Five of the six* closed with zero failures (volumes: 180,070 construction

instances; 550 end-to-end pipeline instances; 876 census graphs; 121 types of the placement family; 2,342 buried-vertex instances; 5,880 alignment instances; plus a dedicated connected-domain cut-vertex check on 106 instances). The sixth group closed with three of its four blocks at zero failures and one block emitting 60 alerts; each of the 60 was examined individually and all 60 are *disconnected* graphs, outside the connected domain on which the budget conjecture is stated — zero connected counterexamples. The account is therefore never "six groups with zero failures": the correct statement is five groups clean, and the sixth clean up to a fully discharged false alarm caused by a missing connectivity filter, with the connected domain covered by the dedicated check.

What was *not* re-verified. The same review registered its limits, which travel with any citation of it: three combinational notes of the chain were logic-checked but not separately machine-checked; machine checks predating the review were not replayed; the reviewers' derivation-before-reading process is not itself reproducible (only its products were reconciled against the originals); the full proof reconstruction of the all-odd branch of Theorem 1.4 remains open (Section C.2); and the domains $n \geq 13$ and the general-graph domain beyond 12 edges at $n \geq 9$ were not enumerated. (One gap registered by the review — two of the five domains below not freshly recomputed — was subsequently closed by a full fresh recomputation of all five domains, 178,860/178,860 instances over 16 fields with zero differences.)

The five exhaustively enumerated domains. The budget conjecture K (Definition 1.17) carries a machine account on five exhaustively enumerated domains: the atlas of connected graphs on $n \in \{4, 6\}$ with $m \leq 13$ together with `geng -c 8 1:12`, and the connected `geng` censuses `8/9/10/11/12 1:12` — in total 39,027 graphs, 178,860 even-degree point instances (a vertex x of even degree in a graph of the domain), of which 43,009 have degree $d \geq 4$. The boundary of the domain is part of the account: only the sparse layers with $m \leq 12$ are enumerated (at $n = 11$ the average degree is ≤ 2.18); the general-graph domain beyond $m \leq 12$ is not covered; and odd and even n are accounted separately throughout, never pooled (the budget ceiling is $\lceil n/2 \rceil$ and the analysis differs by parity).

Table 1: The five exhaustively enumerated domains of the budget conjecture. Columns: graphs; even-degree point instances; with $d \geq 4$; instances dead at the first two insertion depths of the degree-insertion tool; instances where the tool certifies failure; instances attaining the ceiling $\beta = \lceil n/2 \rceil$. In every domain: zero violations of $\beta(G, x) \leq \lceil n/2 \rceil$, and zero mismatches of the insertion criterion's equivalence against an independent tool-truth oracle.

Domain	graphs	instances	$d \geq 4$	two-probe	tool-dead	tight
atlas $n \in \{4, 6\}$, $m \leq 13$, $\cup$ `geng -c 8`	2,933	11,328	4,232	306	136	63
`geng -c 9`	7,654	33,296	10,003	227	12	1,420
`geng -c 10`	11,989	55,664	13,471	783	60	72
`geng -c 11`	10,874	52,126	10,558	1,217	0	990
`geng -c 12`	5,577	26,446	4,745	1,102	0	12
total	39,027	178,860	43,009	3,635	208	2,557

Two readings of Table 1 are fixed by the account. First, *tool-dead is not a counterexample*: the degree-insertion criterion (Section C.2) characterizes one construction tool, and all 208 tool-dead instances still satisfy $\beta \leq \lceil n/2 \rceil$ — indeed the zero-violation verdict of the conjecture itself reads on all 178,860 instances. Second, the witness certificates attached to the account total 1,279 (484 of 500 in the base domain plus 795 of 800 across the extended domains, 5 tool-dead instances skipped

by design, zero failures), each an explicit decomposition verified three ways. The residual proof debts of this layer are collected in Section C.2.

4.4 The Lean development and its trust boundaries

The Lean development (toolchain `leanprover/lean4:4.34.0`) certifies a precise subset of the paper's statements. We describe what is proved, how each class of proof is trusted, and where the boundaries are; the itemized list with file anchors is Table 3 of Section C.

What is proved. Two classification-line theorems are fully formalized from first principles: for every odd $n \geq 3$, $p(K_{4,n}) = (n+1)/2$ (with numeric corollaries $3, 7, 10$ at $n = 5, 13, 19$ following without independent gaps), and for every $k \geq 1$, $p(K_{2k+1}) = k + 1$ (via the edge-count lower bound and an explicit Walecki-type construction; the endpoint $k = 0$, where $p(K_1) = 0$, is a separate fact outside the theorem). Both carry the standard axiom profile `[propext, Classical.choice, Quot.sound]` and no `sorryAx`. On the criterion line, two declarations are formalized without `sorryAx`: the criterion-level bound $g_{13} \leq d_2 + 1$ (`thmA2663_upgraded`, axioms `[propext, Quot.sound]`), where g_{13} is the criterion-layer count functional of the valuation line (in the paper's notation, the number of "leaks" of C_1 — points of the shifted layer F_{13} of Definition 1.11 outside C_1; in the Lean sources it is defined as a count over `List.range n` in the base module and quoted by `ThmA2663.lean`), whose hypotheses — the target side has the prescribed size, and val ≤ 1 — are *assumed, not proved* (it is a bound under a hypothesis, not a feasibility-existence proof; see no-claim (6) of Section 1.3); and the case-2 structural discriminator (`case2_struct_alpha_iff`, axioms `[propext, Quot.sound]`), an if-and-only-if for the machine predicate $C_1((t_1 + g) \bmod N) =$ `true` under six explicit hypotheses. Finally, the four-domain witness accounts of the termination bound are kernel-checked at the level of the instance counts 588/30,760/634,960/7,798,903 and of two fully certified five-column global ledgers: ledger_6 = (5,724; 2,613; 190; 398; 588) (legal pairs; witnesses; min val = 0 bucket; min val = 1 bucket; instances) and ledger_8 = (701,360; 285,372; 20,308; 10,452; 30,760). The body of the termination theorem itself is proved on paper, not in Lean; the accounts are its machine shadow.

The `native_decide` boundary. The finite certificates (including both ledgers and the $N = 10/N = 12$ account modules) are closed by `native_decide`. Such theorems must be quoted as *machine-checked by kernel reduction*: they depend on the instantiation axiom of `Lean.ofReduceBool` (the kernel-reflection axiom of the evaluation mechanism), so they are not unproved assertions, but they do not carry the trust level of pure term-level proofs. Three facts anchor the boundary. The judging-layer semantics (the value functional, the board functionals, the admission predicates) are inherited from the base module with zero redefinition. The $N = 8$ ledger is certified on two mutually pinned routes — whole-domain direct evaluation, and a 56-bucket aggregation with per-bucket certificates — and cross-pinned against a second, independent Python implementation, bucket by bucket. And the trust base is the honesty of the Lean 4.34.0 kernel reflector itself.

The axiom audit. The published declarations carry an axiom audit (`#print axioms` over the audit's probe file against the built library). Re-running the audit against the current sources reproduces the archived audit log entry for entry, and the six audited source files are byte-identical to the review snapshot, so the log applies to the released sources. The audit's complement is equally explicit: the probe covers the declarations named in Table 3 and the five Fan-line declarations discussed next; anything outside the probe (the uniqueness corollary of the Fan-type chain, the $N = 10/N = 12$ account modules, and the auxiliary complete-graph and $K_{4,n}$ modules beyond the

two named theorems) is *not* claimed fully formalized until individually probed. A live comment-stripped scan finds exactly five `sorry` lines: three placeholders of the Fan-type formalization chain (`Astar.lean`, lines 2592, 2849, 2864), one *refuted* research lemma kept as a record only (`Fan.lean`, line 765 — its statement was disproved by a machine counterexample and is never cited; see Remark 3.17), and one unproved research gap consumed by a weakened variant (`Fan.lean`, line 4487). Accordingly the Fan-type theorem chain (Theorems 3.1 and 3.2) is cited as paper proofs with formalization in progress, never as machine-verified (Remark 3.17).

Certificate-grade example: the packing table. The packing residual of Section 3.2.7 carries its own Lean certificates, which illustrate the certificate tier end to end. The main certificate theorem checks, pointwise, all 79,200 points of the table domain (120 canonical orbits × 330 demand vectors × the two board sides): the packing predicate holds exactly when $\Sigma d \le |S|-1$, or $\Sigma d = |S|$ with the demand outside the recorded top-failure set. It is closed by `native_decide` (zero `sorryAx`, with the `ofReduceBool` disclosure attached), and its data anchors are certified alongside it (330 demands; 120 orbits; 1,761 top failures = 204 + 1,557). Nine companion theorems over the same table domain record: the downset property on both sides; the uniform layer ($\Sigma d \le |S|-1$ implies feasibility on both sides); an explicit no-threshold witness (two demands with equal $\Sigma d = 4$, one feasible and one not — refuting any proposed threshold law of the form "feasible $\iff$ $f(\Sigma d)$"); the concentration of all 1,761 failures in the top layer; the top-failure set being an upset; and the failure-orbit counts 110 and 2. All are `native_decide` certificates. Above them sits the paper-level *general* downset theorem (`pkg_downset`, axioms `[propext, Classical.choice, Quot.sound]`, no `native_decide`, no `sorryAx`), quantified over arbitrary inputs under three slot-value hypotheses: if the slot relation $\mathrm{slot}(d', i) + 1 = \mathrm{slot}(d, i)$ holds with all other slots equal and $\mathrm{slot}(d, i) \ge 1$, then feasibility of d implies feasibility of d'. Two boundaries travel with it: the hypotheses are stated as slot-value relations (the subtraction form is not yet formalized; instances are discharged through the slot relations), and the Lean reimplementation of the packing predicate is not proven program-level identical to the reference Python implementation — that identity rests on three-way machine cross-checks (the 79,200-point sweep with zero violations, bit-level reconstruction of the certificate data, and a full Lean-side sweep).

4.5 The key-table ledger for the covering identity

The machine account behind Proposition 3.28 is organized as *key tables*: a key is the datum $(N, W, k, \delta_{12}, \delta_{13})$ up to rotation, and each key summarizes all instances sharing it. Across nine domains $N \in \{6, \dots, 14\}$ the key tables comprise 2,412,436 keys covering 98,602,396 key-weighted admissible positions (the sum over keys of $\binom{|W|}{d_1}$); the per-domain key counts are 271/1,218/4,614/15,627/48,913/144,4 for $N = 6, \dots, 14$. On every key: family\prediction = prediction\family = 0 (both inclusions of the covering identity in key form), and the necessary-side boundary violations are 0.

Four conventions bind every citation of these numbers; they are stated here once and apply paper-wide.

(1) *Key-weighted versus instance-weighted.* The key tables count each key once; instance-weighted accounts (each instance once) exist separately for $N = 8$ (28,056 instances, 221,480 positions) and $N = 9$ (128,016 instances, 1,328,040 positions) and are listed apart. The two accountings are never mixed or compared.

(2) *Two domain conventions.* The domains $N \in \{6, \dots, 13\}$ come from exhaustive enumeration of the instance space, while $N = 14$ comes from an affine-normalized domain (orbit representa-

tives under the normalized action) whose equivalence to exhaustive enumeration is *unproven*; the nine-domain totals are therefore not a union under a single convention.

(3) *The no-V_7 superset reading.* All machine checks of the covering identity are performed with the V_7 branch omitted (Remark 3.29), a conservative superset reading whose zero-counterexample verdict transfers to the true face one-way. The residual V_7-only mass is nonzero and precisely measured: 340 positions at $N = 8$ (1,266 superset-uncovered minus 926 true-face uncovered) and 2,267,655 instance-weighted at $N = 14$.

(4) *No extrapolation.* Nothing in the ledger is extrapolated to $N \geq 15$, and the normalized domain is never read as the exhaustive domain.

The ledger itself passed two independent controls: a re-aggregation with a separate decoder and an independent combinatorial recomputation of the position counts (matching the archived accounts digit for digit), and a three-shard fresh replay with digit-identical accounting fields.

The independent audit. Subsequent to, and separately from, the verification passes above, an independent audit re-derived all twelve supporting lemmas of the covering identity (the nine assembly lemmas and the three micro-lemmas of Remark 3.29) from the definitional layer rather than by checking the write-up, probed each lemma's plausible failure modes (none broke), probed each of the three declared assembly gaps (none blocks the assembly), and re-ran the domain-consistency check with a *third* independent implementation (exhaustive reference-form predicates, tag-level reverse direction) on fresh key tables: four full domains ($N = 6, 7, 8, 10$; 55,016 keys) plus one $N = 12$ shard (50,948 keys) — 105,964 keys in total, six violation classes all zero, with independent aggregation matching the headers and archived accounts digit for digit. The randomized machine checks of the micro-lemmas were re-run in the audit pass with zero violations (4,000/1,109/4,000 instances, after the first pass 12,558/2,500/12,000). The audit's verdict upgrades the package to "paper candidate that has passed an independent audit" — which is still not a mechanized or formalized theorem. The audit did *not* re-run, and these items are carried as archived accounts: the single-swap falsification experiments of Section 3.2.7, the transcription-fidelity anchor (240 instances, 97,848 pairs), the per-instance full-domain enumerations at $N = 8, 9$, and the per-instance glue verification (the tool sources of the latter were lost with no recoverable copy).

A Computational anchors

This appendix lists, for each theorem of the paper, the verification status and the machine anchors. All computations used one pipeline (Section 4.1); randomized steps used fixed seeds; every number below is reproducible from the archived scripts released with the paper. In the table, "exact" means computed by an exact algorithm (dynamic program or exhaustive cover), "witness" means a machine-verified decomposition, and "census" means all graphs in the stated class were processed.

Result	Verification	Anchors (script / certificate)
Theorem 3.1, Theorem 3.2: witness instances	exact $p \leq \lfloor n/2 \rfloor$ on 101 random (A1)–(A2) instances, 50 instances with $\Delta(\mathcal{E}) = 5$, 15 large witnesses; 0 violations	block-reload and witness-upgrade scripts

Result	Verification	Anchors (script / certificate)
Theorem 3.2, Remark 3.3, Proposition 3.9, Proposition 3.10	413 p-anchored (A1)–(A2) instances (13 structured, 400 random), 0 violations; exhaustive $n = 5$ (all 1024 graphs) and $n = 6$ (all 32768 graphs) degree/triangle-layer checks behind Proposition 3.10	structured-fix scripts
Theorem 3.4, Corollary 3.5 (two-high layer)	phantom-step audit 4/4; triangle-forced exits of dirty shared neighbours 34/34; 34 structured chain certificates; random two-high census 144/144 chains verified; exact DP cross-checks 43/43; 0 violations	two-high-layer scripts
Corollary 3.5: the dirty corner	207 dirty-class instances (154 legacy: 34 structured + 120 random; 53 new), plus 16 LB1-tightness hardening runs and 60 two-high regression chains; in every instance a decomposition of $\tilde{G}$ into at most $\lfloor n/2 \rfloor - 1$ paths was produced (constructive refutation of the corner); 0 corner hits, 0 violations	corner-narrowing scripts
Corollary 3.5, Remark 3.8: band-free tightness	125 decision-level instances of the constructed family: all 66 band-free instances sealed constructively (paired-decomposition constructor output verified to partition the edge set into exactly odd/2 simple paths, with per-vertex endpoint assertions); all 59 banded instances rejected with band certificates; constructor success $\iff$ band-free reconciled 125/125; excess excess $\leq \lfloor y/2 \rfloor$ audited on these 125 and the 110 earlier band-level instances of the same pipeline (235 in all), zero violations	band-tightness scripts
Remark 3.8: the band-free statement beyond the constructed family	71 decision-level instance runs: the general band-free statement refuted by cycle-block instances certified three ways (band-free by the criterion, untight by the decision procedure, and an exact combinatorial witness at $p = \text{odd}/2 + 1$ for the cycle-with-one-pendant block and $p = \text{odd}/2 + 2$ for the double-triangle block); 0 failures	relaxed-domain scripts
Proposition 3.14, Proposition 3.15: $W(m)$	parity certificate by hand; decompositions into exactly $\lfloor n/2 \rfloor$ paths verified for $m \in \{5, 7, 9\}$; class memberships asserted programmatically	witness-family scripts
Proposition 3.11: exactness on tested k	DP exact at $k = 2$; verified $(k+1)$-path witnesses for $k \in \{2, 3, 4, 5, 6, 8, 10\}$	witness-upgrade scripts
Theorem 2.5	structured exact search over $K_{4,n}$, $n =$ 3..31, recovering $(n+1)/2$ independently for every n; closed-form construction verified for $k =$ 6..301 ($n =$ 13..603); witness lists machine-verified	$K_{4,n}$ scripts and witness files

Result	Verification	Anchors (script / certificate)
Proposition 2.4	witness verification (edge-exact partition, simplicity)	small-case witness files
Even columns of Case 4 (Section 2.2; Theorem 2.10, Theorem 2.11, Theorem 2.12, Corollary 2.13, Observation 2.14)	$n_2 = 6$: witnesses for $K_{7,6}, K_{9,6}, K_{11,6}, K_{13,6}$ each verified by two independent implementations (pipeline verifier + adjacency-matrix checker); ring construction re-derived for $k = 7..31$ from the base $k_0 = 3$, with overlapping chains from $k_0 = 4, 5$. $n_2 = 8$: bases $K_{9,8}, K_{11,8}, K_{13,8}$ found at 5/7/43 search nodes and verified ($K_{11,8}, K_{13,8}$ re-derived byte-identically); closure sampled at 60 levels $k = 9..30$; negatives on $K_{15,8}, K_{17,8}$: > 14M nodes over all five deficit multisets, zero exhaustions. $n_2 = 10$: bases $K_{11,10}, K_{13,10}, K_{15,10}$ found at 6/13/182 nodes and triple-verified; closure sampled $k = 11..18$; negatives on $K_{17,10}, K_{19,10}, K_{21,10}$, ≈ 0.9M nodes, zero exhaustions	column scripts and witness files
Theorem 2.15 (anchor)	Walecki partition checked edge by edge for $j = 2..16$; formula verified for all odd $j \leq 121$	engine scripts
Theorem 2.18 assembly	14 random chains ($b \in \{2..5\}$) assembled and verified; $b = 4, 5$ chains attain $\sum j_i - 1$	engine scripts
Remark 2.19(2): three K_5's	the 5-path decomposition printed in the text is hand-checkable; contradicts the archived exact value 6 (re-verification requested)	this paper, project archive
Conjecture K (Definition 1.17) evidence	the five exhaustively enumerated domains of Table 1; 0 violations	domain scripts and certificate files
Proposition 3.16: exact values	DP exact: $p(K_{2,5} + 2\ell) = 3$, $p(G') = 4$ (triangle plus two decorated bridges), $n = 9, 11$	witness-upgrade scripts
Lemma 1.12, both orientations (transcription and verification note in Section 1.6)	direction-corrected scan over both ordered (gains-end, budget-end) orientations of every deleted edge: $n \leq 5$ complete, all 482,080 ordered instances (241,040 per orientation), 0 violations; $n = 6$ complete, all 1,701,141,150 ordered instances (850,570,575 per orientation; the fixed-orientation half exactly reproduces the historical census) with 0 violations	direction-scan scripts

B The corner analysis and the band-free tightness theorem

This appendix records, with complete proofs, the theorem-level material behind the band-free tightness statements used in Remark 3.8, in the proof of Corollary 3.5, and in open problem O2. Throughout the appendix, G is a graph of the *dirty class* of Corollary 3.5: G is connected and satisfies (A1)–(A2), has exactly two high vertices u and x with $k = d_{\mathrm{E}}(u)$ and $l = d_{\mathrm{E}}(x)$ (both odd and ≥ 5 by (A2)), $t = t_{ux} \geq 1$ common EV-neighbours $v_1, \ldots, v_t$, every common EV-neighbour of E-degree 3, and no private EV-neighbour of E-degree 1 on either side. We write z_j for the third EV-neighbour of v_j, F for the E-star at u, and $\tilde{G}_j = G - F - \{xv_j\}$, abbreviating $\tilde{G} = \tilde{G}_1$. By (A1) the common EV-neighbours are pairwise non-adjacent inside $\mathcal{E}(G)$ and each z_j is a *far* even vertex of $\tilde{G}_j$ (an even vertex outside $N_{\mathcal{E}}(u) \cup N_{\mathcal{E}}(x)$ that is not a common EV-neighbour); also $d_{\mathrm{E}}(z_j) \leq 3$, since $d_{\mathrm{E}}(z_j) \geq 4$ would make z_j a third high vertex.

Definition B.1 (corridors, terminal units, bands)**.** Fix j and write $\tilde{G}$ for $\tilde{G}_j$. For an even vertex f of $\tilde{G}$, let $Y(f)$ be its set of *corridor* neighbours: the degree-two even vertices y with $N_{\tilde{G}}(y) = \{x, f\}$, where $N(f)$ denotes the neighbour set of f in $\tilde{G}$. The remaining incident edges of f are its *terminal units*, numbering $|N(f)| - |Y(f)|$. Call f a *deadlock band* if $|Y(f)| > |N(f)| - |Y(f)|$. Let Far be the set of far even vertices of $\tilde{G}$ and Y the total number of corridors (each corridor is a private EV-neighbour of x, so $Y \leq l - t$, with equality in the constructed family below).

Definition B.2 (constructed family)**.** G belongs to the *constructed family* if it lies in the dirty class and: (i) every private vertex of E-degree 2 (even in G, on either the u- or the x-side) has its second EV-neighbour in Far, and no EV-edge of $\tilde{G}_1$ joins two far even vertices; (ii) the odd-neighbour block of u — the substructure hanging at u's odd-degree neighbours — is a set of s pendant leaves at u with s odd, or a number of gadgets (u–o, o–a, o–b, with two pendant leaves on each of a and b), possibly mixed; and likewise for x; (iii) every far vertex f of $\tilde{G}_1$ has $d_{\mathrm{E}}(f) \leq 3$ and carries at most one repair leaf, whence $\deg_{\tilde{G}_1}(f) \in \{2, 4\}$; and (iv) every common EV-neighbour of G is adjacent to an odd positive number of pendant leaves and to no odd-degree vertices other than these leaves. (The band condition of Definition B.1 does not depend on j: no far–far EV-edge exists by (i), so deleting xv_j touches no far vertex and none of its neighbours. The machine instances of Appendix A all lie in the family, with the cross-private instances covered by the extension that follows.) *Extension.* Cross-private pairs are admitted: a u-private vertex w may be adjacent to an x-private vertex y, so that $N_{\mathcal{E}}(w) = \{u, y\}$ and $N_{\mathcal{E}}(y) = \{x, w\}$; y then acts as a terminal corridor with $\phi(y) = w$, and wy is a terminal unit of w. The band-free tightness theorem below covers this extension by the identical step (ii) of its proof; the load account and the counting theorem are used only on the family proper, where every private even vertex has its second EV-neighbour in Far.

B.1 The counting gap and the counting theorem

Lemma B.3 (counting-gap ledger)**.** *For any j, let* $\mathrm{odd} := \mathrm{odd}(\tilde{G}_j)$ *and* $\mathrm{gap} := \lfloor n/2 \rfloor - \mathrm{odd}/2$, *where $n = |V(G)|$ and odd_G is the number of odd-degree vertices of G. Then* $\mathrm{odd} = k + 1 + \mathrm{odd}_G$, $n = 2 + t + (k - t) + (l - t) + |\mathrm{Far}| + \mathrm{odd}_G$, *and*

$$\mathrm{gap} = \frac{l - t + |\mathrm{Far}| + 1}{2} \quad (n \text{ even}), \qquad \mathrm{gap} = \frac{l - t + |\mathrm{Far}|}{2} \quad (n \text{ odd});$$

in particular $\mathrm{gap} \geq 1$.

Proof. The odd-degree vertices of $\tilde{G}_j$ are: u, which loses the k star edges (an odd number); x, which loses xv_j; the common EV-neighbours v_i with $i \neq j$ ($t - 1$ of them, each losing uv_i); the private

EV-neighbours of u ($k-t$ of them, each losing uw); and all odd-degree vertices of G (untouched, since the star edges all join even-degree vertices and xv_j joins two even-degree vertices). Hence $\mathrm{odd} = 2 + (t-1) + (k-t) + \mathrm{odd}_G = k + 1 + \mathrm{odd}_G$. The vertex set splits into $\{u, x\}$, the common EV-neighbours, the private EV-neighbours of u and of x, the far even vertices (which include the z_i), and the odd-degree vertices of G, giving the displayed count. For odd n: the remaining terms 2, $k+l$ and odd_G are even, so n is odd exactly when $t + |\mathrm{Far}|$ is odd, which makes $l - t + |\mathrm{Far}|$ even and the second quotient an integer. Finally $l \geq t$ and $|\mathrm{Far}| \geq 1$ (as $z_j \in \mathrm{Far}$): for even n the numerator is ≥ 2; for odd n it is a positive even number, hence ≥ 2. So gap ≥ 1. □

Lemma B.4 (load account)**.** *In the family proper (the constructed family without cross-private pairs),* $\sum_{f \in \mathrm{Far}} d_{\mathrm{E}}(f) = k + l - t$ *(E-degrees read in G). Consequently $d_{\mathrm{E}}(f) \leq 3$ for every $f \in \mathrm{Far}$, and $|\mathrm{Far}| \geq 2$.*

Proof. Each common EV-neighbour v_i has EV-neighbours u, x and $z_i \in \mathrm{Far}$, contributing t EV-edges with one end in Far; each private even vertex (of E-degree 2 in G, on either side) has its second EV-neighbour in Far by Definition B.2(i), contributing the $k-t$ edges at the private EV-neighbours of u and the $l-t$ edges at the corridors of x. By (i) there are no EV-edges between far vertices, and odd-degree vertices carry no EV-edges; so every EV-edge counted has exactly one end in Far, and summing the E-degrees over Far gives $t + (k-t) + (l-t) = k + l - t$. If some f had $d_{\mathrm{E}}(f) \geq 4$, then f would be a third high vertex, impossible; so all E-degrees in Far are ≤ 3. If $|\mathrm{Far}| = 1$, then $k + l - t \leq 3$, so $t \geq k + l - 3 \geq 5 + 5 - 3 = 7 > \min(k, l) \geq t$, a contradiction; hence $|\mathrm{Far}| \geq 2$. □

Proposition B.5 (the counting theorem: small gap forces band-free)**.** *In the family proper (the constructed family without cross-private pairs), if* gap ≤ 2*, then $\tilde{G}_j$ has no deadlock band, for every j.*

Proof. By Definition B.2 the band condition is j-independent, so we argue in $\tilde{G}_1$. Suppose f is a band. Then at least two corridors hang at f, and every corridor is a private EV-neighbour of x; write $y := l - t \geq Y \geq 2$. If n is even, Lemma B.3 gives $y + |\mathrm{Far}| \leq 2\,\mathrm{gap} - 1 \leq 3$, while $y \geq 2$ and $|\mathrm{Far}| \geq 2$ (Lemma B.4) give $y + |\mathrm{Far}| \geq 4$ — a contradiction. If n is odd, gap ≤ 2 gives $y + |\mathrm{Far}| \leq 4$, so $y \leq 4 - |\mathrm{Far}| \leq 2$; equality throughout forces $|\mathrm{Far}| = 2$ and $y = 2$. The load account gives $k + l - t = k + y = k + 2 \leq 3|\mathrm{Far}| = 6$, so $k \leq 4$, contradicting k odd and ≥ 5. So $y = 2$ is impossible, and the bound $y \leq 2$ then leaves $y \leq 1$, i.e. $l - t \leq 1$: no band exists (a band needs two corridors at one far vertex). □

B.2 The pairing framework and the band-free tightness theorem

Proposition B.6 (pairing schemes produce trail decompositions)**.** *Let H be a finite simple graph with a local structure on the edges at each vertex: if* $\deg_H(v)$ *is even, its incident edges are partitioned into pairs; if* $\deg_H(v)$ *is odd, exactly one incident edge is marked free and the rest are paired. Then $E(H)$ decomposes into maximal alternating trails — starting from a free mark, at each internal visit leaving through the mate of the arriving edge — and every edge lies on exactly one trail. Each trail either has two free ends or is closed; the open trails number* $\mathrm{odd}(H)/2$*. If moreover no trail repeats a vertex, then H has a path decomposition D with $|D| = \mathrm{odd}(H)/2$, every odd-degree vertex an endpoint of exactly one path of D and no even-degree vertex an endpoint; in particular $p(H) = \mathrm{odd}(H)/2$.*

Proof. At each vertex the successor of an arriving edge is uniquely determined by the pairing, so the trails are edge-disjoint and exhaust $E(H)$. A free mark can occur only at a trail end (a free

edge carries at most one free mark at each of its ends); an odd vertex has exactly one free mark and is therefore the end of exactly one trail; an even vertex has none and is internal wherever it occurs. The number of open trails is the number of free marks divided by two, i.e. $\mathrm{odd}(H)/2$. If all trails are paths, they form a decomposition D with $|D| = \mathrm{odd}/2$ and endpoint set exactly the odd vertices; the odd-degree lower bound of Lemma 1.1(i) gives $p(H) \geq \mathrm{odd}/2$, so the value is exact. □

Lemma B.7 (odd-neighbour blocks close by themselves)**.** *In the constructed family, the odd-neighbour block of $c \in \{u, x\}$ can be structured as Proposition B.6 requires, with all resulting trails staying inside the block, as follows. A leaf block of s leaves (s odd): mark $c\ell_1$ free at c (the single-edge path ℓ_1–c) and pair $(c\ell_{2i-1}, c\ell_{2i})$ for $2 \leq 2i \leq s$, giving $(s+1)/2$ paths for the $s+1$ odd vertices involved. A gadget: pair c–o free at c (path c–o), $(o\text{–}a, o\text{–}b)$ at o, $(a\ell_1, ao)$ at a with $a\ell_2$ free, and $(bo, b\ell_3)$ at b with $b\ell_4$ free; the resulting paths c–o, ℓ_1–a–o–b–ℓ_3, ℓ_2–a, ℓ_4–b have endpoint multiset covering $\{c, o, a, b, \ell_1, \ldots, \ell_4\}$ exactly once — 8 odd vertices, 4 paths. A mixed block (gadgets and leaves at the same c): the block edges at c number odd in total, so exactly one is free (chosen canonically, say the first), the rest pair arbitrarily among leaf and gadget edges, and a mixed pair $(c\ell, co)$ forms the legal path ℓ–c–o. All trails end at free marks inside the block, so the block never interacts with the rest of the scheme.*

Proof. This is a verification of the pairing tables just displayed: within a gadget the pairs at o, a, b are fixed as shown; leaves pair among the edges at c; and in a mixed block the total number of block edges at c is $s+g$ (s leaves, g gadgets, each gadget contributing a single attachment edge co), which is odd because c has even degree in G and an odd number of EV-neighbours, so the free/paired structure exists; the endpoint count of each displayed path list matches the odd vertices it involves, each exactly once. No trail leaves the block, since free marks sit at block vertices only. □

Lemma B.8 (forced fusions)**.** *In any decomposition of $\tilde{G}$ as in the last sentence of Proposition B.6 (a paired decomposition), the edges at every even vertex are traversed in internal pairs. Hence a corridor y (degree 2) forces its two edges xy and $y\phi(y)$ into one path; and the unit at v_1 is forced: $v_1 z_1$ pairs with $v_1\ell_1$, where ℓ_1 is the leaf at v_1 (any further leaves at v_1 pair among themselves, their number being even by Definition B.2(iv)).*

Proof. At an even vertex v of a paired decomposition, $D(v) = 0$, so no path has v as an endpoint; each visit is internal and uses exactly two edges, so the incident edges split into internal pairs by path. A degree-two vertex therefore has its whole edge set as a single pair on a single path. □

Proposition B.9 (injection at far vertices)**.** *Fix a far vertex f of $\tilde{G}$ with corridor set $Y(f)$, $c(f) := |Y(f)|$, and terminal units $T(f)$, $|T(f)| = \deg(f) - c(f)$. The edges at f can be paired so that every corridor edge fy is paired with a distinct terminal unit if and only if f is not a band, i.e. $c(f) \leq |T(f)|$ (the pairing at f is otherwise unconstrained, so this is a matching condition in a complete bipartite graph between $Y(f)$ and $T(f)$). In the constructed family $c(f) \leq 2$.*

Proof. Only the bound remains: by Definition B.2(iii) $d_{\mathrm{E}}(f) \leq 3$ and f carries at most one repair leaf, so $\deg(f) \in \{2, 4\}$, and $c(f) \leq \deg(f)/2 \leq 2$. □

Lemma B.10 (sorted offset matching at x)**.** *In the constructed family, x is odd in $\tilde{G}$, with exactly one free edge; its non-block edges are the m corridor edges (each labelled by the far vertex it hangs at, with multiplicity $c(f) \leq 2$ per label; a cross-private corridor counts as a label of its own, since no two of them hang at the same w) and the $s = t-1$ bridge edges xv_i ($i \geq 2$), with $m+s = l-1$ even. There is a pairing of these edges in which no two paired corridor edges carry the same label: if m is*

even, sort the corridors $p_1, \dots, p_m$ by (label, index) and pair $(p_i, p_{i+m/2})$ for $i = 1, \dots, m/2$, pairing the bridges among themselves — if $p_i, p_{i+m/2}$ shared the label f, the label f would occupy at least $m/2+1$ of the sorted positions, so $c(f) \geq m/2+1 \geq 3$ for $m \geq 4$, contradicting Proposition B.9; if $m = 2$ with equal labels, pair each of the two corridors with a bridge and the remaining bridges among themselves ($s-2$ is even since $s \equiv m \pmod 2$); the failing configuration $m = 2$, equal labels, $s = 0$ is unreachable, since $s = 0$ forces $t = 1$ and $m = l - 1$, i.e. $l = 3$, contradicting $l \geq 5$; if m is odd (possible only when t is even), extract one corridor of a label of maximum multiplicity together with one bridge into a mixed pair, and apply the even case to the rest.

Proof. The counting arguments displayed are the whole proof: for the even case, if both p_i and $p_{i+m/2}$ carried label f, then all positions from i to $i+m/2$ would carry label f (the sort is by label first), giving $m/2+1$ corridors at f. For $m = 2$ with equal labels and $s \geq 2$, the corridor–bridge pairs leave $s-2 \geq 0$ bridges, even in number. For odd m, extracting one corridor of a maximum-multiplicity label and one bridge leaves an even number of each, and the extracted pair mixes a corridor with a bridge, so the no-equal-label condition is preserved. □

Theorem B.11 (band-free tightness in the constructed family)**.** *Let G belong to the constructed family and fix j. If $\tilde{G}_j$ has no deadlock band, then $\tilde{G}_j$ admits a paired decomposition; consequently $p(\tilde{G}_j) = \mathrm{odd}(\tilde{G}_j)/2$.*

Proof. We define the local structure of Proposition B.6 on $\tilde{G}_j$. (i) The odd-neighbour blocks of u and of x are structured as in Lemma B.7, x's free edge being the canonical first edge of its block. (ii) The degree-two even vertices are fused as in Lemma B.8: each corridor y pairs xy with $y\phi(y)$, where $\phi(y) \in \mathrm{Far}$ is the far vertex at its other end (for the cross-private *extension* of Definition B.2 the same fusion applies with $\phi(y) = w$ a u-private vertex, wy being a terminal unit of w, so the extension is covered by the identical argument); the unit at v_j pairs $v_j z_j$ with $v_j \ell_j$. A shared vertex v_i, $i \neq j$, takes the pair $(v_i x, v_i \ell_i)$, marks $v_i z_i$ free, and self-pairs its remaining leaf edges. (iii) At each far vertex f, Proposition B.9 — applicable because $\tilde{G}_j$ is band-free — pairs every corridor edge fy with a distinct terminal unit; the remaining terminal units number $\deg(f) - 2c(f)$, which is even, and pair among themselves. (iv) At x, the free edge is the block edge of (i), and the remaining edges pair according to Lemma B.10. (v) Every degree-one vertex carries the free mark at its leaf end. The free-mark audit is immediate: each odd vertex of $\tilde{G}_j$ carries exactly one free mark and each even vertex none, by construction.

No trail closes. Consider any trail of the resulting scheme. At a far vertex a corridor end is always paired with a terminal unit, so a trail contains at most two corridors and, having left a far vertex, never returns to it; at x each trail passes exactly once (a single pair); the fusions at v_j and the bridges at the v_i are one-way terminations. Hence every trail has one of the forms

$$\text{terminal} - -f - -y - -x - -y' - -f' - -\text{terminal}', \qquad \text{terminal} - -f - -y - -x - -v_i - -\ell_i,$$

$$\ell_i - -v_i - -x - -v_{i'} - -\ell_{i'}, \qquad \text{terminal} - -f - -\text{terminal}',$$

where in the first form y and y' are corridors of *distinct* far vertices $f \neq f'$, by Lemma B.10; a "terminal" means a terminal unit followed to its free end — the whole segment from f along a repair-leaf edge, along a hook edge to a u-private vertex w, or along the z-role edge $v_i z_i$ read from z_i to its free mark at v_i. When the corridor y' of the first form is cross-private, its right-hand segment y'–f'–terminal$'$ is replaced by y'–w: the trail ends at the free mark of w, an odd vertex of $\tilde{G}$ (so no closed trail arises). In each form the vertices are pairwise distinct, so every trail is a simple path. Proposition B.6 yields a paired decomposition, and the value of p follows. □

Lemma B.12 (a band blocks paired decompositions)**.** *If $\tilde{G}_j$ has a deadlock band f, then $\tilde{G}_j$ admits no paired decomposition; consequently $p(\tilde{G}_j) \geq \mathrm{odd}(\tilde{G}_j)/2 + 1$.*

Proof. Two corridors $y_1 \neq y_2$ of the same band f cannot lie on one path of any decomposition: if a path contained both fy_1 and fy_2, then y_1, being of degree two and internal (even vertices are never endpoints in a paired decomposition), would force xy_1 onto the same path, likewise xy_2, so the path would contain all four edges of the cycle x–y_1–f–y_2–x — impossible for a simple path. Now in a paired decomposition the $\deg(f)$ edges at f split into $\deg(f)/2$ internal pairs. The edge fy of a corridor $y \in Y(f)$ cannot pair with another corridor edge fy' (previous sentence), so it pairs with an edge at a terminal unit; the resulting assignment $y \mapsto$ its paired edge is injective, whence $c(f) \leq |N(f)| - c(f)$, contradicting the band condition. So no paired decomposition exists, i.e. no decomposition with odd/2 paths and endpoint set the odd vertices; with fewer than odd/2 paths excluded by Lemma 1.1(i), this gives $p \geq \mathrm{odd}/2 + 1$. □

Proposition B.13 (one band costs exactly one path)**.** *Let G belong to the constructed family and suppose $\tilde{G}_j$ carries exactly one deadlock band f. Then $p(\tilde{G}_j) = \mathrm{odd}(\tilde{G}_j)/2 + 1$.*

Proof. For a single band the pair $(c(f), |T(f)|)$ is $(2, 0)$ or $(3, 1)$: the band condition $c(f) > |T(f)|$ with $c(f) + |T(f)| = \deg(f) \in \{2, 4\}$ and $c(f) \geq 2$ leaves exactly these two cases, and in both the surplus of corridor ends over terminal units has size $|S(f)| = 2$; write $S(f) = \{a, a'\}$. Modify the scheme of Theorem B.11 in three places only: (1) at f, self-pair the surplus edges (fa, fa') instead of injecting them; (2) at x, spend the free edge on xa rather than on a block edge, marking xa free at both ends (the single-edge path x–a); (3) the remaining injections at f continue as before (in the case $(3, 1)$ the one remaining corridor pairs with the one terminal unit); every other edge follows Theorem B.11, whose offset matching at x keeps distinct labels apart automatically, including among the remaining corridors of the band. The resulting trails are a–f–a'–x–…–terminal (with a' fused, x passed once, f passed once) and the single edge x–a; all are simple, and the 4-cycle x–a–f–a'–x is cut at both x and a by the free marks of xa, so it does not arise as a trail. The endpoint count: a gains two ends (one on the x side, one on the f side) and every other vertex is unchanged, so the number of path-ends is odd + 2 and the number of paths is odd/2 + 1. The reverse inequality is Lemma B.12. □

Proposition B.14 (several bands)**.** *If $\tilde{G}_j$ carries B deadlock bands, then $p(\tilde{G}_j) \leq \mathrm{odd}(\tilde{G}_j)/2 + 2B - 1$; moreover $B \leq \lfloor Y/2 \rfloor$, where Y is the total number of corridors.*

Proof. Start from the scheme of Theorem B.11 and repair the bands one at a time. The first band is repaired as in Proposition B.13 (cost +1, spending x's free edge and one ordinary matching partner). Each further band is repaired by the double-endpoint modification: self-pair the band's surplus pair at its far vertex f and self-pair two x-edges at x, freeing four ends; the two resulting trails a–x–a' and a–f–a' add two ends, that is, one path, per band — an exchange costing two endpoint pairs for one path, relying on no matching partner. Hence $p \leq \mathrm{odd}/2 + 1 + 2(B - 1) = \mathrm{odd}/2 + 2B - 1$. Finally, distinct bands have disjoint corridor sets (each corridor hangs at exactly one far vertex) and each band has $c \geq 2$, so $Y = \sum_{f \text{ band}} c(f) \geq 2B$. □

Corollary B.15 (bands are exactly the obstructions)**.** *In the constructed family: $\tilde{G}_j$ has a deadlock band if and only if $p(\tilde{G}_j) > \mathrm{odd}(\tilde{G}_j)/2$. In particular, on the family proper (the constructed family without cross-private pairs), if the counting gap is at most 2, then $p(\tilde{G}_j) = \mathrm{odd}(\tilde{G}_j)/2 = \lfloor n/2 \rfloor - \mathrm{gap}$ for every j; the cross-private instances at gap at most 2 are covered at probe level by the machine certificates of Appendix A.*

Proof. One direction is Lemma B.12, the other Theorem B.11. If gap ≤ 2, Proposition B.5 gives band-free for every j on the family proper, and Theorem B.11 with Lemma B.3 gives $p = \text{odd}/2 = \lfloor n/2 \rfloor - \text{gap}$. □

B.3 The five escape routes at a common EV-neighbour

Theorem B.16 (escape routes)**.** *Let G be a dirty-class graph, fix a common EV-neighbour v_1 with third EV-neighbour z_1, write $B = \lfloor n/2 \rfloor$ and $\tilde{G} = \tilde{G}_1$, and let Y be the set of private EV-neighbours of x. If any of the following holds, then $p(G) \leq B$:*

(a) *$p(\tilde{G}) \leq B - 1$;*

(b) *$\tilde{G}$ has a decomposition $\tilde{D}$ with $|\tilde{D}| \leq B$ and $\tilde{D}(x) \geq 3$;*

(c) *$\tilde{G}$ has a decomposition $\tilde{D}$ with $|\tilde{D}| \leq B$, $\tilde{D}(x) = 1$ and $\tilde{D}(z_1) \geq 1$;*

(d) *$\tilde{G}$ has a decomposition $\tilde{D}$ with $|\tilde{D}| \leq B$, $\tilde{D}(v_1) \geq 2$ and $\tilde{D}(v_1) > \#\{y \in Y : \tilde{D}(y) = 0\}$;*

(e) *$\tilde{G}$ has at least two components of odd order.*

Proof. The reinsertion skeleton is that of the proof of Theorem 3.4. (s1) Reinsert xv_1 towards v_1 by Lemma 1.12; the condition to check is $\tilde{D}(x) > \#\{w \in N_{\tilde{G}}(v_1) : \tilde{D}(w) = 0\}$. Among the neighbours of v_1 in $\tilde{G}$, the only even-degree vertex is z_1: x is odd in $\tilde{G}$ (it lost xv_1), and every other neighbour of v_1 is an odd-degree vertex of G (the edge uv_1 is deleted, and v_1 has E-degree 3 with EV-neighbours u, x, z_1). So the passing count at v_1 is at most 1. (batch) Lemma 1.14 then absorbs the whole star F towards u, returning $A \subseteq F$ with $|A| \geq \lceil k/2 \rceil$; its hypothesis holds read over G because v_1 is now active and every other star vertex is odd in $\tilde{G}$, hence has $\tilde{D} \geq 1$ by Lemma 1.1(ii). (L) Each edge of $F \setminus A$, $|F \setminus A| = \lfloor k/2 \rfloor$, is reinserted towards its star vertex as in the proof of Theorem 3.1: dead candidates are confined to $N_{\mathcal{E}}(v) \setminus \{u\}$, of size at most $d_{\mathrm{E}}(v) - 1 \leq 2$ by (A2), against a budget of at least 3 at the last step.

(b) $\tilde{D}(x) \geq 3 > 1 \geq \text{passing}(v_1)$, so (s1) applies and activates v_1; (batch) and (L) follow.

(a) Let D^* be optimal with $|D^*| \leq B - 1$. If $D^*(x) \geq 3$, route (b) applies. Otherwise $D^*(x) = 1$ (it is odd by Lemma 1.1(iii), x being odd in $\tilde{G}$). Now $\deg_{\tilde{G}}(x) = \deg_G(x) - 1 \geq l + 1 - 1 \geq 5$ (as x is high, $\deg_G(x) \geq l + 1$), so x is internal to some path of D^*; cutting that path at x raises $|D^*|$ by one — still at most B — and raises the endpoint count at x to 3: route (b).

(c) $\tilde{D}(z_1) \geq 1$ makes the passing count at v_1 equal to 0, so (s1) applies unconditionally; (batch) and (L) as in (b).

(d) Reverse the order of (s1): reinsert xv_1 *towards* x first. The Lemma 1.12 condition is now applied at v_1 as the donor: $\tilde{D}(v_1) > \#\{w \in N_{\tilde{G}}(x) : \tilde{D}(w) = 0\}$, and the dead candidates at x are precisely the dead private EV-neighbours of x (the other neighbours of x, namely $v_2, \ldots, v_t$ and the odd-degree neighbours, all have $\tilde{D} \geq 1$); hypothesis (d) is exactly this inequality. Afterwards v_1 is still active ($\tilde{D}(v_1) - 1 \geq 1$), so Lemma 1.14 absorbs the *full* star F, including uv_1, towards u; and the (L) step runs as above, u's budget being untouched by the first step. The endpoint count at x ends at $\tilde{D}(x) + 1$, which is even, consistent with x being even in G; all $k + 1$ deleted edges are restored and $|D| = |\tilde{D}| \leq B$.

(e) Every component of $\tilde{G}$ lies in the class of Theorem 1.6: it is triangle-free (an induced subgraph of $\mathcal{E}(G)$) and every E-degree is at most 3 (the private EV-neighbours of x by (A2) at x, the far vertices because u and x are the only high vertices, v_1 because $d_{\mathrm{E}}(v_1) = 3$). Componentwise Theorem 1.6 with Lemma 1.3 gives $p(\tilde{G}) \leq \sum_i \lfloor n_i/2 \rfloor = (n - c_{\text{odd}})/2$, where $c_{\text{odd}} \geq 2$ is the number of odd-order components (it is even when n is even, and odd and ≥ 3 when n is odd, giving $(n-3)/2 = (n-1)/2 - 1 = B - 1$). In both cases $p(\tilde{G}) \leq B - 1$: route (a). □

Corollary B.17 (the corner, formally)**.** *Let G be a dirty-class graph with $p(G) > \lfloor n/2 \rfloor$. Then for every common EV-neighbour v_j (with third EV-neighbour z_j and $\tilde{G}_j = G - F - \{xv_j\}$), all of the following hold: $p(\tilde{G}_j) = \lfloor n/2 \rfloor$; $\beta(\tilde{G}_j, x) = \lfloor n/2 \rfloor + 1$; z_j is buried (endpoint count 0) in every optimal decomposition of $\tilde{G}_j$; in every optimal decomposition $\tilde{D}$, either $\tilde{D}(v_j) = 0$, or $\tilde{D}(v_j) \geq 2$ and at least $\tilde{D}(v_j)$ private EV-neighbours of x have $\tilde{D} = 0$; and $\tilde{G}_j$ has at most one component of odd order.*

Proof. This is the contrapositive of Theorem B.16, route by route, for each j. First $p(\tilde{G}_j) \leq \lfloor n/2 \rfloor$ always holds: every E-degree in $\tilde{G}_j$ is at most 3 (as in the proof of route (e)), so component-wise Theorem 1.6 with Lemma 1.3 applies. Route (a) failing then gives $p(\tilde{G}_j) = \lfloor n/2 \rfloor$. Route (b) failing, with endpoint counts at x odd (Lemma 1.1(iii), x being odd in $\tilde{G}_j$), gives $\tilde{D}(x) = 1$ in every optimal decomposition; cutting a path of an optimal decomposition at the internal vertex x (legal since $\deg_{\tilde{G}_j}(x) = \deg_G(x) - 1 \geq 5$) yields $\lfloor n/2 \rfloor + 1$ paths with endpoint count 3 at x, so $\beta(\tilde{G}_j, x) = \lfloor n/2 \rfloor + 1$. Route (c) failing gives $\tilde{D}(z_j) = 0$ in every optimal decomposition. Route (d) failing gives the fourth condition; when $l - t \leq 1$ it degenerates to $\tilde{D}(v_j) = 0$, since at most one private EV-neighbour of x can be dead while $\tilde{D}(v_j) \geq 2$ would need at least two. Route (e) failing gives at most one odd-order component.

(A note on conventions: the β here is the $D(x) \geq 2$ variant of Definition 1.17. The $D(x) \geq 3$ variant takes the same value $\lfloor n/2 \rfloor + 1$: its lower bound comes from route (b) failing exactly as above, and its upper bound from the same cutting construction, which raises the endpoint count at x to 3.) □

C Formalization status, limitations, and open problems

C.1 The Lean formalization, item by item

Table 3 lists every Lean declaration this paper cites that is free of `sorryAx`, with its exact statement, source anchor and axiom profile; every entry of the axiom column was audited by `#print axioms` against the built library, and the audit was re-run against the current sources (identical output; the audited files byte-identical to the review snapshot; Section 4.4). Declarations whose proofs consume `native_decide` carry the kernel-reflection axiom `Lean.ofReduceBool` explicitly and are quoted as "machine-checked by kernel reduction" (Section 4.4); they are not equated with pure proofs.

Not formalized. Three boundary entries complete the picture. Theorem 2.18 (odd clique chains) and its engine Theorem 2.15 have *no* Lean statement: they are paper proofs, and the machine anchors cover the underlying decompositions, not the chaining argument (Section 2.5). The Fan-type chain (Theorems 3.1, 3.2) carries `sorryAx`-dependent declarations — formalization in progress, and never quoted as machine-verified (Remark 3.17). The research lemma `fan6_nf` is never cited: its statement was refuted by a machine counterexample and it survives in the sources as a record only.

The four-domain account layer and its honest debt list. The termination bound's four-domain witness accounts (588/30,760/634,960/7,798,903 instances at $N = 6, 8, 10, 12$; Section 4.4) are kernel-checked equalities at the level of the domain counts and of the two certified ledgers; the theorem body itself is a paper proof with no Lean statement. Five residuals are registered against the account layer, none claimed closed: (R1) the fast≡slow enumeration equivalence has no general-N theorem (it is pinned by double-scale agreement at the $N = 6/8$ anchors and across all bucket tables); (R2) the full witness tables at $N = 10$ (12,211,196 rows) and $N = 12$ (313,315,956 rows) are not certified — bucket-first sampling and the aggregate bucketed ledgers cover them; (R3) the

trailing four digits of the $N = 10/N = 12$ tuples have no kernel-level *third* implementation (the Lean pins and the Python second implementation make two); (R4) the h mod $2g$ keys bridge is proved at zero axioms for its instances, but its general-N derivation is open; (R5) the declaration-listing verifier is a text-level instrument whose statement-level boundary is carried as such.

C.2 Limitations

C.2.1 The budget-conjecture residuals

The degree-two cut-vertex branch. The degree-two layer of the budget conjecture assembles through a case analysis whose cut-vertex branch ($G-x$ disconnected) is closed at the *computational* level only: 1,870 cut-vertex even-degree instances over the base domain of Table 1 (492 with $d = 2$, 1,165 with $d = 4$, 213 with $d = 6$) show zero violations, and a dedicated connected $n \leq 7$ check covers all 106 degree-2 cut-vertex instances, again with zero violations. The theorization is pending, and the drafted splicing argument has a genuine gap: its parity arithmetic assumes $\lceil n_1/2 \rceil + \lceil n_2/2 \rceil = n/2$ for the two components of $G - x$, which holds only when n is even (components of opposite parity). For odd n the components share parity: the same-odd split gives $(n + 1)/2$ and the same-even split gives $(n - 1)/2$, so the drafted equality fails by one — the missing parity case must be supplied before the branch is a theorem. The unconditional ceiling survives unchanged: $\lceil n_1/2 \rceil + \lceil n_2/2 \rceil \leq \lceil n/2 \rceil$ for all parity splits, verified exhaustively for $n \leq 30$.

The degree-≥ 4 layer. The general degree-≥ 4 layer of the budget conjecture is *not* discharged. The exact criterion (Section 1.7) characterizes one construction tool — the degree-insertion operations — and the alternative-construction theory for the 208 tool-dead instances of Table 1 (180 of 208 cut-vertex; only 7 with the reduced case $\sigma = 0$) is not written; the enumeration is confined to the $m \leq 12$ sparse layers. The residual proof obligation is booked, and no closure language is used for this layer.

The recorded dead zone. The 42 recorded dead-zone instances of the base domain are discharged at *witness* level: each admits a machine-verified sharp decomposition with $\beta \leq \lceil n/2 \rceil$ (equality in 41 of 42; the sole strict case has $\beta = 3 < 4$), attributed as 13 two-probe artifacts, 23 cut-vertex witnesses and 6 tight non-cut instances. This is a finite-instance disposition and does not eliminate the corresponding premise of the conditional chain, which remains open as stated (Section 1.8).

Corrected paper-level values of the extended account. The extended-domain verification records corrected values, which supersede the first-pass report: 1,279 witness certificates (484+795; Section 4.3); the tight-layer validation splits as 147 even-n instances for the tight-layer corollary proper and 2,410 odd-n instances for its companion characterization (the literal odd-n proposition has 216 counterexamples and is superseded by the companion form "success $\iff$ $M(u) = d$ or $M(u - 1) \geq d - 2$"); the insertion-depth account volume is 175,244 under the corrected depth formula; and in the base domain's $d = 6$ slice, 157 instances are rescued (156 at insertion depth 2, 1 at depth 3), zero otherwise.

The K_8@28 registered prediction. For the placeability programme: the phrase "surviving regimes only $\{q = 0, q = 7\}$" at K_8@28 is a *registered prediction*, not a measurement (Remark 3.18); the actual census found zero surviving regimes on every line.

The Lovász 1968 external step. The all-odd support consumed by Theorem 1.18 rests on Theorem 1.4, verified at statement level by five concordant published sources and carried by the entire subsequent literature. Its offline proof reconstruction was, however, judged *not achieved*: the skeleton's deletion step inflates the path count and its re-merging is blocked by shared internal vertices, a failure pinned mechanically by 22 witness graphs on $n \leq 5$. Two transcription blemishes are on record alongside (the equality reading "$|D| = \lfloor n/2 \rfloor$" must be "at most"; the "$\leq n-1$ paths" corollary is loose for even n), and the consumed upper-bound-inflation line is cited in its corrected upper-bound form, not as an identity. Page-by-page reading of the 1968 original remains mandatory before the external flag is lifted (Section 1.8).

C.2.2 The generic-account limitations

The key-table ledger of Section 4.5 carries its four conventions as limitations: the $N = 14$ affine-normalized domain equivalence is unproven; key-weighted and instance-weighted accountings are separate; the no-V_7 reading is a one-way superset; and the machine face of the omitted branch is frozen (its legacy candidate-generation sources are lost, with the residual mass precisely measured at 340 positions for $N = 8$ and 2,267,655 at $N = 14$). Two further accounting rules bind quotations: per-instance accounts at $N = 8, 9$ are listed separately from the key accounts, and the two accounting layers of the falsification experiments (12/900 failing instances versus 54/1,501 failing witnesses; Section 3.2.7) are never interchanged. No ledger result is extrapolated to $N \geq 15$ or read across the two domain conventions.

C.2.3 The classification-programme boundaries

The placeability classification programme (Section 3.2) is bounded as follows. The evidence structure of the exception-set candidate (3) is closed to $k \leq 16$ on the empty-side and to $n = 13$ on the tower side, with the $k = 17$ full-type sweep at 53,394,755 types; the layers $k \geq 18$ and the uniqueness side (closed only at $k = 15$, refuted as a literal statement at $k = 21$) are open (Remark 3.19). The prime-family theorem covers the non-infinity regimes only; the infinity segment of the prime family is open for every member (Remark 3.21). The corrected Fourier identity (Theorem 3.22) is the only form of the character equation cited anywhere in this paper, and all machine results of the programme are independent of it (Remark 3.23).

C.2.4 The existence half of the criterion

The general-N existence half of the valuation criterion in its so1 form (Conjecture 3.27) is open: the per-N closures at $N \in \{6, 8, 10, 12\}$ are theorem-form, the general statement is not, and no sentence of this paper states otherwise. Three empirical obstacles are registered against it — the $\lambda = 1$ dead corners (2,528 instances at $N = 10$ with no one-leak witness), the failed-completion triple 31/1,094/13,488, and four falsified discriminators — and the candidate mechanisms (near-saturation normal forms; the leak-repair channel; per-line board normal forms) each carry its own falsification experiment. The associated $\mu\ k = 2$ stratified existence wall is open on its paper face (machine face closed on seven domains; no logical dependency on the family-image characterization; Remark 3.30).

C.2.5 The GEN existence programme boundary

The GEN existence programme carries one further measured boundary. Its zero-dodge complete-walk counts on the $w_1 = r - 5$ flicker line are the six exact values

$$158,\ 1{,}733,\ 11{,}466,\ 81{,}660,\ 484{,}578,\ 3{,}670{,}846$$

for $r = 19, 21, 23, 25, 27, 29$ (the $r = 29$ value exact but class-bound-conditioned, quoted with that caveat) — six tiers, not a closed sequence. One further exact single point, 23,933,468 at $r = 31$, stands on its own at exhaustive grade; at $r = 33$ there is *no* count (a quantified hard wall; a segmented long run was started and reports none). The even-r blocking companion clears the zero-dodge side only: complete walks with a positive zero-defect exist at $r = 20, 22$ in the record and are untouched by it. The existence theorem itself (registry grade, $r \geq 43$) is supported by paper certificates and step-feasibility, with its machine anchors corroborating only.

C.3 Open problems

The open problems below supersede the numbering of any earlier working list; each is stated with what the present account gives and what it withholds.

O1. The remaining even columns of Case 4 (Conjecture 2.9). Is $p(K_{n_1,n_2}) = \lceil n_1/2 \rceil$ for all odd n_1 and even n_2, $2 \leq n_2 < n_1$? The ring-closure theorem (Theorem 2.12) reduces each column to $h + 1$ base instances; the columns $n_2 = 2, 4, 6$ are closed, and five named bases are open: $K_{15,8}$, $K_{17,8}$ (column $n_2 = 8$) and $K_{17,10}$, $K_{19,10}$, $K_{21,10}$ (column $n_2 = 10$). The recorded negatives on these bases are truncations, not exclusions (Observation 2.14): the decompositions may lie deeper in the search tree, or the conjecture may fail. The next methodological step suggested by the data is a constraint-propagation search driven by the deficit ledger of Theorem 2.11 and the endpoint parity, rather than deeper node budgets.

O2. Removing the structural conditions of the Fan-type theorems (Conjecture 3.7). The alignment and separation hypotheses fail exactly at *remote high vertices*; by Proposition 3.9, a minimal counterexample must have a bad partner at every high vertex. The remaining corner of the two-high layer is blocked by the excess bound (zero violations on all 235 computed instances), and the residue of the conjecture is that corner together with graphs of at least three high vertices that are neither separated nor covered by the profile parity. Note that the band-free phenomenon is family-specific: outside the constructed family, the general band-free statement is refuted by certified counterexamples (Section 3.1).

O3. The even-j Walecki residual. Theorem 2.15 is for odd j; for even j (blocks of order 1 mod 4), the chain construction would extend to all odd-order complete-block chains. In-chain existence is machine-verified at the block values K_5 and K_9, but no general construction is known. The natural route is a Hamilton decomposition of $K_{2j+1} - C^*$ for a suitable cycle C^* through the prescribed path, which touches the Oberwolfach problem for odd orders.

O4. The general-N existence half of the criterion (Conjecture 3.27). For all N and all so1 instances with capacity slack $\beta \geq 2$, is there an admissible pair with val ≤ 1? The C_1 side is closed for all N by the constructive one-leak theorem; the obstruction is C_2-side reachability, with the three registered empirical obstacles and the candidate mechanisms of Section C.2.4, each carrying its falsification experiment.

O5. The μ $k = 2$ stratified existence wall. The paper core is the existence of C_1 with $q \leq d_3+1$ and realized. The machine face is closed on seven domains for the $k = 2$ layer (zero violations); the paper face has a recorded obstruction (no base-to-step mechanism); and the problem has no logical dependency on the family-image characterization (Remark 3.30).

O6. The degree-≥ 4 layer of the budget conjecture K (Definition 1.17). Beyond degree 2 the deletion–reinsertion scheme meets its true wall; the computational record shows the wall is an artefact of the method, not of K (Section C.2.1). What is needed is a construction that produces a $\lceil n/2 \rceil$-decomposition with $D(x) \geq 2$ directly, without deleting x.

O7. The exception-set programme. Beyond the closed faces of Remark 3.19: the layers $k \geq 18$; uniqueness of the minimal core for $k \geq 21$; the prime-family infinity segment (Remark 3.21); and the counting-side target of the α'' bridge (one-way and rigidity-free, machine base eight odd points, $p \geq 19$ unenumerated; Section 3.2).

O8. The 4-degenerate residual. The theorem of [21] covers 3-degenerate graphs; the class of 4-degenerate graphs is, as far as we know, entirely open. $K_{4,n}$ shows the interesting behaviour starts immediately: it is 4-degenerate and, by Theorem 2.5, beats its published bounds by one.

C.4 Concluding remarks

What is proved. On the constructive side: the exact value $p(K_{4,n}) = (n + 1)/2$ for odd $n \geq 3$ (two of whose instances are formalized circumstances of one Lean theorem), the full $n_2 = 6$ column and the structural reduction of the remaining even columns of Case 4 of [10]; the odd clique chain theorem (Theorem 2.18) at its all-odd-j domain; and the Fan-type class theorems admitting a block of $\mathcal{E}(G)$ of maximum degree at least 4 (Theorems 3.1 and 3.2), at paper level with formalization in progress. On the conditional side: the budget framework, the implication $K \Rightarrow$ Gallai (Theorem 1.18), the conditional chain, and the descriptive registration of the eight-item premise ledger (Theorem 1.19).

What is conditional. The conditional route is restated one last time in its honest form: the criterion's first conjunct is a machine-closed candidate on explicitly tested domains with a paper status open; every one of the eight premise conjuncts is an open conjecture, an unverified external step, or a paper-open item with machine backing; and no premise of the ledger is claimed, here or elsewhere, to be within reach of the machine layer except as recorded. The Lean layer sharpens checkability, not the claim surface: its `sorryAx`-free theorems are those of Table 3, its `native_decide` certificates are quoted with the kernel-reduction disclosure, and everything outside the axiom probe is explicitly not claimed formalized (Section 4.4).

What is measured. The measured layer is reported at its registered strengths throughout: the five exhaustively enumerated domains of the budget conjecture (Table 1; zero violations, 208 tool-dead instances all satisfying the conjecture); the nine-domain key ledger under its four conventions (Section 4.5); the per-N criterion closures with zero residue on 588/30,760/634,960/7,798,903 instances at $N = 6, 8, 10, 12$; and the classification programme's four complete-graph layers at their stated tiers (Section 3.2.2). The overall conjecture — Gallai's path decomposition conjecture — remains open. What this paper provides is the verified portion of the landscape around it, a conditional route whose every edge is checkable, and a measurement apparatus whose every citation

carries its reading conventions (the four ledger conventions of Section 4.5 and the tier qualifiers of Sections 4 and C.1), so that the paper remains checkable against exactly the evidence it quotes.

The no-claim constraints, restated. The six no-claim constraints of Section 1.3 bind this closing section as they bind the whole account; in the order stated there, and matching the introduction's wording in content: no closure at $L \geq 13$ (candidate level only); no general-N paper closure of the covering identity (the existence half open); no completeness claim for the four-piece (E)+(R) landing; no paper closure of the general-L uniform-construction account (the capacity-starved branch of the channel-or-menu dichotomy); no full closure of that dichotomy programme; and no unconditional theorem-form statement of the valuation criterion — its Lean formalization is a bound under an unproved hypothesis, and a criterion verified over finitely many N is not an existence proof for general N.

References

[1] Erdős problems, problem 583. https://erdosproblems.com/583. Accessed September 2026.

[2] Brian Alspach. The wonderful Walecki construction. *Bull. Inst. Combin. Appl.*, 52:7–20, 2008.

[3] Marthe Bonamy and T. J. Perrett. Gallai's path decomposition conjecture for graphs of small maximum degree. *Discrete Math.*, 342(5):1293–1299, 2019.

[4] Fávio Botler and Márcia Sambinelli. Towards Gallai's path decomposition conjecture. *J. Graph Theory*, 97(1):161–184, 2021.

[5] Fávio Botler, Márcia Sambinelli, Rafael S. Coelho, and Ondrej Lee. Gallai's path decomposition conjecture for graphs with treewidth at most 3. *J. Graph Theory*, 93(3):328–349, 2020.

[6] Xing Chen and Baoyindureng Wu. Gallai's path decomposition conjecture for block graphs. *AIMS Math.*, 10(1):1438–1447, 2025.

[7] Yao Chu, Guangming Fan, and Qinghai Liu. On Gallai's conjecture for graphs with maximum degree 6. *Discrete Math.*, 344(2):112212, 2021.

[8] Yao Chu, Guangming Fan, and Chunxiang Zhou. Path decompositions of triangle-free graphs. *Discrete Math.*, 345(7):112866, 2022.

[9] Yao Chu, Guangming Fan, and Chunxiang Zhou. Gallai's conjecture and the path number of odd semi-cliques. *Discrete Math.*, 349(2):114725, 2026.

[10] Christodoulos K. Constantinou and Georgios Ellinas. Minimal path decomposition of complete bipartite graphs. *J. Comb. Optim.*, 35(3):684–702, 2018.

[11] Nathaniel Dean and Méziane Kouider. Gallai's conjecture for disconnected graphs. *Discrete Math.*, 213(1–3):43–54, 2000.

[12] Guang-Hui Fan, Jia-Feng Hou, and Chun-Xiang Zhou. Gallai's conjecture on path decompositions. *J. Oper. Res. Soc. China*, 11:439–449, 2023.

[13] Guangming Fan. Path decompositions and Gallai's conjecture. *J. Combin. Theory Ser. B*, 93(2):117–125, 2005.

[14] Odile Favaron and Méziane Kouider. Path partitions and cycle partitions of Eulerian graphs of maximum degree 4. *Studia Sci. Math. Hungar.*, 23(1–2):237–244, 1988.

[15] Tibor Gallai. On directed paths and circuits. In Paul Erdős and Gyula Katona, editors, *Theory of Graphs (Proceedings of the Colloquium held at Tihany, Hungary, 1966)*, pages 115–118. Academic Press, New York, 1968.

[16] László Lovász. On covering of graphs. In Paul Erdős and Gyula Katona, editors, *Theory of Graphs (Proc. Colloq., Tihany, 1966)*, pages 231–236. Akadémiai Kiadó, Budapest, 1968.

[17] Changhong Lu and Niping Yi. The problem of path decomposition for graphs with treewidth at most 4. *Discrete Math.*, 347(6):113957, 2024.

[18] Édouard Lucas. *Récréations mathématiques*, volume II. Gauthier-Villars, Paris, 1892.

[19] Tomáš Masařík, Michał Włodarczyk, and Mehmet Akif Yıldız. Optimal path partitions in subcubic and almost-subcubic graphs. In *LIPIcs*, volume 376, 2026.

[20] László Pyber. Covering the edges of a connected graph by paths. *J. Combin. Theory Ser. B*, 66(1):152–159, 1996.

[21] Junhao Zhang, Qinghai Liu, and Yanmei Hong. Gallai's conjecture for 3-degenerated graphs. *Discrete Math.*, 347(7):114057, 2024.

Table 3: The nine `sorryAx`-free Lean items cited by this paper. Axiom profiles: [S] := `[propext, Classical.choice, Quot.sound]` (standard), [P, Q] := `[propext, Quot.sound]`, and nd := the `Lean.ofReduceBool` kernel-reflection axiom instance. Line anchors refer to the released sources where the file is released; `CompleteOdd.lean` and `Cert8.lean` are not part of the released snapshot and their anchors refer to the audited development snapshot of the project.

Declaration	Statement (exactly as proved)	Anchor	Axioms
`pNumK4n_odd_all`	for every odd $n \geq 3$: $p(K_{4,n}) = (n+1)/2$; hypotheses $3 \leq n$, n odd	`T273.lean` :976–978	[S]
`cor_pNumK4n_5/13/19`	$p(K_{4,5}) = 3$, $p(K_{4,13}) = 7$, $p(K_{4,19}) = 10$	`T273.lean` :981–987	[S]
`pCompleteOdd`	for $k \geq 1$: $p(K_{2k+1}) = k + 1$; hypothesis $k \neq 0$	`CompleteOdd.lean` :1590–1596	[S]
`thmA2663_upgraded`	under hypotheses hd2 (target side has the prescribed size) and val ≤ 1: $g_{13} \leq d_2 + 1$ — the val ≤ 1 hypothesis is *not* proved by the theorem	`ThmA2663.lean` :197–200	[P, Q]
`case2_struct_alpha_iff`	under six explicit hypotheses: $C_1((t_1 + g) \bmod N) =$ `true` $\iff t_1 = s$	`Struct2663.lean` :260–265	[P, Q]
`ledger6_ok`	$\text{ledger}_6 =$ (5,724; 2,613; 190; 398; 588) (N=6 five-column global account)	`Cert2663.lean` :113	[P, nd]
`ledger8_ok`	$\text{ledger}_8 =$ (701,360; 285,372; 20,308; 10,452; 30,760) (N=8; dual-route, second Python implementation pinned)	`Cert8.lean` :52	[P, nd]
`l2b_toplevel_certificate`	the 79,200-point packing-table certificate (120 orbits × 330 demands × two sides), pointwise iff	`Pi3L2b.lean` :248–249	[P, nd]
`Pi3Uni` (nine theorems)	downset tables (both sides); uniform layer; the no-threshold witness refuting any "feasible $\iff$ $f(\Sigma d)$" law; top-only failures (1,761); top-failure set an upset; orbit counts 110/2	`Pi3Uni.lean` :57–115	[P, nd]